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Backyard Generating Station

Version 1.5

24 September 2026

One house. One backyard generating station. Electricity and heat, day and night, for 100 years.

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1. The Need

The United States is asking the grid to do two jobs at once, and it is falling behind on both.

It must make a lot more electricity: data centers, factories, electric cars, and houses switching from gas heat to electric heat. Demand that sat almost flat for twenty years is now pointed up. One recent national forecast puts use on the order of **20% higher by 2030** and **about 40% higher by the mid-2030s** ([ICF, 2026](#)). Spare generating capacity above what planners already need is already thin, a few percent of the fleet. Some of the biggest regional grids have little or no extra room.

It must also keep the lights on with wires, poles, and buried cables that were treated as permanent and are not. In 2024 the average US customer sat through about **11 hours** without power, nearly twice the yearly average of the decade before. Major weather did most of that damage. Even in ordinary years, with the big storms stripped out, the average customer still loses about **two hours** ([US Energy Information Administration](#)).

Those two facts meet at the house. The bill is rising because new plants and new wires cost money. The outages are rising because the old plant and the old wires are failing, or because the utility turns the power off on purpose when wind and dry brush make a live line a fire starter. A house that can keep its own power on is how you stay cool in a heat wave, warm in a freeze, and alive if the machine in the bedroom needs a wall outlet.

The Backyard Generating Station is one answer to that need: power made at the house, compared at the same meter to rooftop solar. This chapter is the need, not the machine.

The country is short of new power

For a long stretch, US electricity use barely grew. Efficiency hid new gadgets. That stretch is over.

The new load is lumpy and fast. A single data-center campus can ask for as much power as a small city. Car chargers and heat pumps add house by house. Seattle City Light, to take one utility that has said this publicly, expects electrical demand in its territory to **double in about twenty years**, and it is already shopping for new wind, solar, and geothermal because the old hydro will not cover it ([Seattle Times, Aug. 16, 2026](#)). That utility also needs new substations at a couple hundred million dollars each, and a multi-billion-dollar dam upgrade. The pattern is national: more load, long lead times, and towns that do not want a 1,000 MW plant next door.

When load grows faster than plants and transmission, two household results follow. Rates go up to pay for what does get built. Reserve margin goes down, so a hot afternoon or a cold week has less slack. Chapter 2 puts numbers on the bill. The need underneath those numbers is simple: the country is trying to electrify more of life on a grid that was not sized for that, and that is not being rebuilt at the same pace.

The wires are old, in many cities at once

Outages are not only hurricanes. They are also 1960s cable that was buried as if it would last forever.

Seattle is a clean example because the paper trail is public. More than **300 miles** of underground cable are decades past useful life. Replacing them is pegged at more than **\$2 billion** over at least 15 years. Average restoration time went from **78 minutes in 2014** to **145 minutes in 2024**, even with big weather events taken out, which put that utility in the worst quarter of the country for getting the lights back on. Neighborhoods that buried lines in the 1960s and 1970s for looks now lose power over and over. One Queen Anne household logged more than 20 outages after 2021, including six in the first half of a single year. A “run to failure” habit (keep rates low, fix it when it breaks) is what a former utility chief called the worst reliability she had seen in her career. The cables were never a permanent object. There was no fund waiting when the bill came due ([Seattle Times](#)).

That is one city. The same aging-and-weather pattern shows up across the map:

- **Texas, February 2021.** A winter storm and a grid that could not keep generation online left millions of homes dark in a freeze. The state later counted **246** storm-related deaths. Most were hypothermia. Others were carbon monoxide from grills and generators, house fires from space heaters, and people whose oxygen, dialysis, or other electric medical gear stopped ([Texas DSHS](#); [Texas Tribune](#)).

- **Houston and the Gulf, July 2024.** Hurricane Beryl knocked out more than **2.1 million** CenterPoint customers, about three-quarters of that territory. Restoration ran about **11 days** ([CenterPoint after-action](#)). EIA's national 2024 spike is, in large part, Beryl plus Helene plus Milton ([EIA](#)).
- **Carolinas and Florida, 2024.** Helene and Milton took down transmission, distribution, and substations. South Carolina's average customer lost on the order of **50 hours** that year, an outlier that is still a real year in a real state ([EIA](#)).
- **California, on purpose.** Utilities turn lines off when wind and dry fuel make a spark into a fire. A Public Safety Power Shutoff can run a day or two, sometimes longer, across many counties at once. Oregon, Colorado, Idaho, and Hawaii have copied the tool ([CPUC on PSPS](#)). The wire is aging and the climate is drier. The tool they use is a planned blackout.
- **Portland metro, January 2024.** Ice and wind took out hundreds of thousands of customers. Some sat in the cold for about a week. At least ten deaths were tied to that week: hypothermia, trees, lines, fires ([KGW](#)).

The national scoreboard is the EIA reliability tables: with major events included, the average customer's yearly minutes without power jumped from **342 in 2023** to **611 in 2024**. Strip the major events and you still have about **two hours** a year of ordinary failure, the everyday signature of trees, old cable, and deferred maintenance ([EIA Table 11.3](#)).

The same aging-cable problem is not limited to Seattle. It is what “run to failure” looks like when the failure arrives in a lot of cities in the same decade.

The same story, elsewhere

Aging plants and thin reserve are not only an American problem.

Spain and Portugal, April 28, 2025. A voltage collapse took down most of the Iberian grid, the worst European blackout in about twenty years. Portugal’s transmission was back in about 12 hours. Spain’s took about 16. Elevators, trains, and hospitals ran on whatever backup they had. The official investigation is public ([ENTSO-E expert panel](#)).

South Africa, years on end. Eskom’s aging coal fleet could not meet load, so the country lived on scheduled “load shedding”: hours a day, by neighborhood, for years. When the plants run better, the blackouts ease. When they do not, the schedule returns. That is what a generation shortfall looks like as an everyday schedule, not a one-week storm ([Eskom system updates](#)).

Puerto Rico. After Hurricane Maria and through the years after, an old, storm-hit grid left households on diesel and long dark stretches. It is a US territory living the end state of deferred rebuild.

The point for a US house is that a rich country is not exempt. If the plants are late and the wires are past their life, outages become ordinary. Seattle's 15-year cable program and South Africa's load-shedding calendar are the same problem at different speeds: not enough working infrastructure for the load that is already there.

How uptime saves lives

Electricity is no longer only lights and television. For a growing share of households it is a medical device, a climate-control system, and the pump that keeps water clean. When it stays on, people are less likely to die from what follows the outage: a dark house in a heat wave, a freeze, or a medical device that stops.

Heat. Air conditioning is life-safety equipment in a heat wave. Older adults, infants, and people on certain medicines cannot dump heat. A multi-day outage in July is not an inconvenience. It is a medical event in the living room. The house that keeps 8 kW on through that week is a house that keeps people out of the ER.

Cold. Texas 2021 is the large, counted case: hundreds of deaths, most from the house going as cold as the street. Portland 2024 is the smaller cousin. Space heaters and ovens used as furnaces start fires. The deaths are hypothermia and the fire, not “the storm” in the abstract.

Home medical gear. HHS tracks Medicare beneficiaries who live at home on electricity-dependent equipment: oxygen concentrators, ventilators, BiPAP, home dialysis, infusion pumps, powered beds

and chairs, implanted heart-assist devices. That electricity-dependent group is **over 3 million** people. Count those plus people who need dialysis, tank oxygen, home health, or hospice and the at-risk total is **over 4.6 million**, in every state, down to the ZIP code ([HHS emPOWER](#)). That list is Medicare only. It misses children (the Queen Anne household whose son's air purifier dies in the night), working-age adults on a CPAP or a fridge full of insulin, and anyone whose pharmacy cold-chain sits in a kitchen refrigerator. Hours without power is enough. Days is a hospital trip or worse.

The generator that kills. When the grid dies, people drag out gasoline generators, run cars in garages, and light grills indoors. Texas counted **19** carbon monoxide deaths in that one freeze. Uptime at the panel is how those people never start the motor.

Water and waste. City pumps, well pumps, and lift stations run on electricity. After a long outage you get boil-water notices, dead sewage pumps, and a house that cannot flush. Infection and dehydration follow the dark, especially for the same people already on medical gear.

The rest of the street. Traffic lights, cell sites, and 911 answering all have hours of battery or diesel, not weeks. A house that stays up does not restore the city. It keeps the people in that house off the overwhelmed hospital list while the city is down.

Seattle's story already has the small version: a toddler who coughs when the purifier stops, a parent who cannot keep the house warm, a remote class that assumes the wall outlet works. Scale that by 4.6 million Medicare households and by every heat wave and ice storm, and keeping power on is a public-health issue.

What this means at one house

A typical house uses about 10,000 to 12,000 kWh a year. A high-use house can run \$400 to \$600 a month. Chapter 2 does the bill math, including rates that are already climbing.

The need under that math is uptime and capacity in the same object.

Rooftop solar at that house makes power when the sun is out. In a heat wave that can help in the afternoon. At night, in an ice storm, in a week of coastal weather, or during a Public Safety Power Shutoff, the array is a cold roof. A battery helps for hours, not for a Texas week, unless something is still filling it.

The Backyard Generating Station is sized to keep filling that battery in the dark: 5 to 10 kWe, plus heat the house can use. It does not replace a national rebuild of 300 miles of Seattle cable or a new 1,000 MW plant. It is how one family stays on while that rebuild is 15 years out, and how that family adds a slice of supply the country is short of, at the place the load already is.

If the country is heading into a decade or two of tight reserve and old wires, the house that can run through an outage is the house that does not add a hypothermia death, a heat stroke, a silent oxygen concentrator, or a generator in the garage to the tally.

What this chapter covered

This is the need: more electricity, older wires, longer dark hours, and a medical dependence on the outlet that did not exist at this scale when the 1960s cable went in. The examples are Seattle, Texas, Houston, the Carolinas, California, Portland, and the same failure mode abroad. The life-safety case is heat, cold, home medical gear, carbon monoxide, and water. Chapter 2 is the product and the payback.

2. The Idea

A house should be able to make its own power, keep the lights on in a storm, heat the rooms it already knows how to heat, and sell what it does not use. It should do that without covering the roof in solar panels, without depending on roof seals that can leak for 25 years, and without a fuel truck in the driveway.

That is the Backyard Generating Station: a sealed, factory-built appliance, about the size of a large septic tank, with **ten feet of soil above the lid**. American yards already bury tanks this way (septic tanks, underground propane, heating-oil tanks). Ten feet of cover is on the deep side of that habit, closer to a large buried fuel tank than to a shallow septic lid. The object itself needs about a **10 by 10 foot** patch of yard. How far it sits from the house is a local permit question.

A licensed crew installs it once. Nobody on the property ever opens it. It makes electricity for the house, stores the extra in a wall battery, sends overflow to the grid, and offers leftover heat to the heating system many newer American homes already have.

It is meant to be compared, at the same address, to rooftop solar: what you pay, what you maintain, what you get, and what you can sell back.

What it is

The station is a lead-lined concrete vault, about the size and weight class of the buried tanks already common in American yards. A small excavator digs the hole. A crane sets the vault. The crew puts ten feet of soil back on the lid and the grass goes back on top. It is never serviced in place. It is designed to run for on the order of a century and then be lifted out as one piece.

Above ground you see almost nothing. Heat and a few wires reach the house through a **chase**: a buried pipe run from the underside of the vault to a **utility room**. That room can be in the house (a basement, a garage wall) or a small shed next to it. The house sees four familiar things:

- a circuit that feeds the electrical panel, the same way a backup generator or a solar system does
- a wall battery that stores extra electricity so the house can use more than the station is making at that exact moment (oven, dryer, and air conditioner at once)
- an inverter, the same kind of box a solar install uses, that turns the station's output into ordinary house current and talks to the utility
- a closed water loop, where the house can use it, that hands leftover heat to systems that already run on hot water

The homeowner does not operate a plant. They own an appliance.

How big this is

Power is easier to feel with a few anchors.

A typical American house uses about 10,000 to 12,000 kWh in a year (kilowatt-hours: how much energy you used over time). That is, on average, a little over 1 kW (kilowatts: how hard power is flowing right now), all day, every day, and can spike to 10 or 20 kW when the air conditioner, range, and dryer hit at once.

Some houses use much more. A high-use house can run **\$400 to \$600 a month** for electricity (**about \$5,000 to \$7,000 a year**). That is more kilowatt-hours, a higher price per kWh, or both. At ordinary rates that is on the order of 20,000 to 35,000 kWh a year, an average of about 2.5 to 4 kW, with the same kind of evening spikes. The Backyard Generating Station is sized for that high end as well as for the typical house.

Rooftop solar on that same house is usually an 8 to 12 kW array. On a good sun day it can outrun the house. At night, and through a week of coastal weather, it makes nothing. That is why solar quotes always include the roof, the weather, and a battery if you want the house to keep working after dark.

A utility-scale solar farm is the same idea, multiplied: tens or hundreds of megawatts (1 MW = 1,000 kW), spread over fields, feeding the grid rather than one kitchen.

A typical commercial generating station of the large kind most people picture is about 1,000 MW. That is a million kilowatts, enough, in round numbers, for several hundred thousand homes. It is a campus: cooling towers, switchyards, a security perimeter, a staff.

The Backyard Generating Station sits at the house end of that range. It is sized for **5 to 10 kWe** (kilowatts of electricity: the “e” just means this number is electrical output, not heat) of steady output. Next to a 1,000 MW plant that is about **1/100,000 to 1/200,000** the electrical size. It is a home-furnace-scale heat source that also makes power. That is why it belongs in a yard the way a buried tank does, and why it is not a commercial plant next door.

It is in the same size class as a rooftop solar array, except it keeps making power when the sun is out and when it is not.

It does not try to match the 20 kW spike. The battery does that. The station keeps charging the battery. When the battery is full and the house has what it needs, extra power goes to the grid.

A note on the units. kW is the size of the pipe. kWh is the water that went through it. If the station makes 8 kW for one hour, that is 8 kWh, about what a central air conditioner uses in an hour, or a little less than a day’s worth of a modest house that is not heating or cooling hard. Over a year, 8 kW running most of the time is on the order of 70,000 kWh. A typical house that uses 12,000 kWh has a lot left to sell. A high-use house that uses 20,000 to 35,000 kWh still has tens of thousands of kWh left to sell, or to put into a second car, a workshop, or a storm reserve.

What	Typical size	What that means
House, typical average	~1 kW	About 24 kWh/day, or 10,000 to 12,000 kWh/year. About \$2,000 a year at 18¢/kWh.
House, high-use average	~2.5 to 4 kW	About 20,000 to 35,000 kWh/year. About \$5,000 to \$7,000 a year (\$400 to \$600 a month).
House, peak	10 to 20 kW	Air conditioner, range, and dryer at once. The battery covers this.
Rooftop solar	8 to 12 kW	Same house-scale neighborhood. Makes power only when the sun is out.
Backyard Generating Station	5 to 10 kWe	Same neighborhood as rooftop solar, as a steady output, sun or not.
Utility solar farm	tens to hundreds of MW	Fields. Feeds the grid, not one kitchen. (1 MW = 1,000 kW)
Large commercial station	~1,000 MW	A campus. About 100,000 to 200,000 times the electrical size of this unit.

You do not need to build a 1,000 MW commercial station to serve those several hundred thousand households. You can build one right-sized unit per house and serve the need where the load already is, the same placement idea as a rooftop solar array, without the campus, the cooling towers, or the long wires in between. Rooftop solar already shows that house-scale generation is a normal way to site power. The station is the version that still works when the sun does not.

A day in the life

The station runs as a quiet, steady source. The battery covers the short spikes.

Morning: the house wakes up. The battery covers the coffee-and-shower bump. The station is already making power, so the battery does not drain far.

Afternoon: the house is light. The battery fills. If it is full, the inverter exports to the meter the same way a solar system does.

Evening: oven, laundry, screens. The battery covers the peak. The station keeps topping it off. On the off chance the house wants more than the battery still holds (a rare, hard spike), the panel just pulls the rest from the grid, the same way it does today. The lights do not go out. You buy a bit of power for an hour instead of a whole day's worth.

Night: same as afternoon, except solar is not making power. That is the fair comparison with solar: how many kilowatt-hours this house gets in a year, and what still works when the weather is bad.

Heat is a second product of the same day. The station makes electricity and warmth together. In winter that warmth heats the house. In summer the house may only want showers, or it can send the rest to a greenhouse. Later chapters treat the hardware. The idea is simple: use the warmth where a pipe can take it.

Heat the house already knows how to use

A lot of newer houses in the United States already heat with hot water in pipes, in the floor, in a tank, or in a coil, not only with hot air blown through vents.

The station never mixes its insides with the house. Heat crosses through a sealed metal exchanger: two water loops share warmth through a wall, and the waters never touch. A home boiler already works this way. What the house already has can take that water:

- **Radiant floors.** Plastic heating pipe (PEX) in a slab or under a floor typically wants water in a modest range, roughly 80 to 120°F. That is the temperature leftover warmth can supply. It is not furnace-flame temperature.
- **Indirect tanks and buffer tanks.** Many boiler rooms already have a tank with a coil in it. The station can be the thing that keeps that tank hot, for showers and laundry, and as stored heat for the heating loop.
- **Hydro-air.** Some houses have both a boiler and ducts. A coil of hot-water pipe sits in the air handler (the big box that already blows air through the house). The station heats that coil. The thermostat and the blower stay the same.
- **A high-efficiency boiler loop.** Newer houses with a condensing boiler already have a hot-water piping loop. The station ties in as a second heat source, isolated, and the boiler becomes backup or peak.

- **Air-to-water heat pumps.** A growing share of new work uses a heat pump to warm a tank of water. The station can preheat that tank so the heat pump does less work, or carry the load when that is the better choice.
- **A greenhouse.** If the house has no hot-water heating loop, or if it does not want the warmth in summer, the same loop can run to a greenhouse. Year-round vegetables and fruit become possible in climates and geographies that would not support that on sunlight and a furnace bill alone. The house still gets the electricity. The plants get the heat.

Houses that are all electric baseboard, or mini-splits with no water loop, still get electricity, battery, and grid sale. They can still take a standalone tank for showers and laundry. They do not get whole-house space heat unless someone adds hot-water piping, or they put the warmth into a greenhouse instead. If the house already has a boiler room or a radiant-floor manifold, the station connects to that hardware. If it does not, you are buying a power appliance. Space heat is then a choice: add piping, add a greenhouse, or use the electricity only.

Winter heating for a typical house is often in the 5 to 15 kW range. A station making 5 to 10 kW of electricity will also be making a couple of times that in heat. That is enough, on many days, to run the floors and the water heater without lighting the old boiler.

Living with it

Day to day there is nothing to do. The station keeps making power. The battery tops itself off. Extra goes to the grid. Heat follows the thermostat the house already has. When you leave, the setback you already use is enough.

An app is not required. The thermostat and the utility meter are enough. If you install one on a phone or a tablet, it is only a status window, the same habit solar owners already have: what the station made, what the house used, what was sold, what is in the battery. It can make those numbers easier to see. It does not run the station. It has no modes and no switches that start, stop, or change the vault.

How it makes power

The vault holds a very small nuclear core. Heat comes from splitting atoms in a sealed fuel, at a scale closer to a large home furnace than to a commercial station.

That heat does two jobs for the house at the same time. It makes electricity for the panel. It can also warm rooms, showers, or a greenhouse. Using both is the design. If the house or a greenhouse uses the warmth, most of what the core makes is used, not thrown away.

The numbers stay small on purpose.

A house-scale electrical output of 5 to 10 kW goes with about **20 to 40 kW of heat** from the core. That is furnace scale, not power-plant scale. After the unit is shut down, the leftover heat that has to go

somewhere drops to about as much as a small space heater, then lower over the following hours and days. Dirt, concrete, and time can take that. There is no cooling tower.

Fuel use at this output, over a century, is measured in kilograms, not tons: about **30 kilograms (about 66 pounds) at first start**, a paint-can of heavy metal inside a beer-keg-to-dishwasher-sized core, itself inside the septic-tank vault. By the end of life, on the order of 15 to 25 kilograms of that heavy metal has been used up. Chapter 8 is the arithmetic. Chapter 9 is the fuel, the graphite, the sodium in the pipes, and the physics of how the reaction is meant to stop. The vault is the thing the crane sees. The core is never a backyard project.

Because the core is run cold and slow (a trickle of heat in a lot of mass), it can be built for a very long sealed life. Material mixed into the fuel is used up as the fuel is used up, so the output stays even. There is no refueling truck. There is no hatch for the owner.

The fuel and the core are built so heat itself slows the reaction: hotter means less splitting, in milliseconds, without a person or a motor. That is the ordinary safety case. Chapter 9 is that Doppler brake, the drums, and the last-ditch glass. A last-ditch layer above the core can melt and set into a glass-like mass if the vault ever got far hotter than it should. That is not expected to run. The design is meant to never get there. The unit is too small for leftover heat to run away, it sits in earth, and it does not rely on a pump staying on.

From the factory to first power

The vault is not “on” when it is built, and it is not on when it rides on a truck.

Fresh fuel has never been used, so it is not already hot. In the factory the core is assembled in a shutdown state: locks and control elements are in the position that keeps it from making heat. The vault is sealed around that. What leaves the plant is a heavy, closed object that is not producing electricity and not producing heat.

It is trucked like other permitted heavy freight and set with a crane. The crew connects the house wires, the inverter, the battery, and, if the house will use it, the water loop. Inspectors sign off.

Only after it is buried and connected does a licensed startup happen. The shutdown locks are released in a controlled procedure. That is the first time the core makes heat. Until that step the vault is cold. At the end of life the same locks go back in before the crane returns.

This is also why a size-based path is part of the idea. A 1,000 MW station is a power plant. A 20 to 40 kW heat source in a factory-sealed vault is an appliance, about 1/100,000 the electrical size of that plant. The project’s aim is not to site a campus behind a house. It is to get a manufactured, never-opened unit approved, or exempted, because it is this small, installed by licensed people, and left closed until a licensed crew takes it away.

What it costs

These are high-level bands for a production unit, not a first prototype. Early units cost more. The check you write is in today's dollars. The bill you stop paying will not stay at today's rate. The country is heading into a stretch where most regions will be short of power, and the cents you pay per kWh have been rising and are expected to keep rising. That rising rate belongs in the cost picture.

The station itself. The sealed vault is the manufactured product. In volume, think in the same world as a serious whole-home energy project: tens of thousands of dollars, not hundreds of thousands, once it is a factory object rather than a custom machine. The first ones will not be that.

Setting it. A contractor digs a hole deep enough for the vault plus **ten feet of soil on the lid**, a crane sets the vault, and the crew beds and buries it. The grass footprint is about **10 by 10 feet**. The vault has to be able to sit wet, because a lot of yards will flood in the life of the unit. In ordinary soil that buried-tank job is often **\$10,000 to \$20,000**. If the crew hits rock, or the hole fills with groundwater, the same work can run **\$25,000 to \$40,000**.

Power tie-in. Panel work, the inverter, and the utility hookup: the same class of electrical work as a solar install. Roughly **\$8,000 to \$15,000**, depending on the service and the utility.

Stirling engines. Five sealed 2 kWe convertors in the utility room (10 kWe installed, 8 kWe used, one spare). They sit next to the inverter, not in the vault. Each unit is about the size of a large fire extinguisher (about 12 inches across and 18 inches tall, about 110

pounds). Five of them, plus the oil pump and the heat exchanger, fit a basement utility room or a small shed. They do not fit in a closet next to the panel.

Today, at low volume, that set is about **\$40,000 to \$100,000** (about \$8,000 to \$20,000 per engine). If they become an ordinary factory part, a working target is **\$15,000 to \$30,000** for the set. That first set is part of the install check. They last about **10 to 15 years** and get swapped in that room. The swap is a later check, like a furnace. It is not in the simple-payback tables below.

Battery. Two or three wall units in the Tesla-class size (about 13 kWh each) to cover peaks and a storm night: often **\$20,000 to \$40,000** installed. This is the same kind of line item solar already has if you want the house to work after dark.

Heat tie-in. If the house already has a radiant-floor manifold, a hot-water tank, or a boiler loop, an isolated exchanger and controls are a few thousand dollars to about **\$10,000**. If the house has no hot-water heating pipes, you either skip space heat, add a tank and piping on purpose, or run a loop to a greenhouse.

All-in at a house that can use the heat:

- **Early production: \$130,000 to \$260,000** is the honest band while the vault is still a new manufactured object and the Stirling set is still a low-volume part (**\$40,000 to \$100,000** of that band).

- **Factory scale** (the vault is an ordinary appliance, built in numbers): a working target is **\$70,000 to \$125,000** all-in. The vault gets cheaper. The hole, the Stirling set (**\$15,000 to \$30,000**), the inverter, and the batteries do not vanish.
- **With a government credit like the one solar received** (a federal investment tax credit, often 30% in solar's early years, applied to the generating equipment, including the Stirling set): early production could net in a **\$90,000 to \$180,000** band, and factory scale in a **\$50,000 to \$90,000** band. That credit is a policy path, not a promise. Solar got one because the country wanted more carbon-free generation at the house. Green, in this book, means that: carbon-free electricity and heat at the house, and output that does not stop when the sun goes down. The same argument applies here, with a second one: every house-scale station is a house that does not need a new 1,000 MW plant sited in someone else's town. Communities already refuse those plants. The United States already generates almost all of the electricity it uses; the stronger public case is carbon-free power at the load, fewer giant stations, and a factory product that does not depend on a roof full of imported panels.

What a house uses, and how long the check takes to come back

Household electricity is a range, not one number. Some months are higher (air conditioning, electric heat) and some are lower. The yearly total is what payback uses.

- **Typical house:** about **10,000 to 12,000 kWh a year** (about 900 kWh a month). At a recent US average of about **18 cents per kWh**, that is about **\$2,000 a year**, or about \$160 to \$170 a month. Space heat and hot water are often a second bill of **\$800 to \$2,000 a year** if they are not already on that electric meter.
- **High-use house:** about **\$400 to \$600 a month**, or **\$5,000 to \$7,000 a year**. A \$560 month is about \$6,700 a year. That may be 20,000 to 35,000 kWh (a bigger house, more electric loads), a higher cents-per-kWh rate, or both. On a bill this large, electric heat or a heat pump is often already inside the \$560. Do not add a second heat line on top unless the house still pays a separate gas or oil bill.

The Backyard Generating Station is sized at 5 to 10 kWe of steady output. At 8 kWe, running most of the year:

$$8 \text{ kW} \times 8,760 \text{ hours in a year} \approx 70,000 \text{ kWh}$$

A typical house keeps about 11,000 kWh of that and can sell about **59,000 kWh**. A high-use house keeps about 20,000 to 35,000 kWh and can still sell **35,000 to 50,000 kWh**. Either way, 8 to 10 kWe covers the high end; 5 kWe is a tighter fit if the house averages near 4 kW.

Rooftop solar at the same house is usually an 8 to 12 kW array. It makes about **10,000 to 14,000 kWh a year** on a decent roof, and less on a shaded, cloudy, or low-sun roof. That can match a typical house and not much leftover. On a high-use house it covers only part of the bill. It does not make space heat unless the house already heats with electricity.

So the two systems are not the same size in delivered energy. Solar is built to match a typical house. The Backyard Generating Station, if it runs as a steady 8 kW, makes several times even a high-use house's electric load. Payback then depends on two prices: what you no longer pay the utility (and any leftover heating fuel), and what the utility pays you for the extra.

Simple payback is cost divided by a year's savings **at today's rate**, as if the bill never moved. That is the snapshot in the tables below. It overstates the years if rates keep climbing, which they have been. These are round numbers for a factory-scale Backyard Generating Station after a solar-style 30% credit (**\$50,000 to \$90,000**, including the first Stirling set), next to a complete solar-plus-battery job at the same meter.

Typical house (~\$2,000 electric bill):

	What you pay (net)	Electricity made in a year	A year's savings (round)	Simple payback
Solar + battery, ordinary roof	\$35,000 to \$55,000	10,000 to 14,000 kWh (about what the house uses)	About \$2,000 (the electric bill)	12 to 20 years
Solar + battery, shaded, cloudy, or low-sun roof	\$35,000 to \$55,000, sometimes more	Less than the house uses	About \$1,200 to \$1,800	25 to 30 years
Backyard Generating Station, extra paid near the retail rate (~18¢/kWh)	\$50,000 to \$90,000	~70,000 kWh	About \$12,000 to \$14,000 (bill + heat + sell-back)	4 to 8 years
Backyard Generating Station, extra paid at a lower wholesale-style rate (~5¢/kWh)	\$50,000 to \$90,000	~70,000 kWh	About \$5,000 to \$7,000	7 to 18 years
Backyard Generating Station, extra paid nothing (house offset and heat only)	\$50,000 to \$90,000	~70,000 kWh	About \$2,800 to \$4,000	13 to 32 years

High-use house (~\$5,000 to \$7,000 electric bill, about \$400 to \$600 a month):

	What you pay (net)	Electricity made in a year	A year's savings (round)	Simple payback
Solar + battery, ordinary roof	\$35,000 to \$55,000	10,000 to 14,000 kWh (only part of the bill)	About \$2,000 to \$3,000	15 to 25 years
Solar + battery, shaded, cloudy, or low-sun roof	\$35,000 to \$55,000, sometimes more	Less than that	About \$1,200 to \$2,000	25 to 40 years
Backyard Generating Station, extra paid near the retail rate	\$50,000 to \$90,000	~70,000 kWh	About \$12,000 to \$16,000 (the \$5,000 to \$7,000 bill plus sell-back)	3 to 8 years
Backyard Generating Station, extra paid at a lower wholesale-style rate (~5¢/kWh)	\$50,000 to \$90,000	~70,000 kWh	About \$8,000 to \$10,000	5 to 11 years
Backyard Generating Station, extra paid nothing (house offset only)	\$50,000 to \$90,000	~70,000 kWh	About \$5,000 to \$7,000	7 to 18 years

Solar's problem on a high-use house is simple: the array still makes about 10,000 to 14,000 kWh, so most of a \$6,700 bill remains. The Backyard Generating Station's problem is still the sell-back price, not whether it can cover the house.

The grid is getting short, and the rate is going up

For about twenty years, US electricity use was almost flat. That stretch is over. Data centers, factories, electric cars, and houses switching from gas to electric heat are adding load faster than new plants and transmission are getting built. Demand is on track to be on the order of **20% higher by 2030** and **about 40% higher by the mid-2030s**. The spare generating capacity sitting above what the grid already needs is already thin (a few percent nationally). Some of the biggest regional grids already have little or no extra room. New 1,000 MW stations take years and are the plants communities already refuse. So for the next decade to a few decades, most places in the country are more likely to be short of power than swimming in it.

When the grid is short, two things happen to a household bill.

The cents per kWh go up. They already have. The US residential average has climbed from the low teens (cents per kWh) a decade ago to about **18 cents** now, and recent years have been on the order of **4% to 6% a year**. Forecasts for the rest of this decade often land in a **15% to 40% higher bill by 2030**, depending on the utility: more plants, more wires, and more competition from big new

loads. A \$2,000 typical bill becomes about \$2,300 to \$2,800. A \$6,700 high-use bill becomes about \$7,700 to \$9,400. After that, the climb does not automatically stop.

The kilowatt-hour you do not buy is worth more each year. Simple payback pretends year 10 is still \$2,000 or \$6,700. It is not. At 5% a year, a \$6,700 bill is about **\$10,900 in ten years**. Every year you are not paying it, the savings get larger, so the remaining balance on the \$50,000 to \$90,000 check shrinks faster than the frozen-rate tables show.

A compact way to see it: same \$50,000 to \$90,000 Backyard Generating Station (factory scale, after credit, first Stirling set included), same solar jobs, if the rate rises about **5% a year** (in the middle of the recent pace):

	Frozen-rate payback (tables above)	If rates rise ~5%/year
Solar + battery, typical house, ordinary roof	12 to 20 years	10 to 16 years
Solar + battery, typical house, weak roof	25 to 30 years	18 to 24 years
Solar + battery, high-use house, ordinary roof	15 to 25 years	12 to 20 years
Solar + battery, high-use house, weak roof	25 to 40 years	18 to 30 years
Backyard Generating Station, typical house, extra paid ~5¢	7 to 18 years	6 to 13 years
Backyard Generating Station, typical house, extra paid nothing	13 to 32 years	10 to 20 years
Backyard Generating Station, high-use house, extra paid ~5¢	5 to 11 years	4 to 9 years
Backyard Generating Station, high-use house, extra paid nothing	7 to 18 years	6 to 13 years

Retail-rate sell-back rows are still a handful of years (**4 to 8** typical, **3 to 8** high-use). Rising rates shave those too. The bigger move is in the rows that look slow at a frozen rate. A typical house with no sell-back is **13 to 32 years** if the rate never moves, and about **10 to 20 years** if it rises 5% a year. A high-use house with no sell-back is **7 to 18 years** frozen, and about **6 to 13 years** if rates rise.

Solar also gets a shorter payback when rates rise. It only gets that help on the 10,000 to 14,000 kWh it actually makes. The Backyard Generating Station gets it on the whole house bill, and on whatever extra the utility still pays for. On a high-use house, solar still covers only part of a rising bill. The station can cover the whole bill.

A short grid also changes what extra power is worth. If the country is short, a house that can sell 35,000 to 59,000 kWh is offering power the system needs. That does not guarantee the utility will pay the full retail rate. It does make the “utility pays nothing” row less likely to stay true for decades, and it makes wholesale-style rates more likely to rise along with retail. Later chapters can put a real tariff on that. The frozen-rate tables are the conservative picture.

A short grid also means the house may still have power when the region does not have enough. A Backyard Generating Station plus a battery makes power at the house. It does not wait on a plant that has not been built yet.

What still decides the years is three ordinary things: what the utility pays you for extra power, how large the house bill is, and how fast that rate climbs. Batteries wear out and get replaced. The Stirling set does too, on a 10 to 15 year clock. That is true for solar-plus-battery as well (the battery), and those later swaps are not in the tables above. The first Stirling set is in the \$50,000 to \$90,000 install number. Later chapters can put a real utility price and a real heat load on one address. At \$50,000 to \$90,000, against a range from a ~\$2,000 typical bill to a ~\$5,000 to \$7,000 high-use bill, in a country that is getting short of plants and is already raising rates,

the Backyard Generating Station's payback gets better with time. Solar's payback is still limited by how much of that rising bill the roof can actually cover.

Owner maintenance is in the **utility room**, not in the hole. The battery, the inverter, the Stirling set, and a pump wear on a decade scale and get swapped in that room. The buried vault is never opened or serviced in the yard. There is no filter to change on it, no roof work, and no oil change.

When it is done

Nothing is opened in the yard.

At the end of life, or if a unit is swapped early, a licensed crew cranes the vault out as one piece and sets the next one in the same hole. Digging can take a day or two. During that work an existing house stays on the utility, the same way it did before the station was there. The homeowner is without the station (and, if the crew has to open the panel, without house power) only for the **hours** it takes to lock the old vault, lift it, set the new one, and land the pipes and wires again. The battery can cover part of that window. A first install on a new house can take days. That is a construction week. The house can stay on the utility while the crew works. The old vault goes to a licensed facility. Shielding, metal, and concrete are recycled or reused where that is possible. The homeowner does not store, cut, or dispose of anything.

The fuel that did the work began as about 30 kilograms (66 pounds) of heavy metal. After a century, what cannot be recycled or reused is still in that size class: a paint-can receptacle at a licensed site. It does not stay on the property. The septic-tank vault was almost all shield and box. That is what the crane was for.

A replacement price should be sold as a **swap**: new unit, removal, haul, and processing of the old one, plus the reconnect. It should not be a surprise line item after someone has lived with the appliance for decades. Whether that is a prepaid end-of-life fee, an included core-exchange, or a posted swap tariff is a later-chapter question. For this chapter: replacement means the old station leaves, and the cost of that leaving is part of the replacement price. The household does not dispose of the vault.

Benefits

- **Power that does not depend on the roof or the weather.** Nights, storms, and long gray stretches are ordinary running conditions.
- **A like-for-like bill versus solar.** Same house, same meter, same battery habit, plus heat if the house can take it, compared on delivered energy and sell-back, not on panel count.
- **Heat the house can already use, or a greenhouse if it cannot.** Floors, tanks, air-handler coils, boiler loops, or year-round growing.
- **Sell-back when the battery is full.** The electrical path is the one solar already uses.

- **No holes in the roof.** No array in a wind corridor. No 25-year dependence on roof seals.
 - **No fuel deliveries for the life of the unit.**
 - **No day-to-day operation.** The thermostat and the utility meter are enough. An app, if you want one, is a status screen on a phone or tablet. It does not control the vault.
 - **An install towns already know how to permit.** A buried concrete vault, a crane, backfill.
 - **A size that is meant to be regulated as an appliance.** The whole point of staying at house scale is a path that does not look like siting a 1,000 MW campus.
 - **A path that can sit next to the solar tax credit.** Carbon-free generation at the house, and fewer new commercial stations that communities do not want nearby.
 - **Power at the house while the country is getting short.** Demand is rising faster than plants are being built. Rates have been climbing about 4% to 6% a year. A unit that covers the whole bill, not a slice of it, gets more valuable as that happens.
 - **A clean end.** One lift, one replacement, processing included in the swap. Recycle and reuse where possible. What cannot be processed fits a paint-can receptacle after a 100-year life.
-

Detriments

- **Early units will be expensive.** The first Stirling set is a real line (\$40,000 to \$100,000 today). A complete solar-plus-battery job at the same house is the fair price comparison, not a panel-only ad. Solar's first years were also expensive, and a federal credit (often 30%) brought the net cost down. The same kind of credit could apply here. Until it does, and until the vault and the engines are factory objects, the check is large.
- **The Stirling set wears out.** About every 10 to 15 years the five engines get swapped in the utility room. That later check is not in the simple-payback tables. The first set is.
- **It is not a weekend homeowner project.** It needs permits, a licensed set, a utility hookup, and extra piping if you want the heat in the house or in a greenhouse.
- **Not every house can use the heat indoors without added hardware.** Mini-splits and electric baseboard still get electricity. Whole-house space heat needs hot-water pipes, or the warmth goes to a greenhouse, showers, or a dedicated cooler.

- **In summer the house often does not want the warmth.** The station is still making heat while it makes electricity. If that heat is not sent to showers or a greenhouse, it has to be dumped on purpose through a small outdoor radiator (the same kind of box a heat pump already uses on the side of a house). If it is not dumped on purpose, it soaks into the soil around the vault. That does not heat the neighborhood, but that patch of yard runs warmer than the rest of the lot, and a vault close to the house can make the air conditioner work a little harder. August heat has to be planned. It is not unused warmth you can ignore.
 - **The first approval does not exist yet.** Size-based approval is the plan. No regulator has issued that stamp.
 - **The buried unit is a controlled object.** It is sealed and heavy on purpose. That is a benefit for the household and a responsibility for the people who make, move, and take it back.
 - **End of life is specialized.** The swap includes removal and processing at a licensed facility. Leftover fuel is small (a paint-can class receptacle), but it is not handled like a stack of old solar panels.
 - **Insurance, appraisal, and resale will have to treat this as a normal house feature.** Solar already went through that. Until this one is common, a buried vault plus a utility room will be a new line on a homeowners policy and a new note on an appraisal, the way solar panels once were.
 - **Public comfort will take time.** A septic-tank-sized vault in the yard will need a clear, measurable safety story, not only a specification sheet.
-

What this chapter covered

This is the idea at house scale, and only house scale: one family, one yard, one meter, compared to solar at that same meter. The later chapters are 3 Yard Safety, 4 The House, 5 The Vault, 6 The Approval, 7 The Swap, 8 Vault Spec, and 9 The Fuel. The claim here is that the physics is in the right size class, the house already has ways to use the heat, the owner's life can stay simple, and the end of the story is a licensed crew and a crane.

3. Yard Safety

The claim

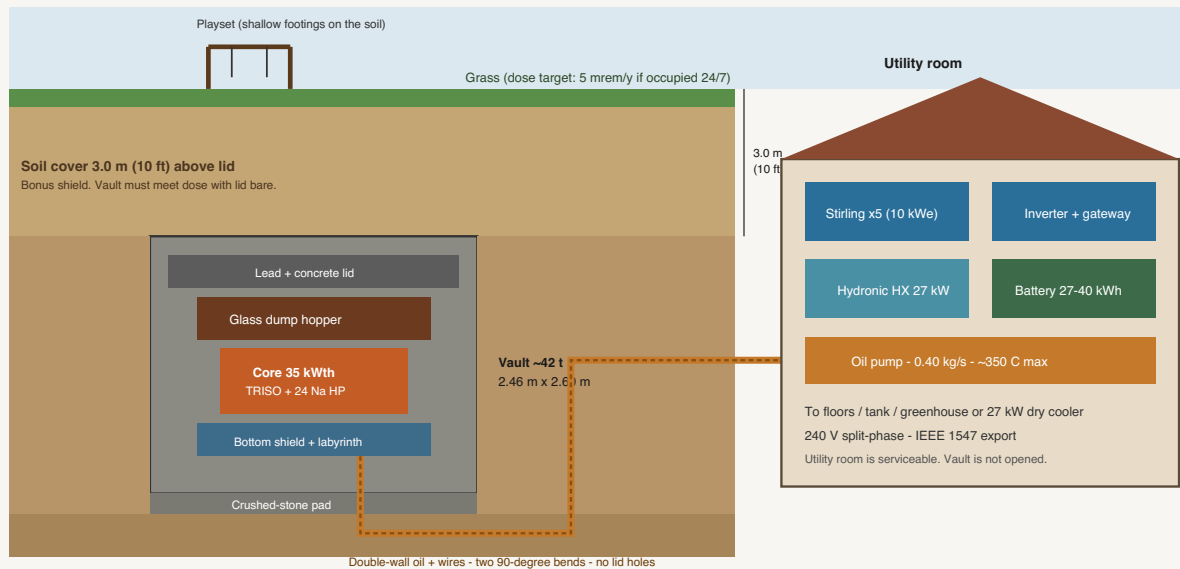
The yard stays usable: grass, a walk, a playset. The Backyard Generating Station sits under **ten feet of soil**, inside a lead-lined concrete vault. That cover is the same idea as other tanks American yards already bury (septic tanks, underground propane, heating-oil tanks). Ten feet on the lid is on the deep side of that habit, closer to a large buried fuel tank than to a shallow septic lid.

The extra dose is meant to be **a few mrem a year if a person lived on the grass over the vault, all day, every day**. That is the conservative number: someone on that spot for every hour of a year. Nobody does that. A playset is an hour here and there. At one hour a day for a year, the extra is a **small fraction of one cross-country flight**, not a flight.

That is the design target. The vault does the stopping. The ten feet of dirt is extra. The numbers in this chapter, and in Chapter 8, are written as if that dirt washed off. If the lawn is still safe with a bare lid, it is safe with a playset on ten feet of soil.

Figure 1. Site section, BGS-8 (not to construction scale)

10 ft (3.0 m) of soil above the lid. Vault 2.46 m OD x 2.60 m H. Chase floor-only, two 90-degree bends. Utility room is replaceable.



What a few mrem means

Dose is how much radiation energy a body absorbs. The everyday US unit is the **mrem** (millirem: one-thousandth of a rem). People already get a dose every year from the sky, the dirt, and the air in the house. That ordinary background in the United States is on the order of **300 mrem a year**. It is higher in the mountains than at the coast. It changes from house to house.

A few mrem a year on top of that, **for a person who lived on the grass**, is about **one percent of background**, or less. That full-time lawn number is in the same size class as **one cross-country flight**. It is far below the usual public limit for a licensed source, which is **100 mrem a year** from that source.

A note on the units. mrem is a yearly kind of number when we talk about a lawn. A survey meter on the grass reads a rate (how hard it is right now). Multiply that rate by the hours you actually stand there, and you get the year's extra. The design uses the hardest case: every hour of the year on that spot. A playset uses a slice of those hours.

One hour a day is 365 hours out of 8,760, about **4% of the year**:

a few mrem/year (if you lived there) $\times (365 / 8,760) \approx 0.15$ to 0.2 mrem/year

That is about one-fifteenth to one-twentieth of one coast-to-coast flight.

What	About how much	What that means
Ordinary US background	~300 mrem/year	Sky, dirt, and house air. Already happening.
Extra if a person lived on the grass over the vault, 24/7	a few mrem/year, if it is measurable at all	The design target. About 1% of background. About one cross-country flight, spread over the whole year.
Extra for a child on a playset ~1 hour a day	~0.15 to 0.2 mrem/year	A fraction of one flight, not a flight.
Public limit from a licensed source	100 mrem/year	A ceiling, not a goal. The vault is built well below it.
Cross-country flight	a few mrem	One trip. Same size as the live-on-the-lawn number, much larger than the playset number.

Nothing shields to zero, and the yard already has background. **The vault is built so the lawn does not become a special place.** You should not need a fence or a keep-off sign.

What has to be stopped

The vault holds a small nuclear core. While it is on, two kinds of radiation try to leave: **gammas** (like very hard X-rays) and **neutrons** (heavy, uncharged particles from splitting atoms). After it is shut down, leftover heat fades and the neutron problem fades with it. Gammas from the used fuel stay inside the same vault. Chapter 9 is what those neutrons are doing in the core, and why graphite and boron are in the story.

Gammas are stopped by dense things. Lead and thick concrete are the point of the lining. A few inches of lead and a thick concrete wall cut gammas by a huge factor. Ten feet of dirt on top cut them again.

Neutrons are the harder yard problem. They do not care much about lead. They slow down when they hit light atoms, especially **hydrogen**. That slowing is called moderation. Once they are slow, boron or more hydrogen can catch them. Wet dirt has water, and water is hydrogen. Dry dirt has much less.

So:

- **The vault** (lead, concrete, and a hydrogen-rich layer such as concrete or a boron mix) is the shield that has to work on its own.
- **The soil** is a bonus layer. Wet soil is a better bonus than dry soil, because of that hydrogen.

- **Dry soil is the worse case.** Neutrons travel farther in dry sand or a drought-baked lot than in damp ground. The dose on the grass must still be a few mrem a year in that worse case, and with the lid bare. Wet years and ten feet of cover are then extra margin, not the thing you counted on.

If someone built a thin vault and counted on ten feet of damp dirt to do the real work, a dry summer or a washed-out lid would show it. That is not this design. The vault is sized as if the dirt dried out and then left.

How it sits in the hole

Ten feet down means **ten feet of soil above the lid**, not ten feet to the middle of the core. The vault is about eight and a half feet tall, about the size of a large septic tank. The hole is deeper than that: vault plus ten feet of cover, on the order of eighteen feet to the pad. Towns already permit deep buried tanks. The grass still only sees a **10 by 10 foot** patch. How far that patch sits from the house is a local permit question, the same as a septic tank or a buried propane tank.

The vault is the object the crane sets. It is sealed. Nobody on the lot opens it. Heat and power leave through pipes and cables, not through a hatch in the lid.

Those pipes and cables do not come out the top, under the playset. They leave through the **bottom** of the vault, then turn and run sideways in a buried **chase** toward the house. A chase is just that buried run: two heat pipes and a few wires in a trench, the same idea as a sewer lateral or a buried propane line.

That path is a design choice, for these reasons:

- Radiation that finds a hole prefers a straight line. A hole in the lid would point at the grass. A hole in the floor points at more earth.
- Earth is a thick, cheap dump. What goes down is swallowed by dirt and water in the soil.
- The chase then bends (down, over, then toward the house) so there is no straight shot back up to the lawn.
- The house still gets heat and power. The connections are at the house end of the chase, in the utility room, where a plumber and an electrician already work.

A particle can bounce. The underside chase makes the easy path point into the ground, and it keeps holes out of the lid. Frost, wet soil, and a path a crew can service at the house end are ordinary buried-line problems. They do not put a hole under a child's feet.

Why a playset is the right test

A playset is hours, not minutes, and it is children, and it is right on the grass. If that is acceptable, a walk across the yard is not a special event.

The dose target is a yearly number on the grass **as if someone lived there**. A child on a playset is a few feet higher than the grass. Air does not shield much, so each hour on the set is about the same as an hour on the lawn, maybe a little less because they are farther from the lid.

They are not living on the set. One hour a day is about 4% of the live-there number. If living on the grass is a few mrem a year (about one flight), the playset year is a **fraction of that flight**. The rest of their year (the house, the car, an actual flight, a town at a different altitude) already moves the number by more than the playset does.

Two practical rules sit next to the physics:

- **Do not pin the playset to the vault.** The swing set sits on the ten feet of soil, on shallow footings or a pad, the way it would over a buried tank. Do not drive long posts down until they hit the concrete lid. The lid is not a footing.
- **Do not make the dirt do the real work.** Ten feet of cover is the story you tell. The vault still has to meet the lawn number if a flood washes that dirt off.

The playset is allowed because the number on the grass is small, the vault does not depend on the swing set being somewhere else, and the pipe openings are aimed at dirt, not at the slide.

If the top two feet go away

Yards get dug. Yards flood. Two feet of soil can leave.

With ten feet above the lid, two feet gone still leaves eight feet of dirt, plus the whole vault. Losing a couple of feet of soil might make the lawn number a few times higher (dirt cuts dose in layers; take away a layer and more gets through). A few times a number that was already “a few mrem, if anything” is still a small add to background, **if the vault was the real shield.**

If the vault was thin and the dirt was doing the work, two feet gone is a different story. That is why dry soil and a bare lid are design cases, not surprises.

Flood water itself is extra shield. The risk in a flood is **scour**: dirt washed off the lid. The vault should still meet the lawn target with the lid bare, sitting in a washed-out hole. Then a playset after a storm is a mud cleanup. The dose target still holds.

If a machine hits the concrete, that person is on the box, not on the lawn. That contact number is a crew-and-design number. It is not the playset number. A vault built to stand alone stays low there too. A vault that needed the dirt would not.

Soil right against the vault can pick up a little short-lived activity if neutrons leak, which is why the vault’s own neutron layer exists. The top two feet of lawn are not that layer. Washing them off does not bring “the hot dirt” to the swing, unless the scour goes all the way to the wall.

What you would measure

On an intact yard, a meter on the grass should read like the rest of the lot, or a faint bump over the lid that is still in the noise of background. Background already wiggles from day to day.

You would not design for a special fence, a keep-off sign, or a rule that the kids play on the other side of the driveway. If those were required, the vault is too thin.

Startup happens after the hole is backfilled and the chase is closed. Until then the core is locked off and is not making this kind of dose. At the end of life the locks go back in before the crane comes. The lawn number is a running, buried number.

What this chapter covered

This is the yard chapter: ten feet of soil above the lid, the vault as the shield, wet dirt better than dry dirt, pipes out the bottom through a buried chase, and a playset as the test. The claim is a few mrem a year extra **if a person lived on the grass**, which is about one cross-country flight; a playset hour a day is a fraction of that flight. Later chapters can put a real shield stack on a drawing. If children cannot play on the grass, the vault is not thick enough.

4. The House

Chapter 3 is the yard. This chapter is how the Backyard Generating Station ties into the panel, the battery, the water heater, and, if you want it, a greenhouse. The vault stays sealed. Everything a plumber or an electrician would recognize happens in a **utility room** at the house end of a buried **chase** (the pipe-and-cable run from the underside of the vault). That room can be in the house or a small shed next to it.

What arrives at the house

Pipes and cables leave through the **bottom** of the vault, bend in the dirt, and come up in the utility room. That is the same idea as a buried propane line or a sewer lateral, with one extra rule: there is no straight shot back to the lid. Chapter 3 explains why. Here, the point is only that people work at the house end, not in the hole.

Figure 3. Heat and power cascade

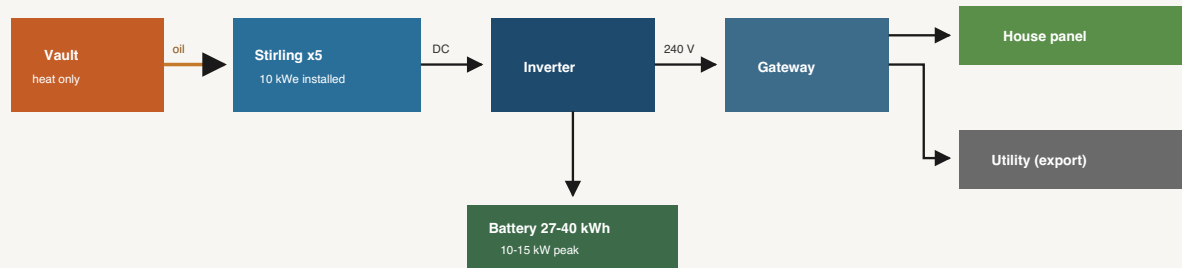
35 kWth in the vault. 8 kWe and 27 kWth out at the utility room. No shaft in the vault.



If the oil pump or Stirling units stop, condenser temperature rises and analog drum springs drive shutdown. Decay heat then goes to the vault mass and earth.

Figure 4. Electrical one-line

Vault is heat-only. Stirling, inverter, gateway, and battery live in the utility room. Gateway required so the house can run when the street is dark.



Without the gateway the inverter trips when the grid dies (anti-islanding). Chapter 1 life-safety case requires the gateway.

Three things come out of that chase:

- **A closed heat loop.** Hot oil (or a similar heat fluid) in a double-wall pipe, cooler fluid back. This is not house water, and it never touches the core.

- **Power cables** from the heat-to-electricity engines in the utility room to an inverter and then to the electrical panel
- **A few status wires** so the inverter and the battery can see that the vault is running

The water in the house never mixes with anything inside the vault. Heat crosses a metal wall in an exchanger, the same way a boiler already hands warmth to a floor loop. If that exchanger ever leaked, you would get a service call in the utility room, not an open vault.

Electricity: panel, inverter, battery

The electrical path is the one a solar-plus-battery job already uses.

There is **no steam turbine** and no spinning plant in the yard. Heat from the vault warms the hot end of a sealed **Stirling** engine in the utility room. A Stirling is a closed box with a hot side and a cool side. The temperature difference moves an internal piston and makes electricity. The same class of machine already runs on wellhead gas and in space tests. Five small units make the house's 8 kW, with a spare. Each is about **12 inches across and 18 inches tall**, about **110 pounds**. Five of them, plus the oil pump and the heat exchanger, sit on a wall or a short rack next to the inverter. They fit a basement utility room or a small shed. They do not fit in a closet. They last on the order of **10 to 15 years** and get swapped in that room, the way a furnace is swapped. They do not live in the vault, because a century-sealed engine inside a buried box has not been

shown. Chapter 2 puts the first set in the install check (**\$40,000 to \$100,000** today, **\$15,000 to \$30,000** as a factory target). The decade swap is a later check. It is not in the simple-payback tables.

The inverter turns that output into ordinary 120/240-volt house current and talks to the utility. When the battery is full and the house is satisfied, it exports. When the house wants more than the battery still holds, it buys a bit from the grid. Chapter 2 is the day-in-the-life. This chapter is the boxes in the utility room.

The battery sits in that room, not in the hole. Two or three wall units in the Tesla-class size (about 13 kWh each) cover the evening spike and a storm night. The vault fills them. They wear out on a decade-scale and get replaced, the same as they would on a solar job. That is owner maintenance. The vault is not.

A backup switch (a gateway) is required if you want the lights on when the street is dark. Ordinary inverters shut off when the grid dies, on purpose, so they do not feed a line a lineman thinks is dead. A gateway isolates the house, then lets the inverter and the battery (and the vault, still making heat) run the panel. Without that box, you have sell-back in good weather and a dark house in a blackout. Chapter 1's life-safety case needs the gateway. Budget it as part of the same electrical job as the inverter.

The utility interconnection is a solar-class permit: an application, a meter that can run both ways, and rules about how much they will buy. The vault does not change that paperwork's shape. It changes how many kilowatt-hours you have to sell.

Heat: use it, or dump it on purpose

Chapter 2 listed what can take the water: radiant floors, a tank, a coil in the air handler, a boiler loop, an air-to-water heat pump, a greenhouse. The house-side hardware is ordinary hot-water heating work.

If the house already has hot-water heat. An isolated exchanger and a pair of pumps (on the house loop, not in the vault) drop in next to the boiler or the manifold. The old boiler stays as backup or peak. The thermostat you already have still calls for heat.

If the house does not. You still get electricity. Showers can take a standalone tank. Whole-house space heat means adding pipes, or sending the loop to a greenhouse, or skipping space heat and using the electricity only.

August. The vault is still making heat while it makes electricity. If the house only wants showers, the rest has to go somewhere on purpose:

- a greenhouse, if you built one
- a small outdoor radiator, the same kind of box a heat pump already uses on the side of a house
- do not leave unused heat to soak into the soil around the vault

If unused heat soaks into the soil, that patch of yard runs warmer. Place the radiator where a condenser already would go. How far the vault sits from the house is a permit question. Those are ordinary siting choices.

Frost on the buried loop is a normal plumbing problem: bury below the frost line, or insulate, or keep a trickle of flow. The chase already runs deep, under the ten feet of cover and over to the house.

Water, flood, and the hole

The vault is built to sit wet. A lot of yards flood in a century. External water is extra shield and extra cooling. The inner vessel stays closed. It is built to live in a wet hole.

The utility-room boxes should sit **above** the flood you actually get, the same as a furnace or a panel should. Where the chase enters the foundation, it gets sealed the way any buried line into a basement gets sealed. Dirt washed off the lid is a Chapter 3 dose case. Water in the basement is a house case. Do not store the battery and the inverter in a pit that fills.

The buried object needs about a **10 by 10 foot** patch of yard. Setbacks from the house, a well, or the property line are the buried-tank kind the town already writes for septic or propane. The drawing says ten feet of soil above the lid. Keep a path a crane can use once at install and once at the swap (Chapter 7).

What the crew does, in order

1. Permits: excavation, electrical, utility interconnection, and plumbing if there is a heat loop. Towns already know most of these as separate stamps.
2. Dig the hole. Bed it. Confirm groundwater and rock before the crane arrives.
3. Set the vault. Do not start it.
4. Run the underside chase to the utility room. Close the trench.
5. Land the Stirling units, inverter, gateway, battery, and (if used) the heat exchanger and radiator.
6. Backfill with **ten feet of soil** above the lid. A playset, if any, sits on that soil, on shallow footings or a pad. It is not bolted to the vault.
7. Inspectors sign the electrical and the burial.
8. A licensed startup (Chapter 5) is the first time the vault makes heat.

A first install on a new house can take days. That is a construction week. The homeowner's job in that list is the same as a solar install plus a buried-tank job: be home for the meter, keep pets away from the work, and decide whether the heat loop is floors, a tank, a greenhouse, or a radiator.

What fails, and what does not

On the house side, the wear items are the Stirling units, the oil pump, the battery, the inverter, a pump on the house heat loop, and the outdoor radiator fan. Those live in the utility room. They are replaceable without opening the vault. A Stirling set or an inverter swap is a decade-class job, in the same cost world as replacing a serious HVAC unit or a home generator, not a new vault. Chapter 2 is the first-set dollar band.

The vault has no owner-service moving parts. Heat inside it moves in sealed **heat pipes** (closed metal tubes, no pump). Chapter 9 is how the sodium in those tubes boils, condenses, and returns in the wick, and why that is not a sodium-cooled plant. There is no turbine in the hole. If a heat pipe fails, that tube freezes in place and the others carry the load. If the oil loop in the chase leaks, the leak stays in the trench or the utility room. It does not open the core. If the utility-room engines stop, the vault overheats a little, the control drums spring shut, and leftover heat soaks into the box and the earth.

If the house is sold, the appliance stays with the lot the way a buried tank does, plus the utility-room equipment. Insurance and appraisal will have to treat that package as a normal house feature, the way they learned to treat solar. Until they do, it is a new line on a policy and a new note on an appraisal.

What this chapter covered

This is the house chapter: a buried chase to a utility room, solar-class inverter and sell-back, a gateway so the house can stay on in a blackout, heat the house already uses or a greenhouse or a radiator, and a vault that can sit wet while the batteries stay dry. Chapter 5 is what is inside the vault. Chapter 9 is the fuel and how heat leaves the core. Chapter 6 is who lets you set it. The utility-room work should look familiar to a solar electrician or a boiler technician.

5. The Vault

The vault is the sealed box in the hole. Chapter 3 is why the lawn stays a lawn. Chapter 4 is how heat and power leave through the floor. This chapter is what is inside, why it can sit closed for on the order of a century, and why it is furnace-scale rather than a commercial plant. Chapter 9 is the fuel itself: composition, core makeup, how sodium moves, first principles, and how the reaction is meant to stop.

A 1,000 MW plant is a million kilowatts. This vault makes **5 to 10 kWe** of electricity and about **20 to 40 kW of heat**. That is about **1/100,000 to 1/200,000** the electrical size of the plant people picture. It is furnace scale, not campus scale.

What the box is

From the outside the vault is a lead-lined concrete object about the size of a large septic tank, about eight feet across and eight and a half feet tall, under **ten feet of soil**. A crane can lift it. Nobody on the lot opens it. The lid has no hatch for the owner. Pipes and cables leave through the bottom only, into the buried chase that runs to the utility room.

Inside, in layers:

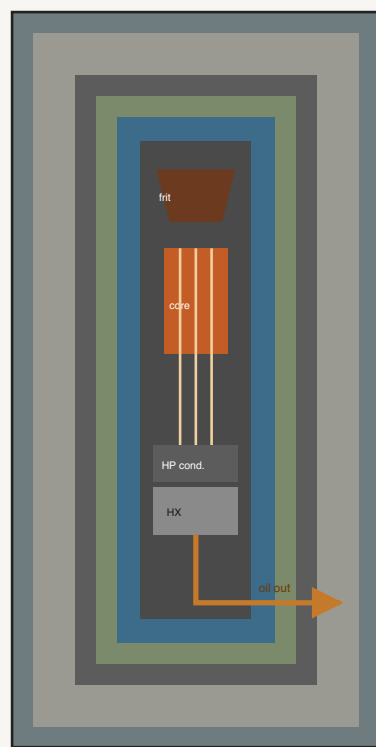
- **A small core** at the center: a beer-keg-to-dishwasher-sized assembly that holds the fuel

- **A hydrogen-rich and boron-bearing layer** that slows and catches neutrons (the particles that do not care much about lead)
- **Lead and thick concrete** that stop gammas (the hard X-ray-like radiation)
- **The sealed outer vault** the crane sees

The dirt above the lid is extra. The vault is sized to do the stopping on its own, including in dry soil and with the lid bare. Chapter 3 is that argument.

Figure 8. Vault vertical cutaway

2.60 m outer height. Same shield above and below the core. Hopper is last-ditch only (plug 750 C). No owner hatch.



Top to bottom

Steel skin 16 mm
Concrete 400 mm
Lead 80 mm
Borated concrete 150 mm
BPE 250 mm
Graphite / vessel

Glass hopper (80 kg frit + 15 kg B4C)
Fusible plug 750 +/- 20 C

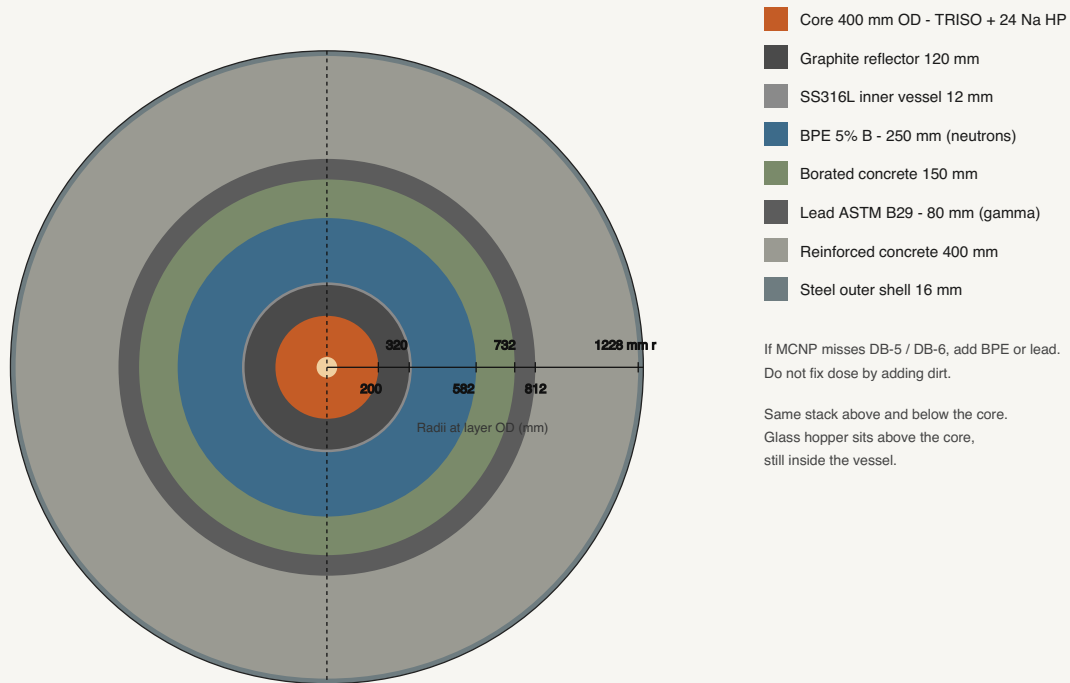
Core 400 x 500 mm - 35 kWth
24 Na heat pipes to condenser
Floor HX - oil out via labyrinth

Mass ~42 t - will not float
No lid hatch

10 ft soil above lid (not shown)
Dose designed with lid bare

Figure 2. Radial shield stack (half-section, type unit)

Outer diameter 2.46 m. Layers ordered so capture gammas born in BPE hit lead next. Soil is not shown; it is bonus above the lid.



How it makes power

The core is a very small nuclear core. Heat comes from splitting atoms in a sealed fuel, at a scale closer to a large home furnace than to a commercial station.

That kit is not invented here. Coated particle fuel (TRISO), heat pipes, and small Stirling engines have been built and tested for space and shop programs. This product would license that class of hardware, or build it from those proven designs. It is not a space reactor placed in a yard, and it is not a new kind of reaction.

That heat does two jobs at the same time. It becomes electricity for the panel. It is also warmth for floors, showers, or a greenhouse. Using both is the design. If the house or a greenhouse uses the warmth, most of what the core makes is used, not thrown away.

A house-scale electrical output of **5 to 10 kWe** (kilowatts of electricity) goes with about **20 to 40 kW of heat** from the core. That is furnace scale. There is no cooling tower. After shutdown, leftover heat drops to about as much as a small space heater, then lower over hours and days. Dirt, concrete, and time can take that.

From heat in the core to electricity in the house

There is no water flowing through the core, and no steam turbine.

1. **The fuel makes heat.** Tiny coated particles of uranium sit in graphite blocks. About 20% of that uranium is U-235, the isotope that splits easily. The rest is U-238. There are no swimming-pool rods and no open water in the core.
2. **Sealed heat pipes move that heat.** Each pipe is a closed metal tube with a little sodium inside. Sodium is a metal that melts and carries heat very well. When the core is warm, sodium vapor travels to the cool end of the pipe and gives the heat up. It then returns on its own. There is no pump in the vault. The sodium never mixes with house water. If one pipe fails, that tube freezes in place. The others keep working. The core does not “dry out” the way a water reactor can.

3. **A sealed exchanger in the vault floor** hands the heat to a double-wall oil loop. That oil runs through the buried chase to the utility room and back. If the oil leaks, the leak stays in the trench or the utility room. It is a hazardous-fluid cleanup, not an open core.
4. **Stirling engines in the utility room** turn the hot oil and a cool water loop into electricity. A Stirling is a sealed engine with a hot side and a cool side. No flame, no steam, no turbine. Five small units make about 8 kW, with a spare. They last on the order of 10 to 15 years. They live in the utility room on purpose, because a 100-year sealed engine inside the vault has not been shown. Replacing them is a room job, not a yard opening.
5. **Leftover warmth** (about 27 kW on the type unit) goes to house floors, a tank, a greenhouse, or a small outdoor radiator in summer.

If the oil pump or the Stirling units stop, heat has nowhere to go except the vault and the dirt. Temperature at the heat-pipe ends rises. Control drums spring shut, without a phone or a motor that has to stay on. Leftover heat then soaks into the 42-ton box and the earth.

The core is run **cold and slow**: a trickle of heat in a lot of mass. That is how a sealed life on the order of a century is even thinkable. The neutron battering that ages metal in a big plant is much gentler here because the power in each liter of core is small.

The paint-can of fuel

Fuel use at this output, over a century, is measured in kilograms, not tons. Chapter 8 works the inventory: about **30 kilograms (about 66 pounds) at first start**, still a paint-can of heavy metal inside the core, inside the vault. By the end of life, on the order of **15 to 25 kilograms has been used up**. Chapter 9 is what that paint-can is made of (coated particles, graphite, about 20% U-235) and why it is not a water-plant fuel rod. The vault is almost all shield and box. The crane is for the box.

Material mixed into the fuel is used up as the fuel is used up, so the output stays even. There is no refueling truck. There is no hatch. At the end, what cannot be recycled is still in that paint-can size class at a licensed site (Chapter 7).

The fuel is assembled in the factory, in a shutdown state, and never handled on the lot.

Why it stays put

A commercial plant needs staff, pumps, and a tower because leftover heat after shutdown is enormous. Here leftover heat is a space heater. The earth is the sink. The vault does not rely on a pump staying on.

If power is pulled faster or slower, the battery and the inverter in the utility room take the jump. The core is not asked to follow the dryer. It makes a steady trickle.

If the core ever tries to run away (a sudden rise in splitting), the fuel and the geometry are chosen so heat itself slows the reaction: hotter means less splitting, in milliseconds, without a person or a motor. That is a materials fact, not a software mode. That passive slowing is the ordinary protection. Chapter 9 draws that heat brake, the drum springs, and the cases that do and do not shut the core down. It is why the last-ditch glass layer below is not expected to run.

If the vault ever got far hotter than it should anyway, a passive layer above the core is meant to melt, flow, and set into a glass-like mass that stills the reaction and locks the insides in place. The whole vault stays one object. That is last-ditch only. The fuel and the drums are designed so the vault does not get there.

Off on the truck, on in the hole

Fresh fuel has never been used, so it is not already hot. In the factory the core is assembled with **locks and control elements in the shutdown position**. The vault is sealed around that. What leaves the plant is a heavy, closed object that is not producing electricity and not producing heat.

It is trucked as permitted heavy freight. A crane sets it. Chapter 4's crew connects the utility room. Inspectors sign.

Only after it is buried and the chase is closed does a **licensed startup** happen. The shutdown locks are released in a controlled procedure. That is the first time the core makes heat. Until that step

the vault is cold. At the end of life the same locks go back in before the crane returns. The vault that rides out is shut down on purpose, the same way it rode in.

The owner does not hold the key to those locks. A phone app, if there is one, does not move them. Chapter 2's app is a status screen only.

What the vault is not

There is no cooling tower, no staff gate, and no refueling floor.

The owner does not open it to service it. There is no filter, hatch, or oil change in the yard.

It is not safe because the dirt is deep. It is safe because the box is thick, the pipes leave through the floor, the power is furnace-scale, and the locks stay in until a licensed person releases them. Ten feet of soil is the yard you see. The vault still has to meet the lawn dose target if that soil leaves.

It is a 20 to 40 kW heat source in a factory-sealed vault, about 1/100,000 the electrical size of a 1,000 MW plant. That size is the approval argument (Chapter 6).

What this chapter covered

This is the vault: layers, furnace-scale heat, two products (electricity and warmth), heat pipes to an oil chase to Stirling engines in the utility room, a paint-can of fuel, no pumps you service in the yard, a glass dump that is not expected to run, and a truck that carries a locked box. Chapter 3 is the grass. Chapter 4 is the utility room. Chapter 6 is the stamp. Chapter 9 is the fuel and the first principles. Chapter 10 is the proof plan. The machine is designed so the yard never needs a hatch.

6. The Approval

No regulator has issued this approval yet. This chapter is the path.

The aim is not to site a 1,000 MW campus behind a house. The aim is to treat a factory-sealed, never-opened, furnace-scale vault as a **manufactured appliance**: built in one place, hauled as a closed object, set by licensed people, started by licensed people, and taken back by licensed people. The homeowner owns the appliance. They do not become a plant operator.

Why size is the argument

A typical commercial generating station is about **1,000 MW**. That is a million kilowatts, a staff, a perimeter, and a town-scale fight.

The Backyard Generating Station makes **5 to 10 kWe** and about **20 to 40 kW of heat**. That heat number is a large home furnace. After shutdown, leftover heat is a small space heater. Next to the 1,000 MW plant, this unit is about **1/100,000 to 1/200,000** the electrical size. Chapter 5 is the vault at that size. Chapter 9 is the fuel physics that makes leftover heat a space heater and keeps a power jump small. The approval claim follows from it:

- The object is too small for leftover heat to run away into the neighborhood.
- It ships and sits as one sealed vault. Nobody on the lot opens it.
- The lawn target is a few mrem a year extra, if anything, including with the lid bare (Chapter 3).

- Install looks like a buried tank plus a solar interconnect (Chapter 4).

A rulebook written for a commercial plant will block a house product even when the physics is friendly. The project's plan is a **size-based approval or exemption**: one certified design, made in a factory, installed many times, rather than a custom power-plant license on every lot.

That approval does not exist yet.

Permits towns already know how to issue

Towns already permit:

- a buried concrete tank (septic, propane vault)
- a solar inverter, a battery, and a two-way meter
- a boiler-room heat exchanger
- heavy freight and a crane in a driveway

The federal piece that does not yet exist is permission for the sealed core inside the vault to be that small, that closed, and that numerous. The closest existing ideas are sealed sources and factory-built packages that move as a unit and come back as a unit. The Backyard Generating Station is asking to be closer to those than to a site-built plant.

The homeowner's permits stay local and familiar: dig, set, interconnect, plumb. How far the **10 by 10 foot** patch sits from the house is that local buried-tank conversation. The core never becomes a backyard license.

Who does what

Who	What they do	What they do not do
Factory	Builds, fuels in a shutdown state, seals, ships one design	Does not open it on the lot
Licensed hauler	Moves a closed, locked vault	Does not start it
Licensed set crew	Digs, cranes, chase, backfill	Does not run the core
Licensed startup	Releases shutdown locks after burial and inspection	Does not hand the owner a key to those locks
Electrician / plumber	Utility room: Stirling, inverter, gateway, battery, heat loop	Do not enter the vault
Utility	Interconnect and sell-back rules	Does not own the vault
Homeowner	Thermostat, bill, optional status screen	Does not operate, refuel, or open
Licensed take-back (Chapter 7)	Locks in, crane out, process	Does not leave parts in the shed

If any of those rows becomes a homeowner weekend job, the product is wrong. The core is never a backyard task.

What the exemption is asking

In plain language, the paper would say:

1. This is one manufactured design, not a custom plant.
2. Thermal power is furnace-scale (tens of kW, not thousands of MW), about 1/100,000 the electrical size of a 1,000 MW plant.
3. The vault is sealed at the factory and opened only at a licensed facility at the end.
4. Transport, set, startup, and take-back are licensed steps.
5. Public dose at the grass is designed to a few mrem a year, with the vault alone as the shield.
6. The owner cannot start, stop the core, or move fuel.
7. A failed unit is a crane event, not a local teardown.

Call that an exemption, a general license, or a product certificate. The name matters less than the shape: **approve the object once, install it many times**, the way a listed inverter is listed once and hung in a thousand garages.

Until that exists, every early unit is a special case and will cost like one. Chapter 2's early-production band is that fact in dollars.

What sits next to the stamp

Utility interconnection is a separate yes. A house that can export 35,000 to 59,000 kWh a year will hit sell-back caps and rate design. That is a price-and-tariff issue. It is separate from approval of the core. Chapter 2 already says payback depends on what the utility pays for extra power.

A federal credit like solar's is a separate yes. Solar's early years used a large investment tax credit. The public case here is carbon-free power at the house, fewer 1,000 MW plants towns refuse, and generation that still works at night. It is a policy path, not a promise.

Insurance, appraisal, and resale will lag, the way they lagged for rooftop solar. Until this is common, a buried vault plus a utility room will be a new line on a homeowners policy and a new note on an appraisal.

Town politics will take time. Chapter 3's playset number has to be true and measurable. If the yard needed a keep-off fence, the vault would have failed its own dose target.

What this chapter covered

This is the approval chapter: size as the argument, a factory-sealed appliance rather than a commercial plant, licensed hands at every step that touches the core, and a first stamp that does not exist yet. Chapter 5 is why the size claim is honest. Chapter 9 is the fuel and the shutdown physics behind that claim. Chapter 10 is what still has to be shown, including which federal rules exist (plant licenses, Type B transport, public dose) and which house-appliance stamp does not. Chapter 7 is the take-back that makes “never opened on the lot” true at the end. The paperwork should look like approving a manufactured appliance, not siting a power plant.

7. The Swap

Nothing is opened in the yard. Replacement means the old vault leaves as one piece and a new one sits in the same hole. The price of that leaving is part of the replacement, not a bill the household discovers decades later.

The day it comes out

On a house that already has a station, the house stays on the **utility** while the crew digs. Digging can take a day or two. The lights stay on. That is not the dark window.

A licensed crew then isolates the station from the house: gateway open, heat loop stopped, inverter off. They put the **shutdown locks back in** the same way Chapter 5 put them in at the factory. The vault stops making heat on purpose. Leftover heat is already small. They wait the hours it takes for that to fall further if the procedure says to wait.

Then a crane lifts the vault, still sealed, onto a permitted truck. The hole that is left is a known shape. The next vault, locked and cold, goes in. Chapter 4's chase lands again in the utility room. Inspectors sign. A licensed startup is the first heat from the new unit.

The homeowner is without the station, and without house power only if the crew has to open the panel, for the **hours** of that lock, lift, set, and reconnect. The wall battery can cover part of that window. A first install on a new house can take days. That is a construction week. The house can stay on the utility while the crew works.

The homeowner does not store, cut, or dispose of anything. There is no pile of old parts by the garage. The core leaves with the vault, on that same truck.

If a unit is swapped early (damage, a recall, a sale that wants a newer vault), the day looks the same. Early or late, it is a crane event.

What happens to the old vault

The truck goes to a licensed facility, not to the town dump.

The box. Lead, steel, and concrete are recycled or reused where that is possible. The vault was almost all shield and box. That mass is ordinary industrial material once it is clean enough for the facility's rules.

The fuel. The work began as about **30 kilograms (66 pounds) at first start** (Chapter 8). Chapter 9 is what that heavy metal is (TRISO-UCO particles in graphite) and what is left in them after a century. After a century, what cannot be recycled or reused is still in that size class. It goes into a paint-can receptacle at a licensed site. It does not stay on the property or in the hole.

The owner never takes title to a waste drum. Take-back is the factory's (or the licensed operator's) job, paid as part of the swap.

What the check includes

A replacement is sold as a **swap**:

- new vault
- removal of the old one
- haul
- processing (recycle what you can, receptacle for what you cannot)
- reconnect of power and heat in the utility room
- licensed shutdown and licensed startup

It should not be a surprise line item after someone has lived with the appliance for decades.

How that is paid can be any of these, or a mix:

- **Prepaid at first install.** A take-back fee sits in the original price, like a deposit on a tank.
- **Included core-exchange.** The company that sold the vault is on the hook to swap it. The homeowner pays a posted swap price that already assumes take-back.
- **A posted price at the time of swap.** Clear dollars, known in advance as a class of job, not a custom demolition bid.

Which of those wins is a business detail. The rule is the same: **replacement means the old station leaves, and the cost of that leaving is inside the replacement.**

The hole, the crane path, and the chase were supposed to stay usable (Chapter 4). If a later owner built a garage on the lift path, that is their extra bill. The vault is still lifted out as one piece. It is not cut open in the yard.

What is not the swap

Batteries, the inverter, and the Stirling units die on a much shorter clock, often 10 to 15 years. Replacing those is a utility-room job, like solar or a furnace. It is not a crane day and not Chapter 7.

The outdoor radiator or a house pump is ordinary hardware. Replacing it is not a vault event.

A playset or a patio on the ten feet of soil is fine if footings stay shallow (Chapter 3). Do not drive long posts into the lid. That makes the later crane lift harder.

The leftover fuel is small, and the process is still specialized. It is not handled like a stack of old solar panels.

A century, or sooner

The vault is designed to run for on the order of a hundred years without a hatch. That is a design life, not a promise that the first owner's great-grandchildren must keep the same serial number. A swap can happen when the unit is done, when a better vault exists, or when the house is reworked.

Until that day, there is no refueling truck and the vault stays sealed. After that day, the story is one lift, one new vault, and leftover fuel in a paint-can-class receptacle at a licensed site.

What this chapter covered

This is the end: locks in, crane out, same hole, recycle the box, leftover fuel to a licensed site, and a replacement price that already includes leaving. An existing house stays on the utility while the crew digs. The dark window, if there is one, is hours. Chapters 3 to 6 are how the yard, the house, the vault, and the stamp make that day ordinary. Chapter 9 is what is inside the paint-can that leaves on the truck. The household does not keep any part of the old vault.

8. Vault Spec

Document type: Preliminary design basis (not a licensed safety analysis).

Unit: Backyard Generating Station, type **BGS-8**. The 8 is the electrical size (8 kWe), not “the eighth try.” This is the first type unit. Family band is 5 to 10 kWe.

Audience: Mechanical, nuclear, shielding, and electrical engineers. Chapter 5 is the homeowner story of the vault. Chapter 9 is the fuel, the core, sodium in the heat pipes, first principles, and the contrast with water plants and sodium fast reactors. Chapter 10 is the proof plan: which claims are closed arithmetic, which are handbook bounds, and which still need codes, tests, or a stamp. This file is what you size, buy, and analyze.

In one paragraph: a sealed heat source in a septic-tank vault, **3.0 m (10 ft) of soil above the lid**, heat pipes to a double-wall oil chase, electricity made in the **utility room** by replaceable Stirling engines. The vault is the shield. Dirt is bonus. Dose is designed with the lid bare. The core kit (TRISO, heat pipes, drums, Stirling) is a known class of hardware, licensed or built from that class, not a new reaction.

An engineer can use this to start a sign-off package. They cannot stamp the vault from this file alone. Required follow-on work is listed in §16. Chapter 10 restates each claim as a proof item (Closed, Bound, or Open) and names the spec or test. Numbers below are closed arithmetic or standard handbook values. Where a transport code (**MCNP**, Monte Carlo N-Particle: the standard program that

tracks neutrons and gammas through the shield to predict dose), SCALE, or an ASME stress report is mandatory, the spec says so and gives the requirement the code run must meet. Do not invent a k-effective in this file or in Chapter 10.

No owner hatch. No in-vault service pumps. Power conversion that wears is on the house side of the chase. The vault is a sealed heat source.

1. Design basis (shall)

ID	Requirement
DB-1	Continuous electrical output 8.0 kWe net at the inverter DC bus, product family 5 to 10 kWe .
DB-2	Core thermal power 35 kWth nominal (32 kWth at 25% conversion plus 10% margin). Family band 20 to 40 kWth.
DB-3	Sealed operating life 100 years at $\geq 90\%$ capacity factor without opening the vault.
DB-4	Heat and electricity are both products. Reject heat from conversion is 80 to 120°F hydronic water (27 to 49°C) for the house, or a greenhouse, or a dedicated dry cooler.
DB-5	Extra dose at the grass, dry soil, vault running: ≤ 5 mrem/year if a person occupied that spot 8,760 h/y. Playset at 1 h/d is then ~ 0.2 mrem/y (Chapter 3).
DB-6	Same 5 mrem/y target with the lid bare (scour / 2 ft of soil gone). Soil is bonus, not the shield.
DB-7	Penetrations floor only , then a bent chase. No lid penetrations.
DB-8	Vault sits flooded . External water is allowed. Inner boundary remains leak-tight.
DB-9	Factory assembled subcritical, shutdown locks in . First heat only after burial and licensed startup. Locks in before crane-out.
DB-10	No vault-side pump required for core cooling or shutdown decay heat.
DB-11	Excess reactivity after shutdown margin shall be under 0.8 dollars so a single failure cannot make the core prompt-critical.
DB-12	Last-ditch borosilicate + B ₄ C dump, fusible, no power required. Not credited for normal operation.
DB-13	All in-vault materials are items a US nuclear or ASME shop can buy or already fabricates. No exotic materials.

2. Power and energy (closed arithmetic)

2.1 Why 35 kWth

House-side conversion baseline is free-piston Stirling, $\eta = 0.25$ (hot end ~ 350 to 600°C , cold end ~ 40 to 50°C). Demonstrated class: NASA KRUSTY / Sunpower-style convertors; commercial wellhead units (Qnergy class) exist at several kWe.

$$\begin{aligned} P_{\text{th}} &= P_{\text{e}} / \eta = 8.0 \text{ kWe} / 0.25 = 32.0 \text{ kWth} \\ P_{\text{th,spec}} &= 32.0 \times 1.10 = 35.2 \text{ kWth} \rightarrow \text{specify } 35 \text{ kWth} \\ P_{\text{reject}} &= 35 - 8 = 27 \text{ kWth} \end{aligned}$$

27 kWth is a mid-size house heating load (typical winter 5 to 15 kW) plus domestic hot water, with margin, or a greenhouse, or a ~ 10 -ton-class dry cooler in summer (1 refrigeration ton ≈ 3.5 kW).

If conversion is only thermoelectric at $\eta = 0.08$, the same 8 kWe needs **100 kWth**. That is a different, larger vault. It is the alternate in §6.3. Baseline is 35 kWth.

2.2 Annual electricity

$$\begin{aligned} 8.0 \text{ kW} \times 8,760 \text{ h/y} &= 70,080 \text{ kWh/y} \\ \text{at } 90\% \text{ CF: } 8.0 \times 8,760 \times 0.90 &= 63,072 \text{ kWh/y} \end{aligned}$$

Typical house 11,000 kWh/y $\rightarrow$ export $\sim 52,000$ to $59,000$ kWh/y.
High-use house 25,000 to 35,000 kWh/y $\rightarrow$ export $\sim 28,000$ to $45,000$ kWh/y.

Average house load 1.1 to 4 kW. 8 kWe covers both. Peaks 10 to 20 kW are the battery (Chapter 4).

2.3 100-year thermal energy and burnup

$$E = 35 \text{ kW} \times 8,760 \text{ h/y} \times 100 \text{ y} = 30,660,000 \text{ kWh} \\ = 30,660 \text{ MWh}$$

$$P = 0.035 \text{ MW}$$

$$E = 0.035 \text{ MW} \times 365.25 \text{ d/y} \times 100 \text{ y} = 1,278.4 \text{ MWd}$$

Heavy-metal mass from burnup B (MWd per tonne):

$$m_{\text{burned}} (\text{t}) = 1,278.4 / B$$

Assumed discharge burnup	Burned HM	Notes
50 GWd/t (50,000 MWd/t)	25.6 kg	Conservative UO ₂ -class
80 GWd/t	16.0 kg	TRISO-UCO, realistic target
100 GWd/t	12.8 kg	Aggressive TRISO

Inventory at first start is larger than mass consumed (need leftover fissile at year 100, burnable poison, and a critical geometry). Specify:

Item	Value
Heavy metal at first start	30 kg (66 lb)
Consumed HM at 80 GWd/t, end of life	16 kg (35 lb)
Residual HM at end of life	~14 kg in the same compacts

Layman Chapter 5's older “~10 kg” was order-of-magnitude at a lower thermal power. This spec supersedes it for engineering.

2.4 Fission rate and U-235 consumed

Energy per fission: $200 \text{ MeV} = 200 \times 1.602 \times 10^{-13} \text{ J} = \mathbf{3.204 \times 10^{-11} \text{ J}}$.

$$R_f = 35,000 \text{ J/s} \quad / \quad 3.204 \times 10^{-11} \text{ J} = 1.092 \times 10^{15} \text{ fissions/s}$$

Per year (365.25 d):

$$\begin{aligned} N_y &= 1.092 \times 10^{15} \times 3.15576 \times 10^7 = 3.447 \times 10^{22} \text{ fissions/y} \\ m_{\text{U235}} &= 3.447 \times 10^{22} / 6.022 \times 10^{23} \times 235 \text{ g} = 13.45 \text{ g/y} \\ 100 \text{ y} &\rightarrow 1.35 \text{ kg U-235-equivalent fissioned} \end{aligned}$$

Some fissions at mid-life are Pu-239. The 1.35 kg is fission-equivalent, not a claim that only U-235 fissions.

Initial U-235 in 30 kg of 19.75 wt% U:

$$0.1975 \times 30 \text{ kg} = 5.93 \text{ kg U-235}$$

Burning 1.35 kg-eq leaves several kg fissile plus bred Pu. End-of-life criticality is a SCALE/ORIGEN depletion problem (§16). It is not a fuel-shortage problem.

2.5 Decay heat (shutdown)

Way / ANS-style: $P/P_0 \approx 0.066 t^{-0.2}$ (t in seconds, infinite prior operation, good enough for sizing).

Time after shutdown	t (s)	$t^{-0.2}$	P/P_0	P (kW)	Household analog
1 s	1	1.000	0.066	2.31	Large space heater
1 h	3,600	0.194	0.0128	0.45	Small space heater
1 d	86,400	0.103	0.0068	0.24	Heat lamp
7 d	6.05×10^5	0.0697	0.0046	0.16	Night-light class, thermal

DB-10 is satisfied: after trip, the vault mass (~42 t, §9) and earth reject <1 kW without a pump. No tower.

Crude earth conduction, treat vault as a sphere $r = 1.4$ m, soil $k = 1.0$ W/m·K, $Q = 0.5$ kW:

$$\Delta T = Q / (4 \pi k r) = 500 / (4 \times 3.1416 \times 1.0 \times 1.4) = 28 \text{ K}$$

Soil far-field plus 28°C at the wall after a few hours. Acceptable. **Do not use this formula for operating 35 kW.** Operating heat is extracted by heat pipes. If 35 kW were dumped to soil: $\Delta T \approx 1,990$ K, which is why August heat must go to a dry cooler or a greenhouse (Chapter 4), not into the dirt.

3. Core

In plain language: the core is a beer-keg-sized graphite block full of coated uranium particles (TRISO), not a pool of water and not a bundle of bare rods. About 20% of the uranium is U-235. Sodium sits only inside sealed heat pipes. There is no water in the core and nothing that can “boil dry.” Heat pipes fail by freezing in place. Chapter 9 is that machine in English: fuel, graphite, sodium in the pipes, and how the reaction is meant to stop.

3.1 Type (baseline)

Heat-pipe microreactor, TRISO-UCO fuel, graphite moderator/reflector, sodium heat pipes.

Heritage: NASA KRUSTY (heat pipes + Stirling), Westinghouse eVinci (heat-pipe microreactor), X-energy / BWXT TRISO. All components exist in some form. This product would license or build

from that class. This unit is smaller than eVinci (that is MWt-class) and longer-lived than KRUSTY (that was a short ground test). It is built from that class of hardware, not a copy of either machine.

Why TRISO, not UZrH, as baseline: TRIGA UZrH has a stronger prompt negative coefficient (good). Hydrogen migration and hydride stability over 100 years are not a signed-off materials program. TRISO-UCO in graphite is the US path that already has a fuel fabrication line. Prompt feedback is Doppler in the kernel (negative, weaker than UZrH). Safety then also depends on **DB-11** (little excess reactivity) plus the glass dump.

Alternate (if a materials program qualifies it): UZrH-LEU 19.75% rods, TRIGA geometry. Same vault, different core cartridge. Note as Option C-2, not baseline.

3.2 Fuel specification

Parameter	Spec	Why it is buyable
Kernel	UCO, 19.75 ± 0.20 wt% U-235	Standard US research / HALEU assay. Not HEU.
Particle	TRISO: IPyC / SiC / OPyC on kernel	BWXT / X-energy class process
Compact	Cylindrical graphite compact, TRISO packed ~35 vol%	Prismatic microreactor practice
HM at first start	30 kg (66 lb)	\$2.3
Burnable poison	Er_2O_3 or Gd_2O_3 integral in compact or as discrete rods	Standard PWR/microreactor practice
Design burnup	80 GWd/t peak compact, 100 GWd/t not required	TRISO has been taken past this in AGR/HTGR tests
Fission-gas	Held in TRISO buffer; no vented fuel	Required for a sealed vault

Enrichment stays **below 20%** so the material is HALEU, not HEU. That is a security and policy choice as well as a physics choice.

3.3 Geometry (type unit)

Active core: 400 mm diameter $\times$ 500 mm height
 Core volume: $\pi \times 0.20^2 \times 0.50 = 0.0628 \text{ m}^3 = 62.8 \text{ L}$
 Power density: $35 \text{ kW} / 62.8 \text{ L} = 0.56 \text{ kW/L} = 0.56 \text{ MW/m}^3$

A PWR is $\sim 100 \text{ MW/m}^3$. This core is $\sim 200\times$ less dense. That is the 100-year fluence trick.

HM loading: $30 \text{ kg} / 62.8 \text{ L} = 0.48 \text{ g/cm}^3$

TRISO compacts routinely hold 0.5 to 1.0 g/cm³ HM. 0.48 is not cramped.

Heat-pipe lattice (plan): 24 sodium heat pipes, 22 mm OD, on a triangular pitch through the graphite, plus ~80 to 120 fuel compact channels. Exact lattice is an MCNP/SCALE item. Pipe count from §4.1.

Reflector: 120 mm radial and axial graphite (or BeO if a later run needs a smaller core; graphite is cheaper and fully supply-chain). Outer reflector diameter $400 + 2 \times 120 = \mathbf{640 \text{ mm}}$. Beer-keg to dishwasher, as Chapter 5 said.

Inner vessel: SS316L, 12 mm wall, ~700 mm OD, helium or low-pressure inert fill around the graphite. Not a PWR pressure boundary. Design pressure $\leq \mathbf{0.5 \text{ MPa}}$ (gas plus heat-pipe fail-in-place). ASME VIII or III as the QA program dictates.

3.4 Neutronics requirements (analysis, not a fake k_{eff})

The following are acceptance criteria for the lattice model. Do not invent a k_{eff} in this file.

State	Criterion
Cold, clean, drums in (factory / truck)	$k_{\text{eff}} \leq 0.95$, including 3σ and tolerances
Hot, xenon eq., drums at power, BOL	$1.000 \leq k_{\text{eff}} \leq 1.005$, drums not on a stop
Hot, EOL, poisons depleted	$k_{\text{eff}} \geq 1.000$ with drums at max worth remaining
Most reactive drum stuck out, others in	$k_{\text{eff}} \leq 0.99$
All drums out, cold, clean (beyond design)	Still < prompt critical; DB-11
Flooded inner vessel (water ingress)	k_{eff} decreases or stays subcritical (void / spectrum shall be analyzed both ways)

Drum / rod worth: **B₄C control drums** in the reflector (eVinci / Kilopower style), 6 drums. Shutdown margin at least 1.0 dollar with most reactive drum stuck withdrawn. Mechanical locks pin drums in the in (absorbing) position for transport. Licensed startup is a keyed, two-person rotation to the operating arc. The owner key does not exist.

Burnable poison designed so drum motion over 100 years is slow (a few degrees per decade), not weekly.

3.5 Fluence (order of magnitude)

Fission density:

$$1.092 \times 10^{15} \text{ /s} \quad / \quad 6.28 \times 10^4 \text{ cm}^3 = 1.74 \times 10^{10} \text{ fissions/cm}^3 \cdot \text{s}$$

If macroscopic fission cross section $\Sigma_f \sim 0.08$ to 0.15 cm^{-1} (to be taken from the lattice), thermal flux is:

$$\phi \sim (1.74 \times 10^{10}) / \Sigma_f \approx 1 \times 10^{11} \text{ to } 2 \times 10^{11} \text{ n/cm}^2 \cdot \text{s}$$

$$t_{100y} = 3.15576 \times 10^9 \text{ s}$$

$$\Phi \sim 3 \times 10^{20} \text{ to } 7 \times 10^{20} \text{ n/cm}^2$$

That is research-reactor cladding fluence, not a 40-year PWR barrel. SS316L, graphite, and TRISO SiC are in a plausible range. **dpa and graphite shrinkage still need a materials assessment** (§16). This is why power density was kept low on purpose.

4. Heat transport (in the vault: no pumps)

In plain language: sodium stays inside 24 sealed heat pipes. Those pipes hand heat to a sealed exchanger in the vault floor. Double-wall oil carries that heat through the buried chase to the utility room. House water never enters the vault. An oil leak is a trench or room cleanup. A sodium leak stays in that pipe and freezes. If the utility-room engines stop, drums spring in and leftover heat goes to the 42 t box and the earth.

4.1 Sodium heat pipes

Parameter	Spec
Working fluid	Sodium, ASTM / nuclear grade
Count	24 operating + 4 spare channels (plugged spares)
Rating	1.6 kW each at 600°C vapor
24 × 1.6	38.4 kW > 35 kW _{th}
Envelope	SS316L or Haynes 230, 22 mm OD, ~1.8 m long
Wick	Sintered or annular-gap, space/heat-pipe vendor practice (ACT, Thermacore class)
Freeze	Na melts at 98°C. Startup from a cold truck uses trace heaters on the condenser block (powered from the house during licensed startup only) or a NaK subset (3 pipes) that are liquid at room temp to bootstrap
Fail-in-place	One pipe fail: $23 \times 1.6 = 36.8$ kW, still above 35. Two fail: 35.2 kW. Three fail: derate electrical output.
Walls	Single sealed envelope. Second barrier is the inner vessel (inert fill). Not a double-wall heat pipe. The oil chase is the double-wall loop, because it leaves the vault.
Design life	100 y sealed, same as the vault. Qualification at that life is a \$16 item. Fail-in-place plus spares is the operational safety net.

Kilopower/KRUSTY ran sodium heat pipes on a real critical core. This is the most manufacturable high-T, no-pump choice.

4.2 Temperature cascade (CHP)

Heat steps down, on purpose, from the fuel to the house. Each row is cooler than the one above it.

Where	About how hot	Why
Fuel compact	650 to 750°C	TRISO ceramic is fine well above this
Heat-pipe vapor	~600°C	Sodium vapor inside the sealed pipes
Primary exchanger	~550°C metal	Vault floor, still inside the box
Oil loop in the chase	~350°C	Double-wall; or 500°C if a later NaK loop
Stirling hot end (utility room)	300 to 500°C	Replaceable engines, not in the vault
Stirling cool end	40 to 50°C water	The leftover heat product
House heat loop	80 to 120°F (27 to 49°C)	After a mixing valve, floor and tank range

A perfect heat engine between those Stirling temperatures (about 600 K hot, 320 K cold) could turn at most about 47% of the heat into electricity. That ceiling is called Carnot. A real Stirling at 25% is about half of that ceiling. That is a normal real engine, not a calculation error.

4.3 Secondary loop (vault floor → chase → house)

Baseline fluid: Dowtherm A or equivalent, double-wall, leak-detected annulus, buried. Max bulk 350°C, well inside the oil's 400°C rating. Industrial solar-thermal / process shops already weld this.

$$\begin{aligned} Q &= 35 \text{ kW} \\ c_p &\approx 2.2 \text{ kJ/kg} \cdot \text{K} \\ \Delta T &= 40 \text{ K} \\ m &= Q / (c_p \Delta T) = 35 / (2.2 \times 40) = 0.40 \text{ kg/s} \approx 6 \text{ to } 7 \text{ gpm} \end{aligned}$$

Pipe: 25 mm (1 in) inner process + 40 mm containment. Two pipes (hot/return) in the chase. Circulation pump is **in the utility room**, not in the vault. Vault-side flow through the primary HX is driven by the heat pipes (vapor to condenser). The oil loop can sit still during shutdown; decay heat does not need it.

Primary HX: printed-circuit or helical-coil SS316L, bolted in a sealed floor compartment of the vault, welded closed at the factory. Oil in, oil out, through the floor only.

Leak: oil stays in the chase / house. It never becomes a core LOCA. Na stays in the heat pipes. A pipe fail freezes sodium in place.

Alternate secondary: NaK (liquid at room temp, better heat transfer, water-reactive). Only if the oil loop's 350°C cap hurts Stirling efficiency. Then an extra isolation HX before any water.

4.4 House reject and summer

27 kWth to:

- hydronic HX (plate, stainless, isolated from potable), or
- greenhouse loop, or
- dry cooler: **27 kW** air coil, ~10-ton HVAC condenser class, COTS (Modine / Guntner / any chiller reject).

If the oil pump stops and Stirlings stop, 35 kW has nowhere to go except the vault and soil. **Overtemp on the condenser block rotates drums in** (analog, spring or gravity, two channels). Core goes to decay heat. That is the protected fault for “utility room offline.”

5. Power conversion (utility room, replaceable)

In plain language: the vault is a sealed heater. Electricity is made in the utility room. A 100-year sealed Stirling inside the vault is not a demonstrated part. Demonstrated engine life is a decade, not a century. So the engines sit where a furnace sits, and they get swapped there.

The vault has no shaft. Conversion is Chapter 4 equipment.

Item	Spec
Convertors	5 × 2.0 kWe hermetic free-piston Stirling (10 kWe installed)
Envelope	~300 mm OD × 450 mm H, ~50 kg each (Microgen 1–2 kWe class). Five plus oil HX and pump fit a utility room, not a closet.
First-cost class	\$8k to \$20k per unit today; \$3k to \$6k factory target. Set: \$40k to \$100k today, \$15k to \$30k factory (Chapter 2).
Dispatch	4 of 5 for 8 kWe; 1 spare online or rotating
Life	10 to 15 years each; swap on the wall, vault stays closed. Decade swap is not in Chapter 2 simple payback.
Output	200 to 400 VDC → hybrid inverter (IEEE 1547 / UL 1741)
Gateway	Required for islanding (Chapter 4). Without it the house goes dark when the utility does.
Battery	2 to 3 × 13.5 kWh (27 to 40 kWh), 10 to 15 kW peak

Alternate 6.3, no moving parts: utility room is a thermoelectric array. $\eta \approx 0.08 \rightarrow$ core must be **100 kWth**, fuel and shield grow (~2.9× burnup, larger vault). Use only if a customer forbids a Stirling set.

6. Shielding and dose

In plain language: the vault, not the dirt, is the shield. Ten feet of soil is the yard you see and extra attenuation. The live-on-the-grass number (≤ 5 mrem/y) must still hold with the lid bare and the soil dry. A playset at one hour a day is then about 0.2 mrem/y. MCNP (defined in the header) has to confirm the stack. Until that run exists, 5 mrem/y is a requirement, not a measured result.

6.1 Sources (running, 35 kWth)

Neutrons: $\sim 2.4 \text{ n/fission} \times 1.092 \times 10^{15} = 2.62 \times 10^{15} \text{ n/s}$

Gamma energy (prompt $\sim 7 \text{ MeV/fission}$):

$$7 \times 10^6 \times 1.602 \times 10^{-19} \times 1.092 \times 10^{15} = 1.22 \text{ kW gamma}$$

Plus capture gammas in the shield (especially 2.2 MeV from H, and Fe if any). Capture gamma is why lead sits **outside** the hydrogenous layer.

Unshielded neutron flux at 2 m:

$$\Phi = S / (4 \pi r^2) = 2.62 \times 10^{15} / (4 \pi \times 4 \times 10^4 \text{ cm}^2) = 5.2 \times 10^9 \text{ n/cm}^2 \cdot \text{s}$$

That is immediately unsurvivable. The shield is not optional.

6.2 Target dose rates

Location	Running	Shutdown (1 d)
Grass, 3.0 m (10 ft) soil above lid, dry soil	$\leq 0.57 \mu\text{rem/h}$ ($\rightarrow 5 \text{ mrem}/8,760 \text{ h}$)	« running
Lid outer surface, no soil (scour)	$\leq 0.57 \mu\text{rem/h}$	same target
Side of buried vault, in soil, 1 m from wall	$\leq 5 \mu\text{rem/h}$ (occupational not applicable; this is dirt)	—
House chase, after two bends	$\leq 50 \mu\text{rem/h}$ contact on the pipe chase (worker, short time)	—

0.57 $\mu\text{rem/h}$ is ~5% of a typical 10 $\mu\text{R/h}$ backyard background. A meter may not see it.

6.3 Layer stack (radial, type unit)

Inside to outside:

Layer	Thickness	Outer diameter	Function
Core	200 mm radius	400 mm	Fuel + heat pipes
Graphite reflector	120 mm	640 mm	Neutrons back, some shield
SS inner vessel	12 mm	664 mm	Seal, not the gamma wall
Borated polyethylene (BPE, 5 wt% B)	250 mm	1,164 mm	Slow and eat neutrons
Borated concrete (2 wt% B)	150 mm	1,464 mm	Neutrons + some gamma
Lead (ASTM B29)	80 mm	1,624 mm	Gamma, including capture
Reinforced concrete	400 mm	2,424 mm	Gamma, structure, missile
Steel outer shell	16 mm	2,456 mm	Skin, lift lugs, waterproof

Outer diameter $\approx$ 2.46 m (8.1 ft). Outer height $\approx$ 2.60 m (8.5 ft).

Large septic / small cask. Not a campus.

Axial: same water-equivalent above and below the core, plus a 250 mm glass-dump hopper **above** the core, still inside the vessel, still under the top shield. Floor chase leaves through the bottom biological shield in an offset steel labyrinth (two 90° turns, 200 mm BPE lining).

6.4 Attenuation (handbook, not a substitute for MCNP)

Tenth-value layers (TVL), approximate, fission spectrum / ~1 to 2 MeV gamma:

Material	TVL neutrons (fission)	TVL gamma (~1.5 MeV)
BPE	~12 to 15 cm	poor (use lead)
Concrete	~20 to 28 cm	~20 to 24 cm
Lead	poor	~4.0 to 4.5 cm
Dry soil	worse than wet by ~1.5 to 2× for n	~25 to 35 cm
Wet soil	closer to concrete for n	~20 to 30 cm

Neutron path (radial): 25 cm BPE (~1.8 TVL) + 15 cm borated concrete (~0.6 TVL) + 40 cm concrete (~1.6 TVL) $\approx$ **4 TVL $\approx 10^4$** before soil. Dry-soil bonus is then extra. **Not enough alone** for 10^8 to 10^9 . The reflector, BPE boron capture, and the fact that the core is small ($1/r^2$ and self-shielding) have to be in the transport model.

Preliminary stack is sized to give MCNP a geometry that can meet DB-5 and DB-6. If it misses, add BPE or lead, do not add “more dirt” as the fix.

Gamma path: 8 cm lead $\approx$ 1.8 TVL (×63) plus 40 cm concrete $\approx$ 1.8 TVL (×63) $\approx$ **×4,000** plus soil. Capture gammas born in the BPE are what the lead is for. Order the layers as drawn. Do not put lead inside the BPE.

Sign-off run: MCNP6 or equivalent, running and shutdown, dry sand (5 wt% water) and wet (15 wt%), lid buried and lid bare, chase streaming, skyshine with lid bare. Until that run exists, DB-5 is a requirement, not a demonstrated number.

6.5 Playset (from Chapter 3, numbers)

5 mrem/y $\times$ (365 h / 8,760 h) = 0.21 mrem/y at 1 h/d
 Coast-to-coast flight $\approx$ 3 to 5 mrem
 Playset year $\approx$ 1/15 to 1/25 of one flight

7. Passive safety

7.1 Normal

- Low power density, heat pipes, no pump.
- House load steps are the battery. Core is steady.
- Doppler in UCO kernels: negative. Graphite: slight positive moderator temp in some lattices; **the lattice shall be designed so the net power coefficient is negative** from 20°C to 750°C. That is a hard SCALE/MCNP criterion.

7.2 Anticipated faults

Fault	Response
Oil pump off / Stirling off	Condenser T rises. Analog bi-metal or fusible link releases drum springs in . Decay heat to earth.
One or two heat pipes dead	Margin in §4.1.
Grid loss	Gateway islands. Vault does not care.
Chase shear (excavator)	Double-wall oil spill is a hazmat event, not a core event. Drums in on overtemp.
Flood	External water: more shield, more cooling. Inner vessel designed not to flood. If it does, k_{eff} criterion in §3.4.
Dry soil / drought	Shield still meets DB-5 without wet dirt.
Scour to lid	DB-6.

7.3 Glass dump (last ditch, not credited in the licensing basis until tested)

Item	Spec
Mass	80 kg borosilicate frit + 15 kg B ₄ C powder
Hold-up	Hopper above core, 304 SS, fusible plug
Plug	Au-Ge or Al-Si eutectic, melt 750 ± 20°C (above normal HP 600°C, below vessel limits)
Action	Gravity pour into core channels, boron kills reactivity, glass wets and sets
Power	None
Expected use	Once in never. Qualification: furnace test on a dummy lattice.

7.4 Why a “runaway” is not a plant-scale event

Stored energy in 35 kW for 10 s = 350 kJ (one-tenth of a kWh)

A prompt jump that self-limits in milliseconds dumps a small energy into a large graphite/steel mass. There is no 3 GWth inventory. Decay heat is a space heater. That is the size argument in Chapter 6, in numbers.

8. Mechanical envelope, mass, soil, flood

8.1 Mass build (type unit)

Item	Volume or note	Mass
Concrete (shield + structure)	$\sim 10.5 \text{ m}^3 \times 2,400 \text{ kg/m}^3$	25,200 kg
Lead	$\sim 1.05 \text{ m}^3 \times 11,340 \text{ kg/m}^3$	11,900 kg
BPE	$\sim 0.85 \text{ m}^3 \times 940 \text{ kg/m}^3$	800 kg
Steel shells, vessel, lugs	—	2,800 kg
Graphite + TRISO + Na + drums	—	1,400 kg
Total		$\sim 42,100 \text{ kg}$ (92,800 lb, 46.4 short tons)

Lead volume check: cylindrical shell ID 1.464 m, $t = 0.08$ m, $h = 1.80$ m:

$$\begin{aligned} V_{\text{wall}} &= \pi h (R^2 - r^2) = 3.1416 \times 1.80 \times (0.812^2 - 0.732^2) = 0.70 \text{ m}^3 \\ V_{\text{ends}} &\approx 2 \times \pi \times 0.77^2 \times 0.08 = 0.30 \text{ m}^3 \\ V_{\text{Pb}} &\approx 1.00 \text{ m}^3 \quad (\text{table uses } 1.05 \text{ with laps}) \end{aligned}$$

Concrete: outer cylinder 2.42 m OD, 2.60 m tall, minus inner cavity ~1.62 m, plus extra in the floor labyrinth. 10.5 m^3 is a rounded shop number; the civil drawing shall close the takeoff.

8.2 Soil bearing and buoyancy

Footprint $\approx \pi \times 1.23^2 = 4.75 \text{ m}^2$.

$$q = 42,100 \text{ kg} \times 9.81 \text{ N/kg} / 4.75 \text{ m}^2 = 87 \text{ kPa} \approx 1.8 \text{ ksf}$$

Most residential soils are 1.5 to 3 ksf allowable. Spec a **200 mm compacted crushed-stone pad**. If the geotech says no, a thin concrete mat. Not a pile foundation.

Displaced volume $\approx \pi \times 1.23^2 \times 2.60 = 12.3 \text{ m}^3$. Buoyant force in full flood:

$$12.3 \text{ m}^3 \times 1,000 \text{ kg/m}^3 = 12.3 \text{ t} \ll 42 \text{ t}$$

The vault **does not float**. No hold-down straps required for buoyancy. Still grout the pad so it does not walk in a scour hole.

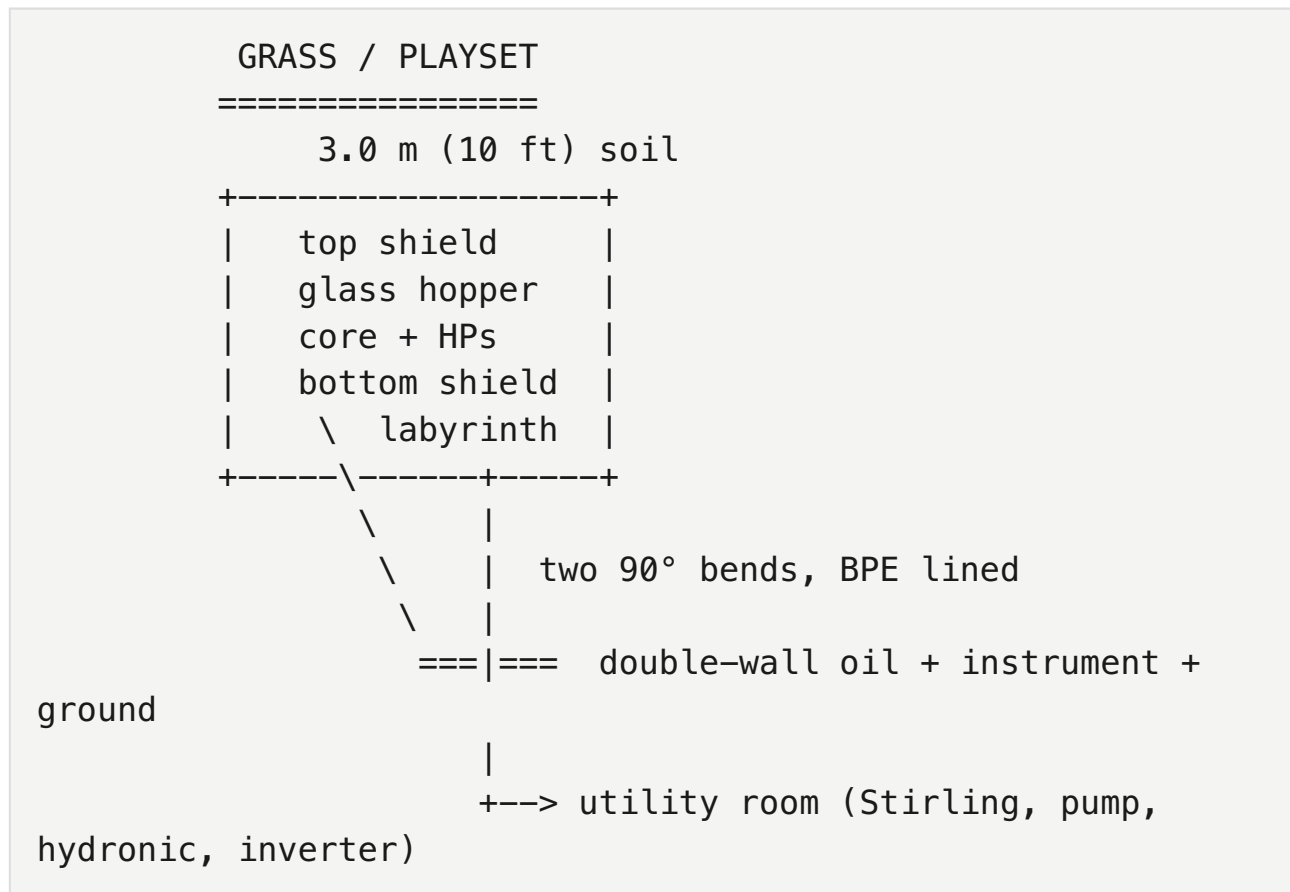
8.3 Lift and transport

- Four trunnions on the steel shell, 2 × 25 t WLL each, redundant pair.
- Highway: overweight / overwidth permit, lowboy. 2.46 m wide is legal in most states with a permit (8.5 ft wide is the usual unpermitted cap; this is ~8.1 ft **body**; add 6 in lugs and you need a permit). Height on a 0.5 m deck ≈ 3.1 m, OK.
- Crane: 80 to 100 t class at 6 to 8 m radius (ordinary house-move / septic crane).
- Package: analyze as a **Type B** transport (10 CFR 71) for the fueled vault. That is a real cost and a real test (drop, fire, immersion). Do not skip it.

8.4 Lid-to-grass

3.0 m (10 ft) of backfill above the lid. Dose still designed to DB-5 / DB-6 with the lid bare. Soil is bonus. Vault height 2.60 m + 3.0 m cover → excavation ~5.6 m (about 18.5 ft) to the pad. That is a deep buried-tank hole, shored, in the same class as a large underground fuel tank or cistern, not a shallow septic lid. Grass footprint of the object is about **10 × 10 ft**. Setbacks from the house are a local buried-tank permit, not this spec.

9. Penetrations and chase



Through the floor only:

- 2 × thermal-oil process + containment
- 1 × instrument bundle (MI cable, Class 1E-equivalent for drum trip)
- 1 × startup heater power (used once, then locked out)
- 1 × ground

No air intake. No stack. No lid penetration. Streaming: each line offset so there is no straight line from the core to the grass.

Acceptance: MCNP chase model $\leq 50 \mu\text{rem/h}$ on the house-end pipe.

Frost: bury below local frost line or heat-trace the oil (oil is already hot when running; shutdown in a Minnesota January needs freeze protection on the oil, pour-point of Dowtherm A is fine, viscosity is the issue). House pump and a small trickle or a tank heater.

10. I&C and states

10.1 States

```
FACTORY_SHUTDOWN → (seal) → TRANSPORT → SET → CONNECT
    → INSPECT → LICENSED_START → RUN → (overtemp |
command)
    → SHUTDOWN → COOL → LICENSED_LOCK → LIFT
```

State	Drums	Heat	Oil pump	Notes
Factory / transport / set	Locked in	0	off	Fresh fuel, no decay heat
Licensed start	Stepped out	rising	on	Trace heat on Na if needed
Run	Fine arc	35 kWth	on	Owner cannot move drums
Shutdown	In	decay	off	Analog trip or licensed
Lift	Locked in	decay only	off	Wait until P_decay < 0.4 kW (~1 d)

10.2 Trip (analog, two channels, no phone)

- Condenser / HP vapor T high
- Drum position disagree

- Seismic switch (optional, cheap)
- Manual licensed trip at the house panel (key, not an app)

App / inverter: status only (Chapter 2). Not a drum control. Not a storm mode.

10.3 Instruments (ex-core)

- 2 × fission chambers or Compensated ion chambers in the reflector (startup + run)
 - 6 × Type K or N thermocouples on heat-pipe condensers
 - Oil T and flow at the HX
 - He / moisture in the inner vessel
 - Lead-door / seal wires (IAEA-style) for tamper
-

11. Environments

Load	Spec
Flood	Full immersion, indefinite. IP68 / submarine weld on outer shell.
Dry sand	Shield analysis at 5 wt% water.
Design soil T	−20 to 40°C far field
Seismic	ASCE 43 or IBC site class D, 0.3 g, vault as a rigid box on a pad. No in-core mechanisms except drums (latched).
Tornado / missile	Buried. Lid under 3.0 m soil. Bare-lid case is flood/scour, not a missile.
Fire (transport)	10 CFR 71 800°C / 30 min. Glass dump may fire; that is acceptable if k _{eff} stays down.
Design basis excavation	Assume a backhoe can hit the shell. Contact dose on bare shell still a worker number, not a playset number. Keep it low enough for a 1-hour unplanned exposure < 100 mrem. That is another MCNP tally.

12. Materials and manufacturing (all real parts)

Part	Material / spec	Who already makes this class of thing
TRISO-UCO	19.75% U, ASTM / NRC fuel QAP	BWXT, X-energy line
Graphite	Nuclear grade (NBG-18 or equivalent)	GrafTech / Toyo Tanso class
Heat pipes	SS316L or Haynes 230, Na	ACT, Thermacore, space vendors
Inner vessel	SS316L, ASME IX weld, He leak $\leq 1 \times 10^{-8}$ mbar·L/s	Any N-stamp or high-spec vessel shop
BPE	5% boron polyethylene	King / Nuclear-shielding catalogs
Lead	ASTM B29, cast or brick, no voids (X-ray)	Shielding foundries (cask vendors)
Concrete	4,000 psi, 2% borate in inner pour, rebar epoxy	Precast nuclear / cask plants
Outer shell	Carbon steel, coated, or 316	Tank / cask shops
Drums	B ₄ C in SS cladding, geared, lockable	Control-drum vendors, research-reactor shops
Frit dump	Borosilicate + B ₄ C, eutectic plug	Glass + specialty metals
Oil loop	Double-wall A312, Dowtherm A	Process / CSP contractors
Stirling (utility room)	Hermetic free-piston, 2 kWe class	Sunpower / Qnergy class
Inverter / gateway	UL 1741, IEEE 1547	Tesla / Schneider / SolarEdge class
Dry cooler	27 kW air coil	Any HVAC OEM

Factory sequence: fuel compacts → core + pipes + drums → inner vessel (He leak) → BPE pour / tiles → lead pour → concrete precast or poured in a pit form → outer shell → trunnions → seal wires → cold critical (zero-power) at the factory site in a cell → drums locked → ship.

Cold critical at the factory is a real nuclear operation (a licensed facility). The backyard is not that facility.

13. House interfaces (Chapter 4), numbers

Interface	Spec
Electrical	8 kWe nominal, 10 kWe installed convertors, 240 V split-phase after inverter
Battery	27 to 40 kWh, 10 to 15 kW
Hydronic	27 kWth, 80 to 120°F, isolated plate HX, ~7 gpm house water
Dry cooler	27 kW, outdoor
Oil loop	0.40 kg/s, 350°C max, double-wall
Clearances	Grass patch ~10 × 10 ft; setbacks are a local buried-tank permit; dry cooler as a condenser; crane path kept for Chapter 7
Flooded house	Inverter, battery, Stirling above design flood. Vault may be underwater.

14. Safety case in one page (for the reviewer)

1. **Inventory is small.** 35 kWth, 30 kg HM, 1.35 kg fission-equivalent U-235 in 100 years.

2. **Decay heat is a heater.** 2.3 kW at 1 s, 0.3 kW at 1 d, into 42 t and earth.
3. **No pump in the vault.** Heat pipes + earth.
4. **Cannot go prompt on a single fault.** DB-11 + Doppler.
5. **Dose at the grass is a living-on-it number of ≤ 5 mrem/y,** including dry soil and a bare lid, pending MCNP. A playset hour a day is ~ 0.2 mrem/y.
6. **Holes point down.**
7. **Fresh core is cold on the truck.**
8. **Conversion failures trip the core;** they do not open it.
9. **46 t, no float, crane-only access.**
10. **Take-back is a sealed lift** (Chapter 7). Residual HM is still tens of kilograms at a licensed site.

What this does **not** claim: that MCNP has been run, that 10 CFR 71 is passed, that HALEU is contracted, or that a Stirling has run 100 years. Those are §16.

15. Security and material control (high level)

- 30 kg of 19.75% U is HALEU. Attractive. Not direct-use HEU.
 - Access: 42 t monolith, IAEA-style seals, no owner hatch, drums locked.
 - Startup source and drum key: two-person, licensed, logged.
 - GPS / tamper on transport.
 - This section is a requirement list, not a how-to.
-

16. What an engineer must still run before a stamp

Chapter 10 is the readable version of this table: every design claim, what would count as shown, and which rows need specialized software or equipment. This table stays the engineer checklist.

Work	Code / method	Pass / fail
Lattice, drums, flood, temperature coefficients	MCNP6 or Serpent + SCALE/KENO	§3.4 table
Depletion 100 y	ORIGEN / Serpent depletion	$k_{\text{eff}}(\text{EOL}) \geq 1$ with drums; peak burnup ≤ 80 GWd/t
Shield, chase, bare lid, dry sand	MCNP6	DB-5, DB-6, chase 50 $\mu\text{rem/h}$
Heat pipes (startup, fail, freeze)	Transient HP model + 1 failed-pipe test	35 kW with 2 pipes out
Vessel, pad, lift, seismic	ASME VIII or III, ASCE 7/43	Stress < allowables
Type B package	10 CFR 71	Drop, fire, immersion
Oil loop fire / leak	NFPA / local	Double-wall, detected
Stirling + inverter	UL / IEEE 1547	Listing
TRISO qualification	Existing AGR/HTGR data + lot tests	Failure fraction vs source term
Glass dump	Furnace mockup	Flows, B worth, no vessel cut
Cold critical	Licensed cell	Matches model
Environmental (NEPA etc.)	Legal	Not this file

Until those rows are green, this is a **design specification**, not a signed reactor. Chapter 10 expands each row (method, regulation, pass/fail, what you cannot substitute).

17. Risks that are real

Risk	Why it is real	Mitigation in this spec
100-year convertor	Not demonstrated	Convertor is in the utility room
100-year graphite / TRISO in this flux	Data exist for shorter, hotter cores	Low power density; still need a materials memo
HALEU supply	US capacity is thin	30 kg/unit is small; still a contract risk
Type B + first-of-a-kind license	Cost and years	Chapter 6; size argument
Oil at 350°C in a yard	Fire	Double-wall, buried, trip on leak
Sell-back	Utility, not physics	Chapter 2
Public dose if someone cheats the BPE	Dirt is not the shield	DB-6 bare lid

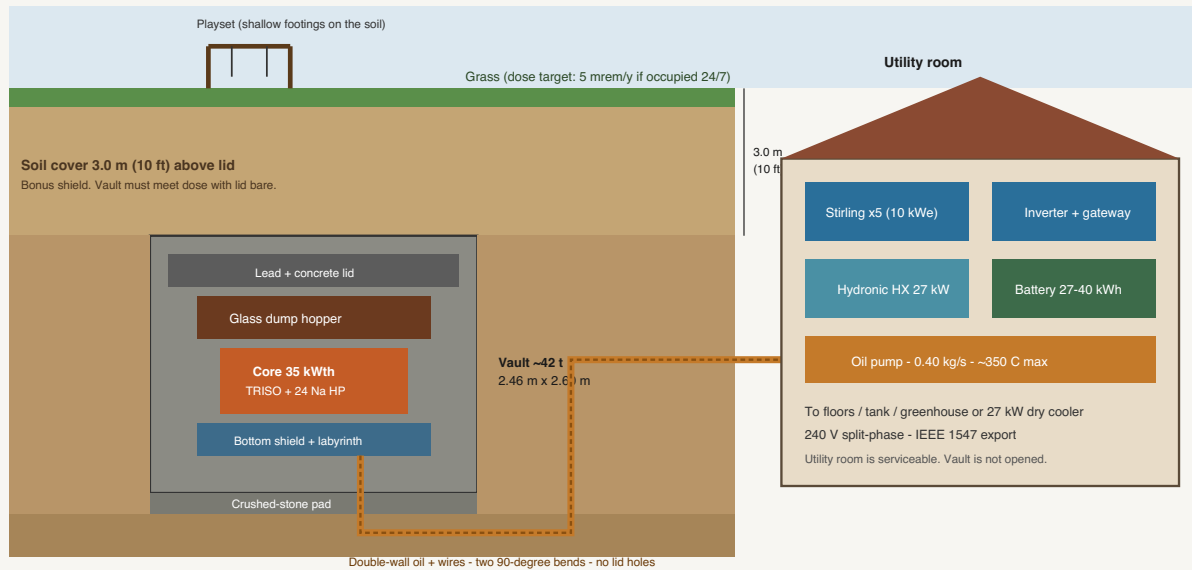
18. Diagrams

Vector drawings live in /diagrams/. They are type-unit schematics, not fabrication drawings. Figures 1 to 8 match §§3, 6, and 8. Figures 9 to 21 are the Chapter 9 first-principles set. Figures 22 to 26 are the Chapter 10 proof-plan set.

18.1 Site section

Figure 1. Site section, BGS-8 (not to construction scale)

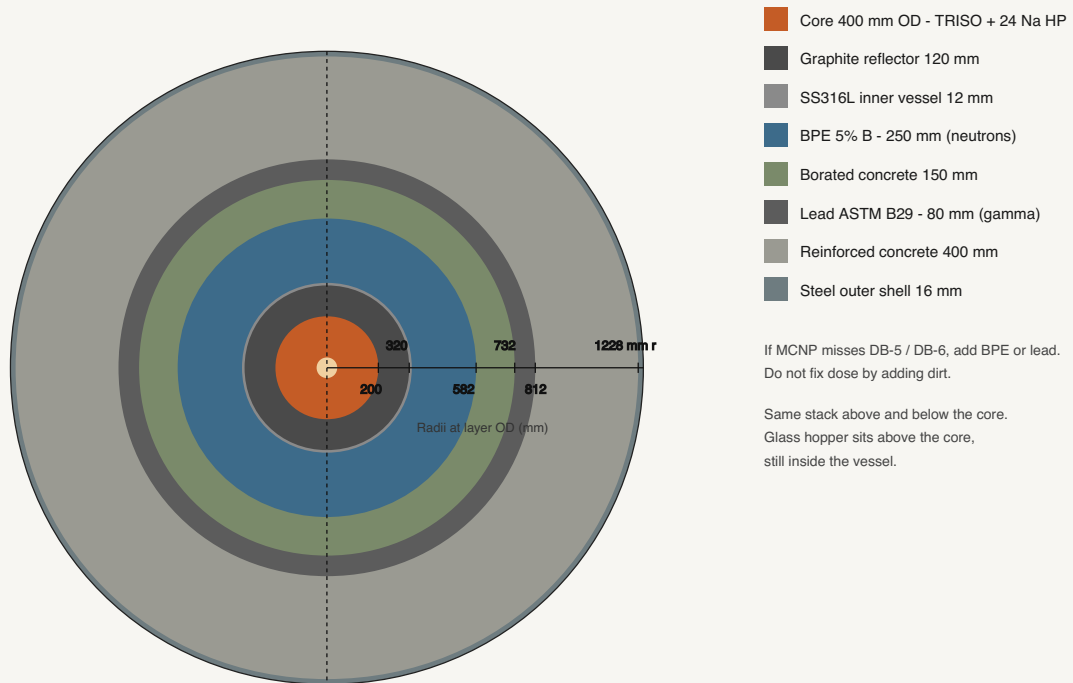
10 ft (3.0 m) of soil above the lid. Vault 2.46 m OD x 2.60 m H. Chase floor-only, two 90-degree bends. Utility room is replaceable.



18.2 Radial stack

Figure 2. Radial shield stack (half-section, type unit)

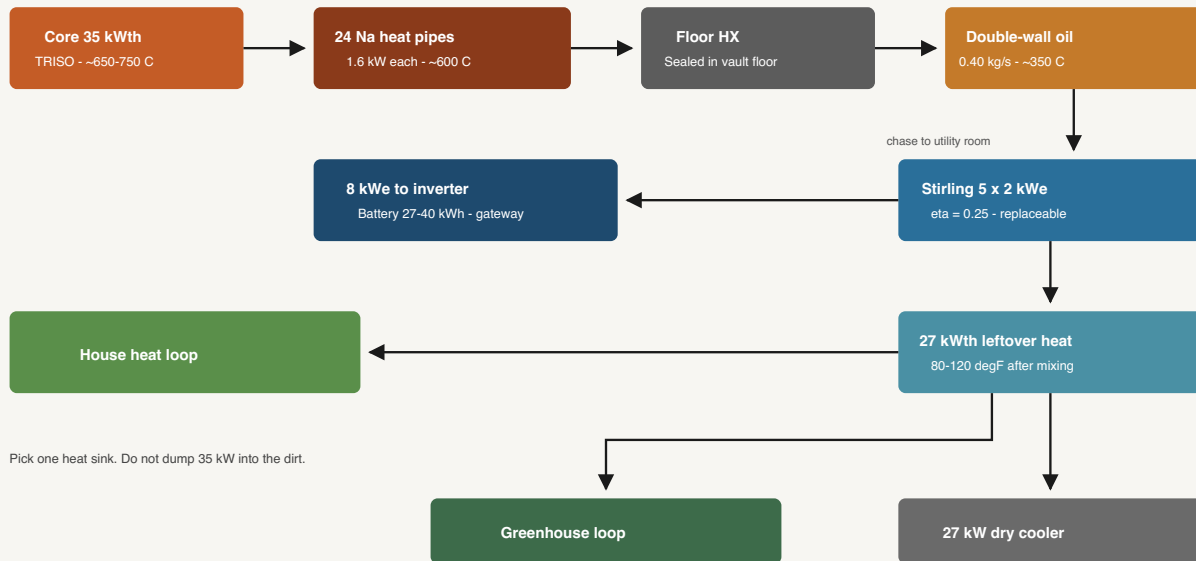
Outer diameter 2.46 m. Layers ordered so capture gammas born in BPE hit lead next. Soil is not shown; it is bonus above the lid.



18.3 Heat and power

Figure 3. Heat and power cascade

35 kWth in the vault. 8 kWe and 27 kWth out at the utility room. No shaft in the vault.

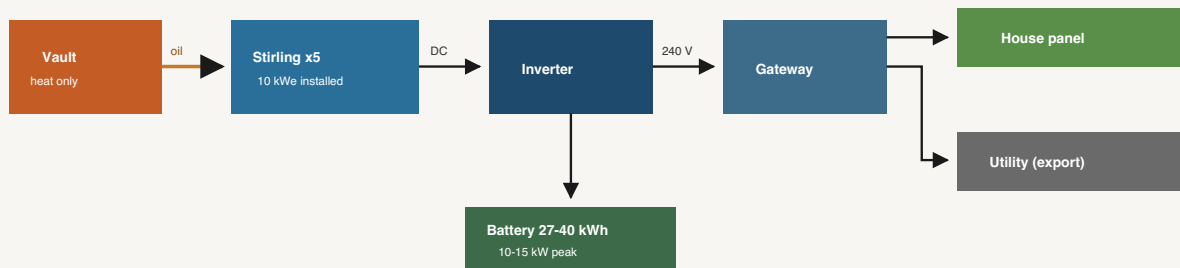


If the oil pump or Stirling units stop, condenser temperature rises and analog drum springs drive shutdown. Decay heat then goes to the vault mass and earth.

18.4 Electrical one-line

Figure 4. Electrical one-line

Vault is heat-only. Stirling, inverter, gateway, and battery live in the utility room. Gateway required so the house can run when the street is dark.

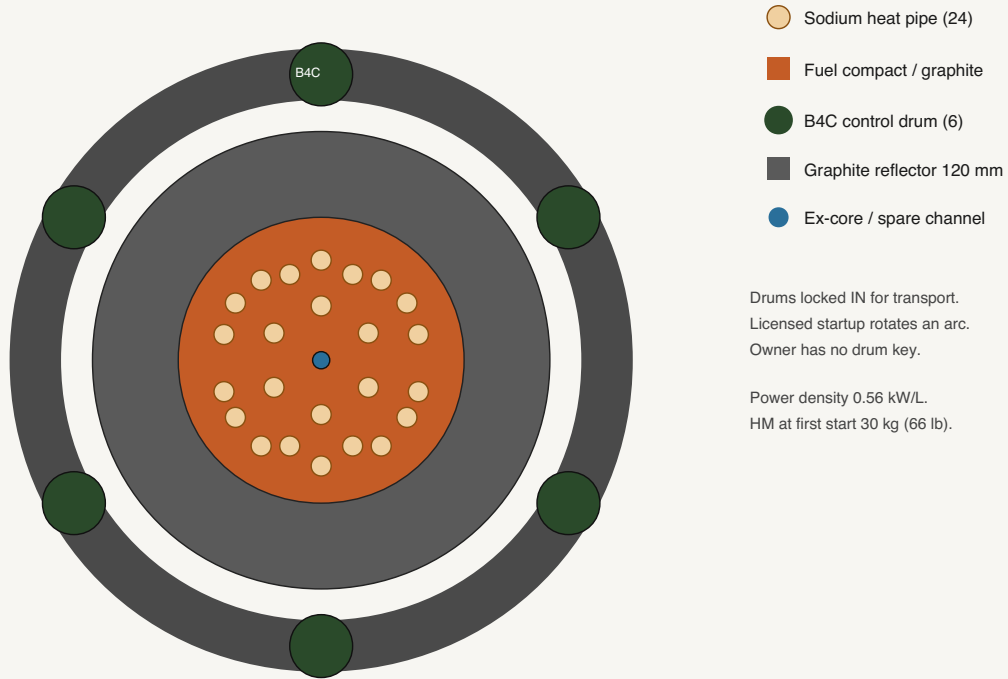


Without the gateway the inverter trips when the grid dies (anti-islanding). Chapter 1 life-safety case requires the gateway.

18.5 Core plan

Figure 5. Core plan (schematic, not a fabrication drawing)

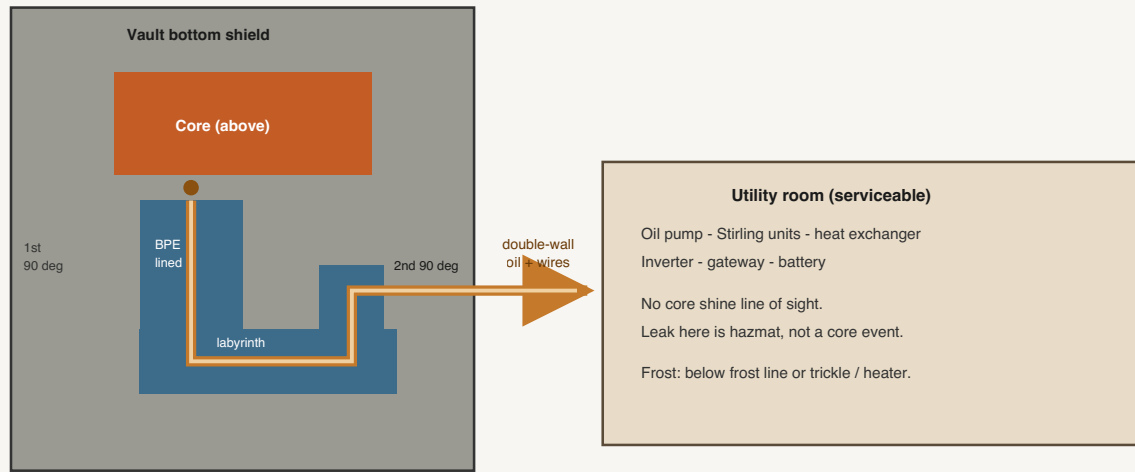
400 mm active OD. 24 sodium heat pipes. Fuel compacts in graphite. 6 B4C drums in the reflector. Exact lattice is an MCNP item.



18.6 Floor chase

Figure 6. Floor chase labyrinth

No straight line from the core to the grass. Two 90-degree bends, BPE-lined. Streaming is an MCNP tally (50 microrem/h or less at the house-end pipe).



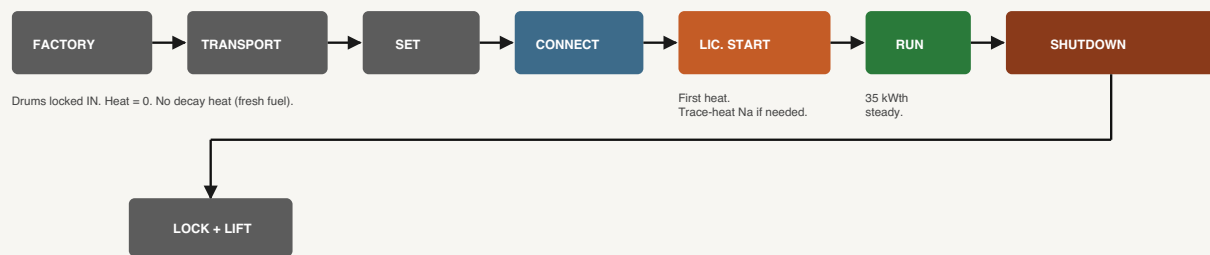
Forbidden: any hole through the lid (would point at the playset).

Allowed through floor only: 2 x oil (process + containment), instrument bundle, startup-heater power (locked out after start), ground.

18.7 States

Figure 7. Vault states

Fresh fuel is cold on the truck. First heat is licensed startup after burial. Locks in before the crane returns.

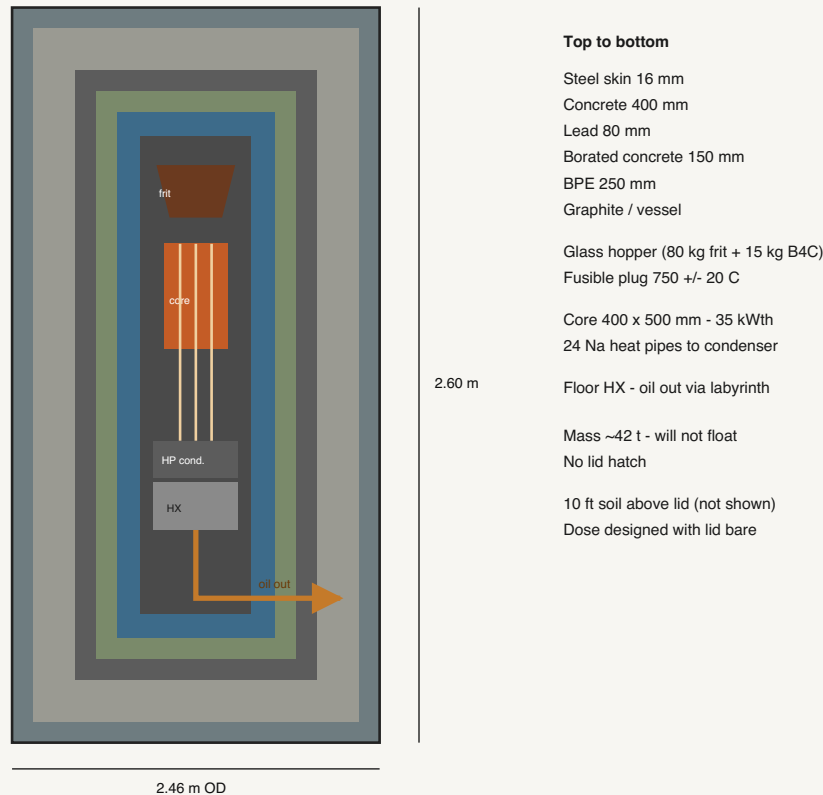


Overtemp or licensed trip drives drums IN (analog, two channels). Wait until decay heat < 0.4 kW (~1 day) before lift. Owner phone cannot move drums.

18.8 Vertical cutaway

Figure 8. Vault vertical cutaway

2.60 m outer height. Same shield above and below the core. Hopper is last-ditch only (plug 750 C). No owner hatch.



18.9 Fuel, neutrons, sodium, shutdown (Chapter 9)

Figures 9 through 21 are the fuel-and-physics set. They live in Chapter 9 so this file can stay the tables you size and buy against. Types: TRISO particle, particle-to-core, core makeup, heat pipe, one fission, neutron life, thermal vs fast, chemistry, shutdown ladder, PWR/BWR/BGS, two sodiums, scale, Doppler.

18.10 Proofs (Chapter 10)

Figures 22 through 26 live in Chapter 10: three kinds of proof, decay-heat fit, shield bounds versus MCNP, Carnot versus Stirling, Doppler chain without a coefficient.

19. Type-unit data sheet (BGS-8)

Electrical, net	8.0 kWe (family 5 to 10)
Thermal	35 kWth
Conversion	Utility-room Stirling, $\eta = 0.25$, 5×2 kWe
Reject	27 kWth
Fuel	TRISO-UCO, 19.75% U-235, 30 kg HM (66 lb)
Burnup target	80 GWd/t
Life	100 y sealed, $\geq 90\%$ CF
Core	400 mm $\times$ 500 mm, 0.56 kW/L
Heat pipes	24 $\times$ 1.6 kW, sodium
Vault OD $\times$ H	2.46 m $\times$ 2.60 m
Mass	~ 42 t (93,000 lb)
Cover	3.0 m (10 ft) soil above lid (dose still lid-bare)
Dose, live-on-grass	≤ 5 mrem/y (MCNP to confirm)
Dose, 1 h/d playset	~ 0.2 mrem/y
Flood	Immersion OK, no float
Penetrations	Floor only
Owner hatch	None

20. What this chapter covered

This is the vault and core **design specification**: power, fuel, pipes, layers, mass, dose targets, states, and buyable parts. Chapter 5 is the same machine in English. Chapter 9 is the fuel and the first principles. Chapter 10 is the proof plan. Chapter 3 is the grass. Chapter 4 is the utility room. Chapter 6 is the stamp. Chapter 7 is the crane.

The arithmetic in §2, §3.3, §4.1, §4.3, §6.1, §6.5, §8 is closed. Rechecked in version 1.0: $8 \text{ kWe} / 0.25 = 32 \text{ kWth}$, specified 35 kWth; $8 \times 8,760 = 70,080 \text{ kWh/y}$; $35 \text{ kW} \times 100 \text{ y} = 1,278 \text{ MWd}$; 16 kg consumed at 80 GWd/t; 30 kg at first start; $24 \times 1.6 \text{ kW} = 38.4 \text{ kW}$ pipes; oil 0.40 kg/s; vault ~42 t, will not float; playset $5 \times 365/8,760 = 0.21 \text{ mrem/y}$. The dose **stack** in §6.3 is an engineering starting geometry. The **dose number** on the lawn is a requirement until MCNP says it is a result.

If a line in this file and a line in Chapter 5, Chapter 9, or Chapter 10 disagree, this file wins for engineering, and those chapters should be updated to match (heavy metal at first start is 30 kg, not 10 kg).

9. The Fuel

Chapter 5 is the buried box: how big it is, how heat and power leave through the floor, and why the yard never gets a hatch. The engineering tables that size the parts live in Chapter 8. This chapter is why the insides are the machine they are: what the fuel is made of, how the graphite core is put together, how sodium moves without a pump, what a neutron does when it hits an atom, how the chemistry changes over a century, and how the reaction is meant to stop when something goes wrong.

A homeowner should be able to follow the story. A physicist should see the actual mechanism, not a slogan. Where a number is a requirement still waiting on a computer model of the core, the text says so. If this chapter and the engineering tables disagree, the tables win.

This is the design of the core, in ordinary language:

The fuel is tiny coated particles of uranium, packed in graphite. Graphite slows the neutrons so the next split is likely. Neutrons that have been slowed this way are said to travel in the **thermal** range: they have about the same energy as ordinary heat motion in the room. That is not a **fast** reactor. A fast reactor has no graphite (and no water) on purpose, so the neutrons stay at high speed. This vault also does not use water to slow the neutrons. Water is what almost every US power plant uses. Graphite is the solid that sits in this core instead.

What the fuel is

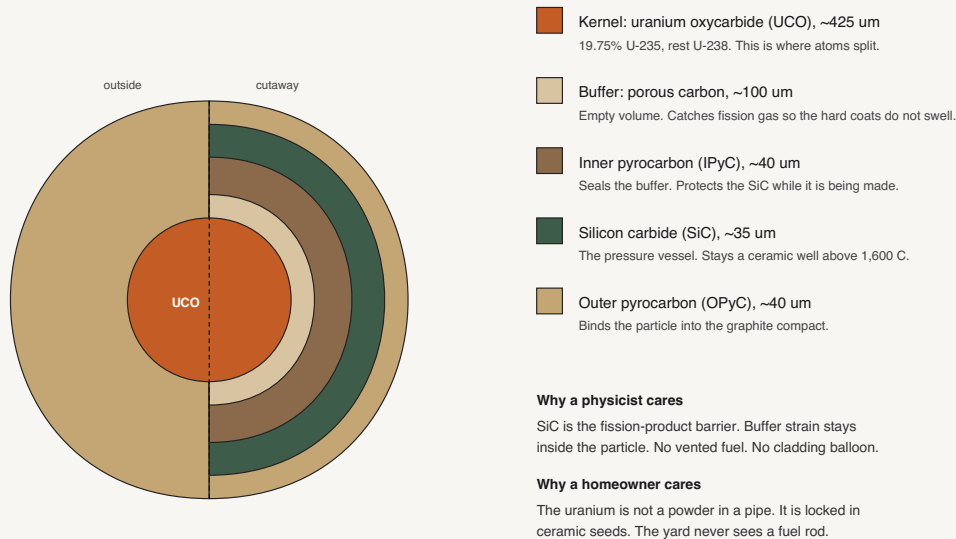
The useful uranium in this vault starts as about **30 kilograms (66 pounds)** of heavy metal. That is a paint-can, inside a beer-keg-to-dishwasher core, inside a septic-tank vault. The crane is for the vault. Nobody on the lot ever handles the fuel.

About **20%** of that uranium is **U-235**, the isotope that splits easily when a slow neutron hits it. The rest is **U-238**, which mostly sits there. (Some of it slowly turns into another isotope that can also split. That comes later.) Ordinary US water plants use about 4 to 5% U-235. This mix is richer than that, and still **below 20%**. Below 20% is the civilian line. Weapons uranium is a different number, far above 20%. Staying under 20% is a security choice as well as a physics choice. The name for this band is high-assay low-enriched uranium. The short label is HALEU.

The center of each particle is **uranium oxycarbide**: uranium mixed with oxygen and carbon, not metal shavings and not a powder in a pipe. Oxygen and carbon together keep the kernel from making too much carbon monoxide as it is used, which would over-pressure the coats. That is why this kernel is not plain uranium oxide.

Figure 9. One TRISO particle (typical of the class, not a shop drawing)

About 0.9 mm across. Each coat is a sealed can. Fission gas stays in the buffer. There are tens of thousands of these in the core.



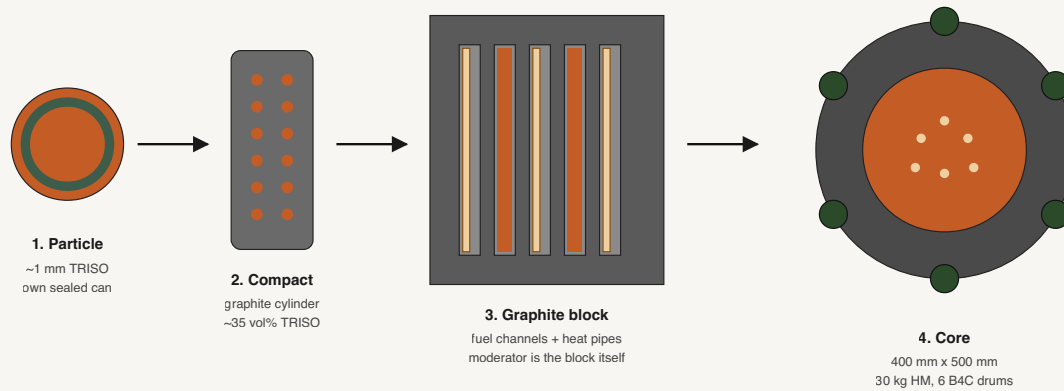
Layers and thicknesses are typical AGR / X-energy / BWXT class. Exact kernel size is a fuel-spec item in Chapter 8.

Each kernel is wrapped in three hard ceramic coats, plus a soft carbon layer that acts as a cushion. The stack is called **TRISO** (a name for those three structural coats). Typical of the class, one particle is about **0.9 mm** across, a large grain of sand. The silicon carbide coat is the pressure vessel. The soft buffer is empty volume so gas from splitting has somewhere to sit. There are tens of thousands of these particles in the core. Each one is its own sealed can.

They are pressed into graphite cylinders, about **35%** particles by volume. Those cylinders sit in channels in a graphite block. That block is both the furniture and the thing that slows neutrons.

Figure 10. From one seed to the core

Particles are pressed into graphite cylinders (compacts). Compacts sit in channels in a graphite block. Heat pipes and control drums share that block.



Heavy metal at first start is 30 kg (66 lb) inside that orange disk. The septic-tank vault around it is almost all shield. This figure is the idea, not a shop drawing.

Mixed into the compact, or as separate rods, is a **burnable poison**: a neutron sponge (erbium oxide or gadolinium oxide). It is used up as the fuel is used up, so output stays even for decades and the control drums barely have to move. There is no refueling truck. There is no hatch.

This fuel is not invented here. The same class of coated particles is already made in the United States for other small reactors. This product would license that class, or build from it, at house power. It is not a space machine lowered into a yard.

The core around it

The active core is **400 mm across and 500 mm tall** (about 16 inches by 20 inches). Volume is **62.8 liters**. At **35 kW of heat** that is **0.56 kW per liter**. A typical US water plant packs about **100 kW into each liter**. This core is about **200 times** less dense. That is how a

sealed life on the order of a century is even thinkable: the neutron battering that ages metal is much gentler because the power in each liter is small.

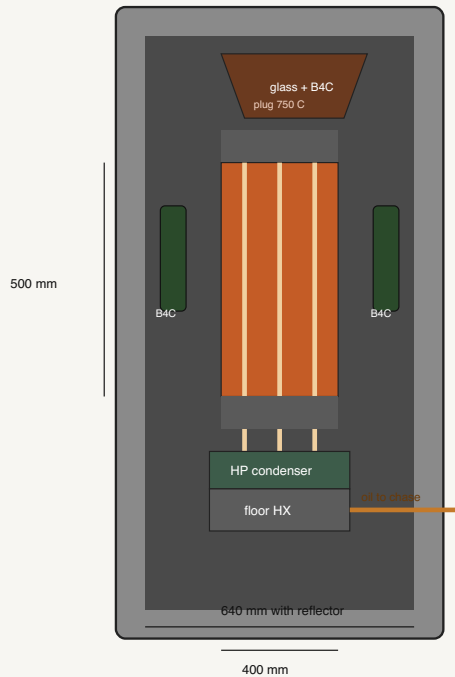
Around the fuel is **120 mm** of extra graphite, so the cartridge you would recognize is about **640 mm** across: beer-keg to dishwasher. That extra graphite bounces leaking neutrons back in. Six drums sit in it. Each drum has a face loaded with boron carbide, which eats neutrons. Rotate the absorbing face in, the reaction dies. Rotate it partly out, the reaction holds. Mechanical locks pin the absorbing face in for the truck. Licensed startup is a keyed, two-person rotation to the operating position. The owner key does not exist. A phone app, if there is one, does not move them.

24 sodium heat pipes run through the graphite, each about **22 mm** across (a fat marker), down to a cool block in the vault floor. Four spare channels are plugged. A stainless can around the graphite (about 700 mm across, 12 mm wall) holds helium or another inert gas. Design pressure is low, **under 0.5 MPa** (about 70 psi). A US water plant runs about **15 MPa** (about 2,200 psi) in the primary loop. This can is not that kind of pressure cooker.

Above the core, still inside the can, is a hopper of glass grit and boron carbide. That is last-ditch only. It is described with shutdown, not with Tuesday.

Figure 11. Core makeup (vertical cut, cartridge only)

Beer-keg to dishwasher. Graphite all the way through. Sodium only inside the 24 pipes. No water. Hopper is last-ditch, not day-to-day.

**Inside the inner vessel**

Helium or inert fill around the graphite.
Not a pressure cooker. Design pressure d 0.5 MPa.

Active core

400 mm diameter x 500 mm tall. 62.8 liters.
35 kW / 62.8 L = 0.56 kW per liter.
A US PWR is about 100 kW per liter. This core is about 200 times less dense. That is the 100-year fluence trick.

24 sodium heat pipes + 4 plugged spares

22 mm tubes through the graphite, down to the condenser in the vault floor.

6 B4C drums in the reflector

Locked IN on the truck. Licensed startup rotates an arc. Owner has no key.

Graphite is moderator and reflector

It slows neutrons so U-235 will split, and bounces leaking neutrons back in.
That is why this is not a fast reactor.

There is no water in the core. There are no swimming-pool rods. Nothing in the core can boil dry. Heat leaves through the sealed pipes. Electricity is made later, in the utility room. The vault is a sealed heater.

How sodium moves, and how the pipes are built

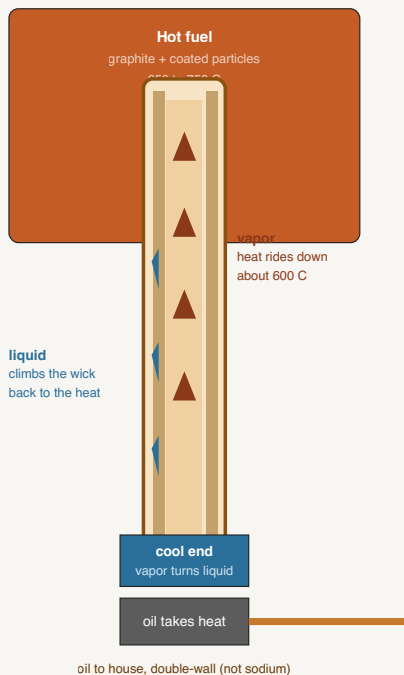
Sodium does not circulate around the core the way water circulates in a water plant. There is no sodium pump and no sodium pool.

Each heat pipe is a closed metal tube with a little sodium inside. The middle of the pipe sits in the hot fuel (**about 650 to 750°C** at the compact). The lower end sits on a cool block in the vault floor. Sodium boils at the hot section. Vapor moves to the cool end (**about**

600°C) and gives up its heat. Liquid soaks into a wick and returns to the hot section by capillary action, the same kind of pull that makes a paper towel drink a spill. No motor. Sodium never leaves that tube. It never mixes with the oil that runs to the house, and it never mixes with house water.

Figure 12. One sodium heat pipe (no pump)

A closed metal tube with a little sodium inside. Heat rides as vapor to the cool end. Liquid climbs back in the wick. Sodium never leaves this tube.



What you are looking at

Each of the 24 pipes is one sealed metal tube, about as thick as a fat marker. A few milliliters of sodium live inside. There is no pump and no sodium loop.

The four-step cycle

1. Heat from the fuel boils the sodium at the hot end.
2. Vapor shoots to the cool end. The heat rides along.
3. Vapor condenses. Heat leaves the pipe into the oil.
4. Liquid soaks into the wick and climbs back up.

Two walls, different from the oil

The heat pipe itself is one sealed tube (first wall). The steel can around the whole core is the second wall. If a pipe leaks, sodium stays in that can, in inert gas, and freezes (sodium melts at 98 C). The oil pipes that leave the vault are a separate double-wall loop.

If one pipe dies

That tube is out. The other 23 still carry 35 kW. Two dead: still enough. Three dead: make a little less electricity. There is no water in the core to lose.

Life

Designed to sit closed for on the order of 100 years. Space tests ran years, not a century. Spare pipes are why one frozen tube is an ordinary fault, not a crisis.

Are the sodium pipes double-walled?

The **oil** that leaves the vault is double-walled. That oil run goes through the yard in a buried chase, so a leak there is a trench cleanup, and the second wall is there to catch it.

The **sodium** pipes are not double-walled in that same way. Each heat pipe is one sealed metal tube. That is the first wall. The **steel can around the whole core** is the second wall. If a heat pipe leaks,

the sodium stays inside that can, in inert gas, and freezes. Sodium melts at **98°C**. A leak is a dead tube, not a sodium fire in the yard, and not sodium in the house pipes.

A second wall on each heat pipe would make the tube worse at dumping heat and is not the baseline. The oil needs its own double wall because it leaves the vault. The sodium is not supposed to.

How long they last, and whether that is safe

The vault is specified to sit closed for on the order of **100 years**. That is the design life of the heat pipes too. Tests of this class of sodium heat pipe have run for years, not a century in a buried house appliance. That gap is real. It is why there are 24 pipes plus spares, and why a dead pipe is designed as an ordinary fault:

- One pipe dead: the other 23 still carry 35 kW.
- Two dead: still enough.
- Three dead: make a little less electricity.
- The core does not uncover. There is no water to lose.

A frozen pipe is the failure mode. The sodium stays in that tube, or freezes in the inert gas inside the steel can. House water never enters the vault, so the classic sodium-and-water reaction is not the yard accident. An oil leak in the chase is a hazardous-fluid cleanup, not an open core.

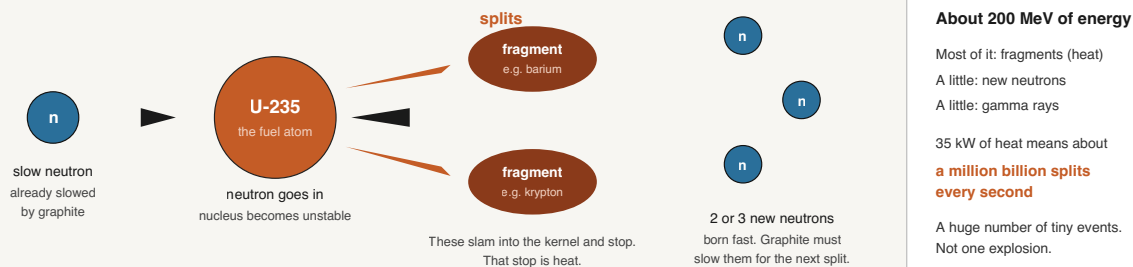
On the truck, sodium is a solid until 98°C. Licensed startup uses a little heat on the cool end, or a few pipes filled with a sodium-potassium mix that is already liquid at room temperature, to get the first heat moving. After that the cycle runs itself.

What a neutron does

Heat comes from **fission**: a slow neutron hits a U-235 nucleus, the nucleus splits into two lighter pieces, a few extra neutrons come out, and about **200 MeV** of energy is released. Most of that energy is the fragments flying apart inside the kernel. They stop in a fraction of a millimeter. That stop is heat.

Figure 13. One split: where the heat comes from

A slow neutron hits a U-235 atom. The atom splits. Heat, two fragments, and a few new neutrons come out. That stop-in-place of the fragments is the furnace.



How that becomes a steady furnace, not a runaway

If, on average, exactly one of those new neutrons causes another split, heat stays even.
If fewer than one, the reaction dies. If more than one, heat rises.

Almost all of the new neutrons come out immediately. A tiny share (about 65 in 10,000) come out a beat later, from the fragments. Those slow extra neutrons are why drums can steer the core.

If you ever added more splitting-power than that tiny share all at once, the reaction could speed up faster than the drums can move. The design keeps leftover splitting-power smaller than that.

A bomb is a different geometry with no heat sink. This core is built not to do that.

The two fragments drawn are one common pair (barium and krypton). Real fuel makes a spread of fragments. They stay inside the coated particle.

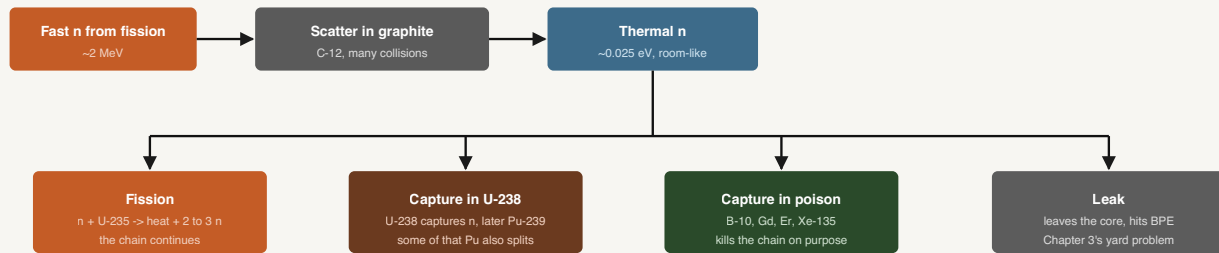
At 35 kW, that is about a million billion splits every second. A huge number of tiny events, not one explosion.

The extra neutrons are born **fast**. U-235 is much more likely to split if the neutron is **slow**, in the thermal range described at the start of this chapter. So the neutron has to bounce off carbon atoms in the graphite many times, losing speed each hit, until it is slow. Then four things can happen:

1. **Split another U-235**. The chain continues. (Later in life, some splits are a different isotope made from U-238.)
2. **Get caught in U-238**. That path slowly makes a little plutonium. Some of that also splits. This is not a breeder. The core is not trying to make more fuel than it uses.
3. **Get caught on purpose** in boron in the drums, in the burnable poison, or in xenon that builds while the core is running.
4. **Leak** out of the core into the shield. Neutrons that leave are the yard problem in Chapter 3. A hydrogen-and-boron layer slows them and eats them. Lead then stops the gamma rays that catching a neutron can create. Lead does almost nothing to neutrons by itself.

Figure 14. What a neutron can do in this core

Born fast (~2 MeV). Graphite knocks it down to thermal (~0.025 eV) in many collisions. Then it either splits U-235, is captured, or leaks into the shield.

**In the yard**

Graphite is the traffic cop. It slows neutrons so they are likely to split U-235 instead of flying out. Boron in the drums and in the shield is the trap: boron-10 eats a neutron and stops that chain. Lead does almost nothing to neutrons. That is why the vault has a hydrogen-and-boron layer, then lead outside it.

Why graphite, not water

A neutron hitting carbon keeps most of its speed, so it takes many hits to slow down. Hydrogen would slow it in fewer hits, which is why the shield uses plastic with hydrogen, not the core. The core needs a solid that can sit hot for a century without boiling. Graphite is that solid. A fast reactor skips this slowing step.

Physicists name the average **k-effective**: if exactly one of those new neutrons causes another split, k-effective is 1, and heat holds steady. Below 1, the reaction dies. Above 1, heat rises.

Almost all of the new neutrons come out immediately. A tiny share, about **65 in 10,000**, come out a beat later from the fragments. Those slow extra neutrons are why drums can steer the core. If you ever added more splitting-power than that tiny share all at once, the reaction could speed up on the immediate neutrons alone, faster than the drums can move. The design rule is: even after the shutdown drums have done their job, leftover splitting-power stays **under eight-tenths of that delayed share**, so a single failure cannot get there.

This is a thermal machine because graphite is there on purpose. A fast machine has no moderator. Neutrons stay fast. Different fuel, different coolant story, different size. Sodium in a heat pipe does not make this core fast. The graphite does the slowing. Sodium only moves heat.

Figure 15. Thermal core vs fast core

Same element (sodium) can show up in both. The spectrum is what differs. This vault is the left column. TerraPower Natrium is the right.

Thermal (this vault)	Fast (Natrium-class SFR)
Moderator: graphite Neutrons slowed to ~0.025 eV before the next split. Fuel: TRISO-UCO in graphite 19.75% U-235. Ceramic particles, not metal pins. Sodium: inside 24 heat pipes A few milliliters per tube. No pool. No pump. Sodium is a heat taxi, not the neutron story. Power: 8 kWe / 35 kWth House furnace scale. One meter. If you remove the graphite Most neutrons stay fast. U-235 fission drops. This lattice is not designed to run that way. Family: Kilopower / eVinci-class heat-pipe microreactor, scaled to a house, not a copy.	Moderator: none Neutrons stay ~0.1 to 1 MeV. That is the design. Fuel: HALEU metal (typical SFR) Pins in a sodium pool. Different can, different spectrum. Sodium: the primary coolant Tons of liquid metal, pumped through the core, then an intermediate loop, then steam or salt. Power: 345 MWe (Natrium) ~40,000 times BGS-8. A plant, with staff. Why go fast Better breeding, some waste-actinide arguments, high power density, no water-pressure vessel. Family: BN-800, CFR-600, PRISM / Natrium. Same sodium word. Different machine.

What happens to the atoms over a century

Almost all of the following stays inside the TRISO coats. The lot never sees it.

At first start: 30 kg of heavy metal. About 6 kg of that is U-235. About 24 kg is U-238. Plus the burnable poison.

Splitting: over 100 years at 35 kW, about **1.35 kg** of U-235-equivalent actually splits. Some of those splits, later in life, are plutonium. The 1.35 kg is “the energy of that much U-235,” not a

claim that only U-235 splits. The fragments include noble gases (xenon, krypton) held in the buffer, iodine, cesium-137, strontium-90, and a spread of others. Xenon-135 is a strong poison while the core is running; it builds and is used up as power holds. These fragments are why used fuel is still “hot” after neutrons stop: leftover heat and gamma rays from the fragments, fading on a known curve.

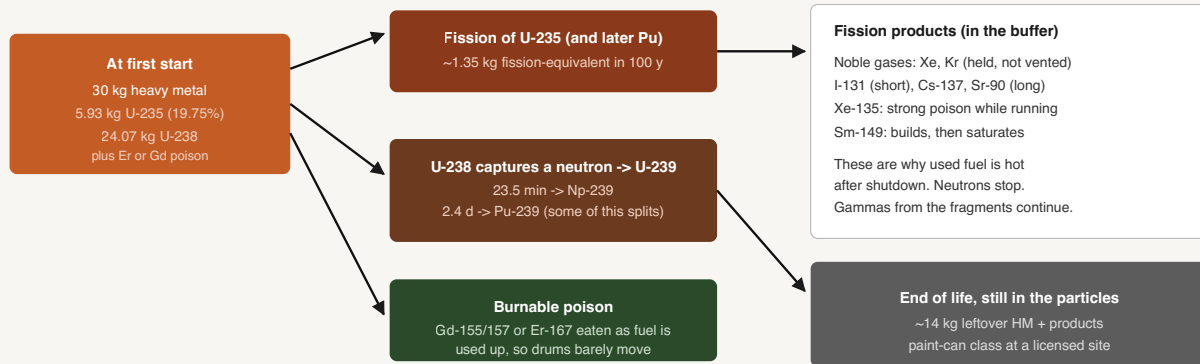
U-238’s side path: U-238 catches a neutron, becomes U-239, becomes neptunium-239 in about 23 minutes, then plutonium-239 in about 2.4 days. Some of that plutonium-239 splits and helps the century.

Poison: the gadolinium or erbium is eaten as the fuel is used, so drums move a few degrees per decade, not weekly.

At end of life: about **16 kg** of heavy metal has been consumed (the engineering target). About **14 kg** remains in the same particles, plus fission products. Several kilograms of material that can still split are still there. End of life is a geometry and poison problem, not an empty tank. What cannot be recycled is still paint-can class, at a licensed site (Chapter 7).

Figure 16. Nuclear chemistry over a century

Almost all of this stays inside the TRISO coats. The paint-can at end of life is the same heavy metal, rearranged.

**What is not happening**

The 30 kg does not vanish. About 16 kg of heavy metal is consumed. That means those atoms have split or changed. About 1.35 kg is the U-235-equivalent that actually split. Several kilograms that can still split remain at year 100. End of life is not an empty tank.

Kernel chemistry

UCO is uranium oxide plus uranium carbide. Oxygen and carbon together keep the kernel from making too much CO, which would over-pressure the particle. That is why UCO, not plain UO₂, is the US coated-particle kernel.

A little radioactivity is also made in the stainless. It rides out with the vault.
None of this chemistry is handled on the lot. Factory in, crane out.

A little radioactivity is also made in the stainless of the can and the heat-pipe walls. It is real and small at this low power density. It rides out with the vault. It is not a yard waste stream.

Why these choices are the safety case

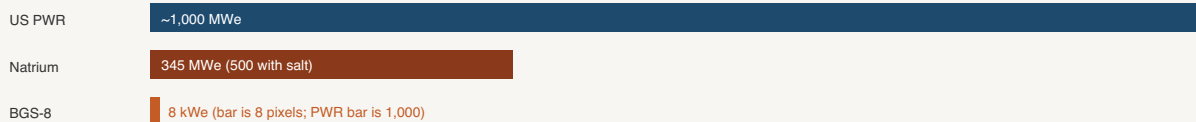
Nobody has stamped this vault “the safest option possible.” What the design does is stack choices that remove the accident paths a water plant and a sodium-pool plant have to manage at plant scale. Each layer is a materials fact or a size fact, not a software mode.

1. Size. 8 kW of electricity is about 1/100,000 the electrical size of a 1,000 MW plant. Leftover heat after shutdown is a space heater, then less. A 10-second jump at 35 kW is **350 kJ**, a tenth of a kilowatt-hour, into a lot of graphite and a 42-ton box. There is no plant-scale inventory of leftover heat to dump into a neighborhood.

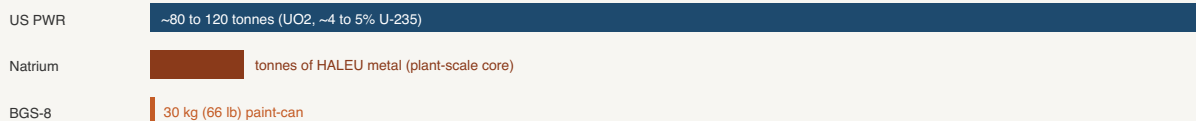
Figure 20. Why size is a safety number

Electrical output and fuel inventory, log-friendly bars. Decay heat and accident energy scale with the machine.

Electricity (log-ish bars, labeled true values)



Heavy metal at first start (order of magnitude)



What that means after shutdown

A US water plant still has megawatts of leftover heat. It needs water, pumps or passive water systems, and a site. This vault: about 0.45 kW after an hour, 0.24 kW after a day. A 42-ton box and the earth can take that without a pump.

Accident energy scales the same way. 35 kW for 10 s is 350 kJ. That is not a neighborhood event. Size does not replace the ceramic coats, drums, or the shield. It does put a ceiling on how bad a bad day can be.

2. Ceramic cans, many of them. Each TRISO particle is its own pressure vessel. Silicon carbide stays a ceramic well above 1,600°C. The compact runs at 650 to 750°C. Gas from splitting stays in the buffer. There is no zirconium tube to balloon, and no open fuel.

3. No water in the core. The core cannot lose its water, because there is none. There is no steam explosion from a flooded hot core, and no zirconium-steam hydrogen. Flood water outside the vault is

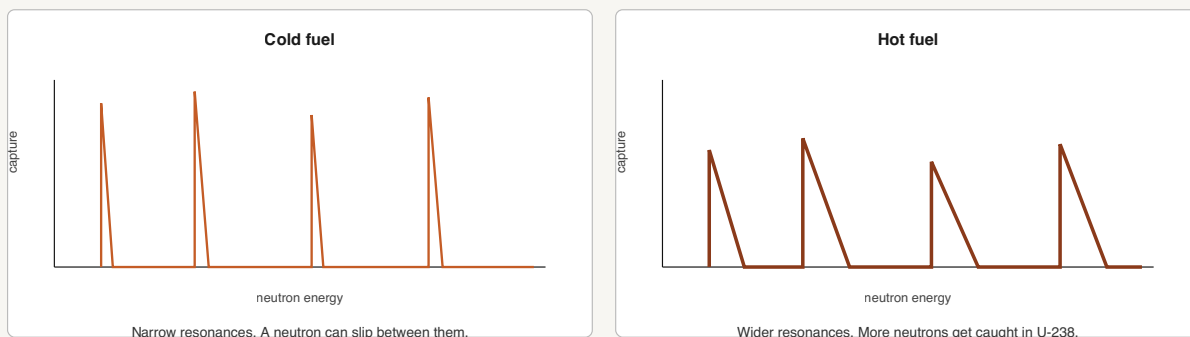
extra shield and extra cooling. If the inner can ever flooded, the engineering requirement is that the reaction dies or stays dead. That is an analysis item, not a slogan.

4. Heat pipes fail in place. A dead pipe is a frozen tube. Spare capacity covers one or two. No pump in the vault is required to cool the core or to take leftover heat after shutdown.

5. Heat itself applies a brake. When the fuel gets hotter, U-238 becomes better at catching neutrons (the energy traps in the nucleus look wider). Fewer neutrons are left to split. Milliseconds. No moving part.

Figure 21. Doppler: why hotter means less splitting

U-238 has narrow energy traps (resonances). Heat jiggles the nuclei. The traps look wider to a moving neutron. More neutrons fall in. Fewer are left to split U-235.



House sentence

If the core tries to run away, heat itself applies the brake, in milliseconds, without a person or a motor. That brake lives in the uranium, not in an app.

Physics sentence

Doppler broadening of U-238 (and U-235) capture resonances. Negative fuel-temperature coefficient. Graphite moderator-temperature coefficient can be slightly positive; net power coef

Graphite, getting hotter, can go slightly the other way in some layouts. The core must be laid out so that from room temperature to 750°C, hotter still means less splitting overall. That is a hard requirement on the computer model of the core, not a hope.

6. Little leftover splitting-power. After the shutdown drums have done their job, leftover splitting-power stays under eight-tenths of the delayed-neutron share, so one failure cannot run away on the immediate neutrons. The burnable poison holds the long century so drums are not chasing a large swing.

7. Drums, then earth. If the oil pump dies, or the engines in the utility room stop, or someone shears the chase, the cool end of the heat pipes gets hotter. A spring or a melt-link, with no software, drives the drums in. Leftover heat soaks into the 42-ton box and the soil. The owner phone is not in the chain.

8. Last-ditch glass. If the vault ever got to about 750°C at a fusible plug, glass and boron carbide pour and set. That layer is not counted on until a furnace test exists. The design is meant never to get there.

What this stack does **not** claim: a 100-year sealed yard life is already proven. It is not. Coated particles, heat pipes, and drums are proven in pieces, at other sizes and other lives. Graphite shrinkage and the ceramic coats at this exact neutron dose still need a materials assessment. The lawn dose has a target; the transport model has not yet signed it. Those are real. They are not the same as a water-plant loss-of-coolant accident at a gigawatt.

The safety case is: the accident energy is small, the fuel is already in ceramic cans, there is no water to lose from the core, sodium cannot drain the core, leftover heat is a household object, and heat itself plus drums plus size are the brakes. That is as far as first principles will take a design that has not yet been built.

How the reaction stops, and when it would not

Four layers, fastest first.

Figure 17. How the reaction is meant to stop

Four layers, fastest first. The owner phone is not on this list. Glass dump is last-ditch and is not credited until it is tested.

1. Doppler milliseconds	Fuel gets hotter: U-238 traps widen, more neutrons get caught, fewer left to split. No moving part. No software. This is the ordinary protection against a sudden rise. Graphite can go slightly the other way in some layouts. The core must be laid out so that, from room temperature to 750 C, hotter still means less splitting overall.
2. Drums seconds	Six B4C drums in the reflector. Springs or gravity drive them IN on overtemp. Analog. Two channels. Loss of oil pump or Stirling set raises condenser T and trips them. One drum stuck out: the others still hold the core below a self-sustaining rise. Leftover splitting-power stays small, so one failure cannot run away on the fast neutrons.
3. Size + earth hours to days	After drums IN, leftover heat is a space heater, then a heat lamp, then less. No pump required. 42 t of vault plus soil take < 1 kW. A 1,000 MW plant cannot say this. A 10-second prompt blip at 35 kW is 350 kJ: a tenth of a kilowatt-hour into a lot of graphite. There is no 3 GWth inventory to dump. That is the size argument, in numbers.
4. Glass dump last ditch	Hopper above the core: 80 kg glass frit + 15 kg boron carbide. Plug melts at about 750 C. Gravity pour. Boron kills reactivity. Glass wets and sets. No power required. Expected use: once in never. Not credited for normal operation until a furnace test exists. The fuel and drums are designed so the vault does not get here.

Ordinary Tuesday. The core is held just barely self-sustaining, with drums on a slow arc. House loads (dryer, air conditioner) are the battery's job. The core is not asked to follow the dryer.

Sudden rise in splitting. Heat itself applies the brake, in milliseconds. The delayed neutrons keep the rise slow enough to steer. Leftover splitting-power is small by requirement.

Loss of the heat path (oil pump off, engines off, excavator shears the chase). The cool end of the pipes gets hotter. A mechanical link releases the drum springs. Leftover heat goes to the box and the earth. An oil leak is a trench or utility-room cleanup, not an open core.

One or two heat pipes dead. Margin in the pipe count. The core stays covered because there is nothing to uncover.

Grid loss. The vault does not care. The house needs the backup switch (the gateway) or it goes dark. That is Chapter 4, not a core event.

Flood. External water: more shield, more cooling. The inner can is supposed to stay dry. If it does not, the reaction must die or stay dead.

One drum stuck with the absorbing face out. The other five still hold the core below a self-sustaining rise.

All drums out, cold, never-used fuel (beyond design). Still below the point where the immediate neutrons could run away, by the leftover-splitting-power rule. That is a requirement on the model of the core, not a measured number in this chapter. Do not invent one.

What would not shut it down by itself: a phone app. A thermostat. A utility outage. Those are not in the chain, on purpose. **What is not a shutdown:** “the dirt will save us.” Dirt is bonus. The vault is the shield. Drums and the heat-brake are the reaction brakes.

Glass dump runs only if the plug reaches about 750°C. Normal vapor in the pipes is about 600°C. Expected use: once in never.

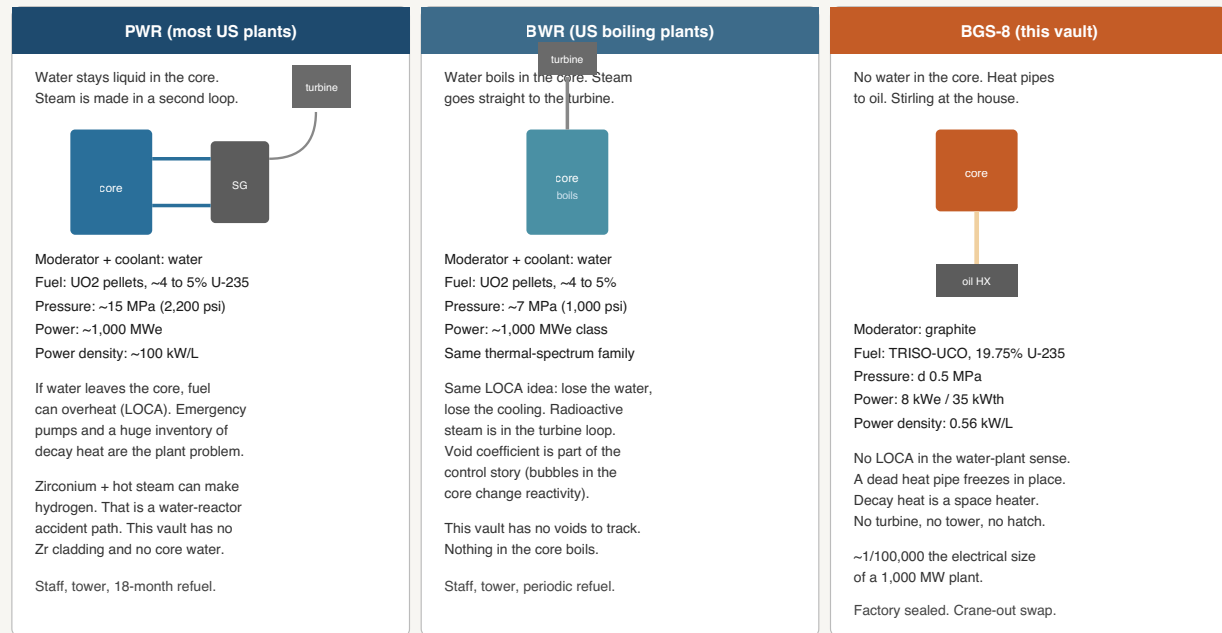
How this is not a US water plant

Almost every commercial reactor in the United States uses water as both the thing that slows neutrons and the thing that carries heat. Two families.

A **pressurized-water reactor** keeps water liquid in the core at about 15 MPa and makes steam in a second loop. A **boiling-water reactor** boils in the core and sends that steam to the turbine. Both use uranium-oxide pellets in zirconium tubes at about 4 to 5% U-235. Both are about 1,000 MW of electricity. Both have a large leftover-heat problem after shutdown. Both have to design against losing the water: if the water leaves, the fuel can overheat. Hot zirconium and steam can make hydrogen. That is a water-reactor path. It is not this vault's path.

Figure 18. US water plants vs this vault

Same physics family (thermal fission). Different fluid, pressure, and size. Not to scale.



	Typical US pressurized-water plant	Typical US boiling-water plant	This vault
Job	Grid plant	Grid plant	One house
Electricity	about 1,000 MW	about 1,000 MW	8 kW
Neutrons	Slowed by water	Water	Slowed by graphite
Heat carrier	Pressurized water	Boiling water	Sodium in sealed pipes, then oil
Fuel	Oxide pellets, about 4 to 5% U-235	Same idea	Coated particles, about 20% U-235
Heavy metal	about 80 to 120 tonnes	similar	30 kg
Core pressure	about 15 MPa	about 7 MPa	under 0.5 MPa
Heat per liter	about 100 kW per liter	high, similar class	0.56 kW per liter
Electricity made by	Steam turbine	Steam turbine	Stirling engines in the utility room
After shutdown	Megawatts of leftover heat	Megawatts of leftover heat	A space heater, then less heat
Opened on site	Refuel floor, about every 18 months	Same idea	Never. Crane-out swap

This vault is furnace-scale fission in ceramic particles. Neutrons are in the same slow range as a US water plant. It does not share the water, the pressure, the pellet-in-tube fuel, or the leftover-heat problem.

How this is not a sodium-pool plant

Sodium shows up in the next plants being built, and that word will get this vault lumped in with them. It should not. The rest of this section is for readers who already know those machines, or who will hear the names.

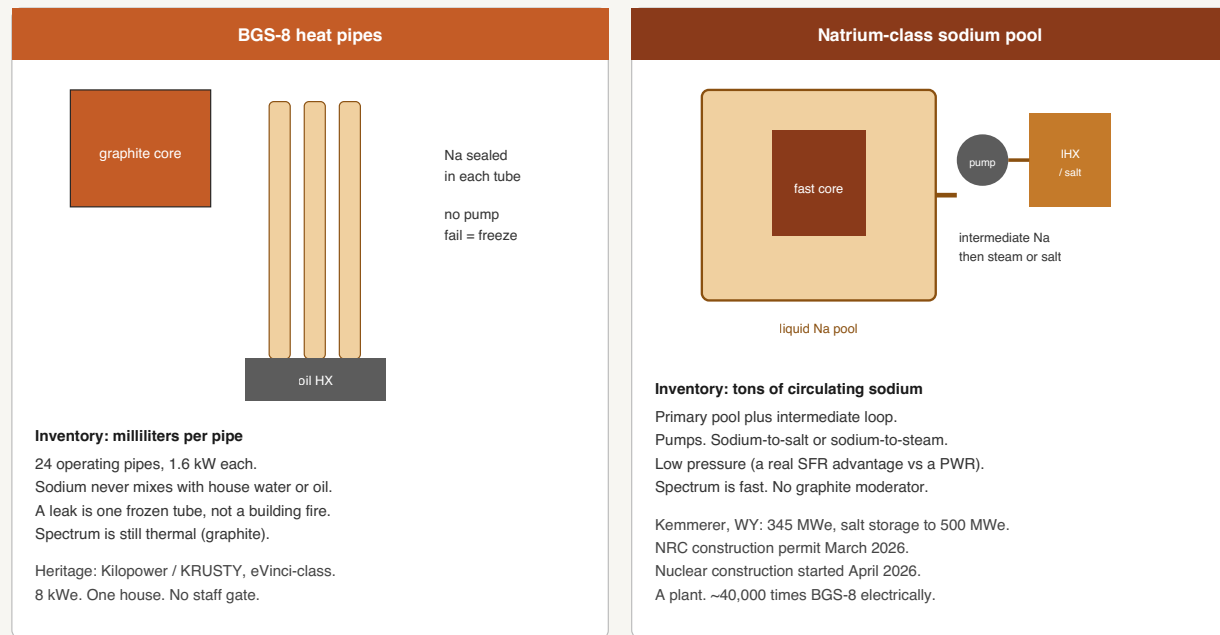
NASA ran a small heat-pipe core on the ground (Kilopower / KRUSTY). Westinghouse is designing a larger heat-pipe machine (eVinci) for mines and towns. Those are the family this vault is built from: sealed sodium tubes, no vault pump, drums, coated-particle or similar high-temperature fuel. This product is that class pointed at one house, not a copy of either machine.

TerraPower Natrium, under construction at Kemmerer, Wyoming (construction permit March 2026, nuclear construction started April 2026), is a different family. It is a **345 MW sodium-cooled fast reactor** with molten-salt storage that can push output to **500 MW** for hours. The fuel is HALEU metal. Neutrons stay fast on purpose. There is no graphite. Sodium is the primary coolant: **tons**, in a **pool**, with pumps, then an intermediate sodium loop, then salt and steam. Staff on the order of hundreds. It is a plant on a retiring coal site. Electrically it is about **40,000 times** this vault.

That design has real advantages at plant scale: low pressure versus a water plant, a large sodium pool as a heat sink, no water in the primary. It also has plant-scale sodium inventory, pumps, intermediate loops, and a fast-spectrum core. Those are solved with plant engineering. They are not the house problem, and this vault does not take them on.

Figure 19. Two uses of sodium

Heat-pipe milliliters vs a pumped pool of tons. Same metal, different job. Natrium (Kemmerer, Wyoming) is the pool.



The same sodium-pool family, elsewhere: Russia's BN-800 (operating), China's CFR-600, India's long-delayed PFBR. All are fast, hundreds of megawatts. Hyperion's old "hot tub" pitch was also a small plant (25 to 30 MW), not this object.

	Natrium (Kemmerer, Wyoming)	This vault
Customer	Grid, coal-site replacement	One house
Electricity	345 MW (500 with salt)	8 kW
Neutrons	Fast (no moderator)	Slowed by graphite
Sodium	Tons, pumped pool plus another loop	Milliliters, sealed heat pipes
Fuel	HALEU metal, plant core	Coated particles, 30 kg
Electricity made by	Steam plus salt storage	Stirling engines in the utility room
Staff	about 250 full time (plant figure)	None on the lot
Approval	NRC construction permit, 2026	Size-based appliance path, does not exist yet

If someone asks whether this is "the Wyoming reactor in a backyard," the answer is no. Same metal in a sentence. Different speed of neutrons, different amount of sodium, different job.

What this chapter covered

The fuel is coated uranium particles in graphite, about 30 kg at first start, about 20% U-235. Neutrons are slowed by graphite, not by water, so they travel in the thermal range. This is not a fast reactor. Sodium lives in 24 sealed tubes. Each tube is one wall; the steel can around the core is the second. The oil that leaves the vault is the loop that is double-walled. A dead heat pipe freezes in place. The pipes are specified for a century; tests so far are years; spares are why that gap is a materials job, not a yard accident.

The chain is held by heat itself, by drums, by little leftover splitting-power, and by size. Chemistry stays in the particles. US water plants and sodium-pool plants are different machines. Chapter 5 is the vault as a box. Chapter 8 is the engineering tables. Chapter 10 is what still has to be shown before a stamp. Chapter 3 is the grass. Chapter 7 is the paint-can at the end. The yard never needs a hatch, and the owner never holds a drum key.

10. The Proofs

This chapter does not prove the machine. It says, for each design claim, what would count as shown, how you would show it, and what number means pass. Where ordinary arithmetic already closes the claim, the work is on the page. Where a transport code, a furnace, a drop pad, or a licensed cell is required, the row stays **open** and names the spec or test.

Chapter 8 is the design basis you size and buy against. Chapter 9 is why the insides are that machine. This file is the proof plan those two chapters were missing. If a number here and a number in Chapter 8 disagree, Chapter 8 wins, and this file should be updated to match.

Three kinds of statement appear below.

Closed. The arithmetic is finished in this book. Rechecking the algebra is enough. No MCNP, no drop test, no k-effective.

Bound. A handbook estimate that sizes the part or shows the claim is in a plausible range. It is not a signed result. The usual mistake is treating a tenth-value-layer stack as a lawn dose.

Open. Needs specialized software, a test article, a listed product, a licensed facility, or a regulation that does not yet have a house-appliance stamp. The row names what to run and what pass looks like.

Nothing in the **Open** column is filled in with a made-up computer output. Chapter 8 already forbids inventing a k-effective. The same rule holds here for dose, drum worth, burnup at year 100, and heat-pipe life.

Figure 22. Three kinds of proof in this chapter

Closed arithmetic is on the page. Handbook bounds size parts. Open rows need codes, tests, or a stamp. Do not mix them.

Closed	Bound	Open
Ordinary algebra from the type-unit numbers. 8 kWe / 0.25 = 32 kWth Specify 35 kWth 24 x 1.6 kW vs 35 kWth 42 t will not float Playset 365 / 8,760 Recheck the algebra. Not a license.	Handbook estimate. Sizes the part. Not a signed result. Tenth-value layers Inverse square (1/r squared) Carnot 47% vs Stirling 25% Way decay-heat fit Fluence order of magnitude Do not treat TVLs as lawn dose.	Needs a code, a test stand, or a regulation. k-effective (MCNP / SCALE) Lawn dose (MCNP) Type B (10 CFR 71) ASME stresses, heat pipes Cold critical, glass furnace Do not invent the output.

House sentence

The book can finish the kitchen-table math. It cannot stamp a core. A later revision turns Open rows green when the runs and tests exist.

Engineering sentence

Chapter 8 remains the design basis. This chapter is the proof plan. If the two disagree, Chapter 8 wins.

How to read a proof item

Each proof item is one claim from the design.

Column	Meaning
Claim	What the book asserts, in one sentence
Status	Closed, Bound, or Open
Method	Arithmetic, handbook, named code, named test, or named rule
Pass	The number or behavior that means the claim is shown
Home	Where the requirement lives in Chapter 8

A **Closed** row can still fail if someone changes 8 kWe, 25% conversion, 35 kWth, 30 kg, or 100 years. Recheck the algebra. Do not treat Closed as a license.

Closed arithmetic

These are the pieces that do not need a neutron-transport code. They use the type-unit numbers: **8.0 kWe**, $\eta = 0.25$, **35 kWth**, **30 kg** heavy metal at first start, **80 GWd/t** target, **100 years**, vault **~42 t**.

Power: why 35 kW of heat

House-side conversion baseline is a free-piston Stirling at **25%** of heat to electricity (hot end about 350 to 600°C, cold end about 40 to 50°C).

$$\begin{aligned} P_{th} &= 8.0 \text{ kWe} / 0.25 = 32.0 \text{ kWth} \\ P_{th,spec} &= 32.0 \times 1.10 = 35.2 \text{ kWth} \rightarrow \text{specify } 35 \text{ kWth} \\ P_{reject} &= 35 - 8 = 27 \text{ kWth} \end{aligned}$$

Pass: 8.0 kWe at the inverter DC bus with 35 kWth in the core and 27 kWth leftover as house heat, greenhouse heat, or dry-cooler reject. If conversion is only 8% (thermoelectric), the same 8 kWe needs 100 kWth. That is a different vault.

Status: Closed for the energy balance. **Open** for a listed Stirling that actually holds 25% at this temperature pair for a decade (that row is later).

Annual electricity

$$8.0 \text{ kW} \times 8,760 \text{ h/y} = 70,080 \text{ kWh/y}$$

$$\text{at 90\% capacity factor: } 8.0 \times 8,760 \times 0.90 = 63,072 \text{ kWh/y}$$

Pass: nameplate energy is 70,080 kWh/y. Design energy at 90% capacity factor is 63,072 kWh/y. Peaks of 10 to 20 kW are the battery, not the core.

Status: Closed. Sell-back and the bill are Chapter 2, not physics.

100-year heat and heavy metal consumed

$$E = 35 \text{ kW} \times 8,760 \text{ h/y} \times 100 \text{ y} = 30,660,000 \text{ kWh} = 30,660 \text{ MWh}$$

$$P = 0.035 \text{ MW}$$

$$E = 0.035 \text{ MW} \times 365.25 \text{ d/y} \times 100 \text{ y} = 1,278.4 \text{ MWd}$$

Heavy metal consumed at discharge burnup B (megawatt-days per tonne):

$$m_{\text{consumed}} (\text{t}) = 1,278.4 / B$$

Assumed discharge burnup	Heavy metal consumed	Notes
50 GWd/t	25.6 kg	Conservative oxide-class
80 GWd/t	16.0 kg	TRISO-UCO target
100 GWd/t	12.8 kg	Aggressive TRISO

Inventory at first start is larger than mass consumed (leftover fissile at year 100, burnable poison, critical geometry):

Item	Value
Heavy metal at first start	30 kg (66 lb)
Consumed at 80 GWd/t, end of life	16 kg (35 lb)
Residual heavy metal at end of life	about 14 kg in the same compacts

Pass: 16 kg consumed at the 80 GWd/t target is algebra from 1,278.4 MWd. **30 kg at first start is a specification**, not a depletion result. Whether 30 kg with poisons and drums still holds k-effective at year 100 is **Open** (SCALE/ORIGEN).

Status: Closed for energy and the consumed-mass table. **Open** for end-of-life criticality.

Fission rate and U-235-equivalent consumed

Energy per fission: 200 MeV = $200 \times 1.602 \times 10^{-13} \text{ J} = 3.204 \times 10^{-11} \text{ J}$.

$$R_f = 35,000 \text{ J/s} \quad / \quad 3.204 \times 10^{-11} \text{ J} \quad = \quad 1.092 \times 10^{15} \text{ fissions/s}$$

Per year (365.25 days = $3.15576 \times 10^7 \text{ s}$):

$$\begin{aligned} N_y &= 1.092 \times 10^{15} \times 3.15576 \times 10^7 = 3.447 \times 10^{22} \text{ fissions/y} \\ m_{\text{U235-eq}} &= 3.447 \times 10^{22} / 6.022 \times 10^{23} \times 235 \text{ g} = 13.45 \text{ g/y} \\ 100 \text{ y} &\rightarrow 1.35 \text{ kg U-235-equivalent fissioned} \end{aligned}$$

Some fissions at mid-life are Pu-239. The 1.35 kg is fission-equivalent, not a claim that only U-235 fissions.

Initial U-235 in 30 kg of 19.75 wt% U:

$$0.1975 \times 30 \text{ kg} = 5.93 \text{ kg U-235}$$

Pass: 1.35 kg fission-equivalent over 100 years at 35 kWth. Several kilograms of fissile remain. End of life is not an empty tank.

Status: Closed for the fission count. **Open** for the mix of U-235 vs Pu-239 vs leftover (ORIGEN).

Decay heat after shutdown

Way / ANS-style fit for sizing, infinite prior operation: $P/P_0 \approx 0.066 t^{-0.2}$ with t in seconds.

Time after shutdown	t (s)	$t^{-0.2}$	P/P_0	P (kW)	Household analog
1 s	1	1.000	0.066	2.31	Large space heater
1 h	3,600	0.194	0.0128	0.45	Small space heater
1 d	86,400	0.103	0.0068	0.24	Heat lamp
7 d	6.05×10^5	0.0697	0.0046	0.16	Night-light class, thermal

Lift waits until decay heat is under about 0.4 kW (about a day).

DB-10: no vault-side pump for leftover heat. Vault mass (~42 t) and earth take less than 1 kW.

Crude earth conduction, vault as a sphere $r = 1.4$ m, soil $k = 1.0$ W/m·K, $Q = 0.5$ kW:

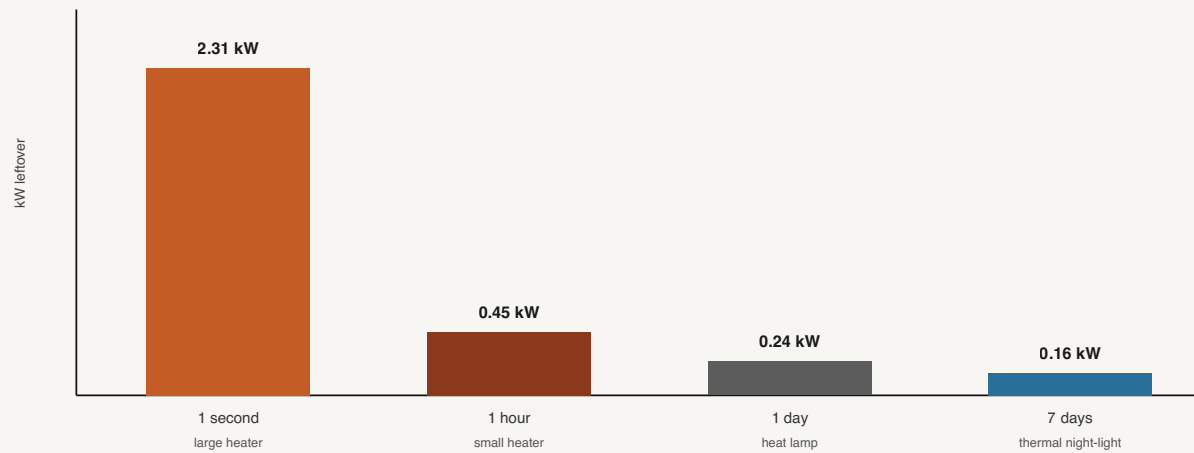
$$\Delta T = Q / (4 \pi k r) = 500 / (4 \times 3.1416 \times 1.0 \times 1.4) = 28 \text{ K}$$

Soil far-field plus about 28°C at the wall after a few hours.

Acceptable for decay heat. **Do not use this formula for operating 35 kW.** If 35 kW were dumped to soil: $\Delta T \approx 1,990$ K. Operating heat leaves through heat pipes, then oil, then a dry cooler or a greenhouse.

Figure 23. Leftover heat after shutdown (Way fit, 35 kWth)

Sizing curve only. Sign-off uses ANSI/ANS-5.1 from a real isotope inventory. $P/P0 = 0.066 t^{-(0.2)}$, t in seconds.



DB-10: no vault pump. 42 t and earth take less than 1 kW. Lift waits until about 0.4 kW (about a day).

Do not dump operating 35 kW into soil. That path is about 2,000 K on the same crude conduction formula.

Pass (sizing): leftover heat is heater-class in an hour and lamp-class in a day. **Pass (sign-off):** a decay-heat curve from a depletion inventory (ANSI/ANS-5.1 or equivalent, built from ORIGEN isotopes), not this fit alone.

Status: Bound for the table (good enough to size the box). **Open** for the isotope-true curve.

Pipe count versus 35 kW

Count	Capacity at 1.6 kW each	Versus 35 kWth
24 operating	38.4 kW	3.4 kW margin
23 (one dead)	36.8 kW	still above 35
22 (two dead)	35.2 kW	still above 35
21 (three dead)	33.6 kW	derate electrical output

Four spare channels are plugged. Rating 1.6 kW each at about 600°C vapor is a **vendor rating to be shown**, not a number this chapter can certify.

Pass (count): $24 \times 1.6 \text{ kW} \geq 35 \text{ kW}$ th with two pipes out. **Pass (hardware):** a heat-pipe test that holds 1.6 kW at the design temperatures, plus a failed-pipe test on a representative block.

Status: Closed for the count. **Open** for the 1.6 kW rating and 100-year life.

Oil flow at 35 kW

Dowtherm A class, $\Delta T = 40 \text{ K}$, $c_p \approx 2.2 \text{ kJ/kg}\cdot\text{K}$:

$$m = 35 / (2.2 \times 40) = 0.40 \text{ kg/s} \approx 6 \text{ to } 7 \text{ gpm}$$

Pass: process flow about 0.40 kg/s in 25 mm inner pipe plus 40 mm containment, two pipes (hot and return) in the chase. Pump in the utility room, not in the vault.

Status: Closed for the flow. **Open** for listed pipe, leak detection, and fire (NFPA row later).

Vault mass versus float

Type-unit mass ~42,100 kg (46.4 short tons). Footprint $\approx \pi \times 1.23^2 = 4.75 \text{ m}^2$.

$$q = 42,100 \times 9.81 / 4.75 = 87 \text{ kPa} \approx 1.8 \text{ ksf}$$

Most residential soils are 1.5 to 3 ksf allowable. Spec a 200 mm crushed-stone pad.

Displaced volume $\approx \pi \times 1.23^2 \times 2.60 = 12.3 \text{ m}^3$. Full flood:

$$12.3 \text{ m}^3 \times 1,000 \text{ kg/m}^3 = 12.3 \text{ t} \ll 42 \text{ t}$$

Pass: vault does not float. No buoyancy straps. Still grout the pad so it does not walk in a scour hole. Lead and concrete takeoffs on the shop drawing must close to this mass.

Status: Closed for buoyancy with the type-unit takeoff. **Open** for the civil drawing and soil report.

Playset arithmetic

Live-on-grass target is **$\leq 5 \text{ mrem/year}$** if someone sat there 8,760 hours. Playset at 1 hour a day:

$$5 \text{ mrem/y} \times (365 \text{ h} / 8,760 \text{ h}) = 0.21 \text{ mrem/y}$$

A coast-to-coast flight is about 3 to 5 mrem. A playset year is about 1/15 to 1/25 of one flight.

Hourly rate that yields 5 mrem in 8,760 h:

$$5 \text{ mrem} / 8,760 \text{ h} = 0.00057 \text{ mrem/h} = 0.57 \text{ } \mu\text{rem/h}$$

Pass (arithmetic): 0.21 mrem/y at 1 h/d follows from 5 mrem/y.

Pass (dose): MCNP shows $\leq 0.57 \text{ } \mu\text{rem/h}$ at the grass, dry soil, lid buried and lid bare.

Status: Closed for the 365/8,760 ratio. **Open** for the 5 mrem/y itself.

Core volume, power density, order-of-magnitude fluence

Active core: 400 mm diameter x 500 mm height
 Volume: $\pi \times 0.20^2 \times 0.50 = 0.0628 \text{ m}^3 = 62.8 \text{ L}$
 Power density: $35 \text{ kW} / 62.8 \text{ L} = 0.56 \text{ kW/L}$

A typical US water-plant core is about 100 kW/L. This core is about **200 times** less dense.

HM loading: $30 \text{ kg} / 62.8 \text{ L} = 0.48 \text{ g/cm}^3$

TRISO compacts routinely hold 0.5 to 1.0 g/cm³ heavy metal. 0.48 is not cramped.

Fission density:

$$1.092 \times 10^{15} \text{ /s} \quad / \quad 6.28 \times 10^4 \text{ cm}^3 \quad = \quad 1.74 \times 10^{10} \text{ fissions/cm}^3 \cdot \text{s}$$

If the macroscopic fission cross section Σ_f is about 0.08 to 0.15 cm⁻¹ (from the lattice, not from this page):

$$\begin{aligned} \phi &\sim (1.74 \times 10^{10}) / \Sigma_f \approx 1 \times 10^{11} \text{ to } 2 \times 10^{11} \text{ n/cm}^2 \cdot \text{s} \\ t_{100y} &= 3.15576 \times 10^9 \text{ s} \\ \Phi &\sim 3 \times 10^{20} \text{ to } 7 \times 10^{20} \text{ n/cm}^2 \end{aligned}$$

That is research-reactor cladding fluence, not a 40-year water-plant barrel. Stainless, graphite, and TRISO silicon carbide are in a plausible range. **Displacements per atom and graphite shrinkage still need a materials memo.**

Pass (geometry): 0.56 kW/L and 0.48 g/cm³ are algebra. **Pass (materials):** a fluence and dpa assessment against AGR/HTGR and graphite data at this flux and 100 years.

Status: Closed for volume and power density. **Bound** for fluence (Σ_f is assumed). **Open** for dpa and shrinkage.

Source terms for shielding (running, 35 kWth)

Neutrons: $\sim 2.4 \text{ n/fission} \times 1.092 \times 10^{15} = 2.62 \times 10^{15} \text{ n/s}$
 Gamma energy (prompt $\sim 7 \text{ MeV/fission}$):
 $7 \times 10^6 \text{ eV} \times 1.602 \times 10^{-19} \text{ J/eV} \times 1.092 \times 10^{15} = 1.22 \text{ kW gamma}$

Unshielded neutron flux at 2 m, isotropic point source, no shield:

$$\Phi = S / (4 \pi r^2) = 2.62 \times 10^{15} / (4 \pi \times 4 \times 10^4 \text{ cm}^2) = 5.2 \times 10^9 \text{ n/cm}^2 \cdot \text{s}$$

That is immediately unsurvivable. The shield is not optional. Capture gammas in hydrogen and iron are extra, and are why lead sits **outside** the hydrogenous layer.

Status: Closed for the source magnitude. **Open** for dose after the stack (MCNP).

Stored energy in a short jump

$$35 \text{ kW} \times 10 \text{ s} = 350 \text{ kJ} \quad (\text{one-tenth of a kWh})$$

A prompt jump that self-limits in milliseconds dumps a small energy into a large graphite and steel mass. There is no 3 GWth inventory.

Status: Closed for 350 kJ. **Open** for the actual prompt-energy deposition from a reactivity insertion (that needs the lattice and delayed-neutron fraction).

Handbook bounds that are not a substitute for transport

These estimates size the shield and the engine. They do not replace MCNP, and they do not replace a measured Stirling efficiency.

Tenth-value layers

A tenth-value layer (TVL) is the thickness that cuts a given radiation to one-tenth, for a stated spectrum. Approximate, fission-spectrum neutrons / ~1 to 2 MeV gamma:

Material	TVL neutrons (fission)	TVL gamma (~1.5 MeV)
Borated polyethylene	~12 to 15 cm	poor (use lead)
Concrete	~20 to 28 cm	~20 to 24 cm
Lead	poor	~4.0 to 4.5 cm
Dry soil	worse than wet by ~1.5 to 2× for neutrons	~25 to 35 cm
Wet soil	closer to concrete for neutrons	~20 to 30 cm

Neutron path (radial, type-unit stack): 25 cm BPE (~1.8 TVL) + 15 cm borated concrete (~0.6 TVL) + 40 cm concrete (~1.6 TVL) $\approx$ **4 TVL $\approx 10^4$** before soil.

Gamma path: 8 cm lead $\approx$ 1.8 TVL ($\times 63$) plus 40 cm concrete $\approx$ 1.8 TVL ($\times 63$) $\approx$ **$\times 4,000$** plus soil.

Unshielded flux at 2 m was 5.2×10^9 n/cm²·s. Four TVL of neutrons is a factor of 10,000: still on the order of 10^5 n/cm²·s if you treated the core as a point and ignored everything else. Live-on-grass needs something more like 10^8 to 10^9 of total attenuation, plus a gamma story, plus no streaming hole. **4 TVL is not enough alone.** The reflector, boron capture in the polyethylene, self-shielding in a small core, and $1/r^2$ as you move out through thick layers have to be in a transport model.

If the model misses DB-5 or DB-6, add BPE or lead. Do not add “more dirt” as the fix. Dirt is bonus. Dose is designed with the lid bare.

Inverse square ($1/r^2$)

For a point source in empty space, flux falls as $1/r^2$. The core is not a point, and the yard is not empty space. Inverse square is why a beer-keg source is easier to bury than a plant core, and why a 2 m unshielded number is already a crisis. It is **not** why the grass is 5 mrem/y. Most of the drop is absorption and scatter in BPE, concrete, and lead.

Skyshine with the lid bare is a scatter path that inverse square does not capture. That is an MCNP tally.

Figure 24. Inverse square and TVLs do not replace MCNP

Unshielded 2 m flux is unsurvivable. Four neutron TVLs are about 10,000. Live-on-grass needs more, plus no streaming hole. Dirt is bonus.

1. Inverse square

Point source, empty space.
Flux falls as $1/r^2$.

$S = 2.62 \times 10^{15}$ n/s
at 2 m: 5.2×10^9 n/cm².s

Immediately unsurvivable.

Why a beer-keg source is easier than a plant core.

Not why the grass is 5 mrem/y.
Skyshine is not $1/r^2$.

2. Tenth-value layers

25 cm BPE ~ 1.8 TVL
15 cm borated conc. ~ 0.6
40 cm concrete ~ 1.6

 ~ 4 TVL neutrons $\sim 10,000$
Lead + concrete $\sim 4,000$ gamma

Not enough alone for 10^8 to 10^9 .
If MCNP misses, add BPE or lead.

Do not add more dirt as the fix.
Dose is designed lid-bare.

3. MCNP (still Open)

Real geometry: stack, chase, labyrinth, skyshine, lid bare.

Running and shutdown.
Dry sand 5 wt% water, wet 15%.

Pass: ≤ 0.57 urem/h grass
= 5 mrem/y live-on-grass
Chase: ≤ 50 urem/h

Until that run exists, 5 mrem/y is a requirement, not a result.

House sentence

The vault is the shield. Ten feet of soil is the yard you see. A playset hour a day is then about 0.21 mrem/y, if the 5 mrem/y target holds.

Physics sentence

Most of the drop is absorption and scatter, not distance. Capture gammas are why lead sits outside the hydrogenous layer. Streaming holes are why penetrations go through the floor with two 90-degree bends. 10 CFR 20.1301 is 100 mrem/y to the public; this design asks 5 mrem/y extra.

Pass (handbook): the stack is a geometry that can plausibly meet DB-5 and DB-6 once transport is run. **Pass (sign-off):** MCNP6 or equivalent, running and shutdown, dry sand (5 wt% water) and wet (15 wt%), lid buried and lid bare, chase streaming, skyshine with lid bare, contact on bare shell for a 1-hour worker exposure under 100 mrem.

Status: Bound for TVL and $1/r^2$. **Open** for every dose number on the lawn.

Carnot versus 25% Stirling

A perfect heat engine between two temperatures cannot beat Carnot:

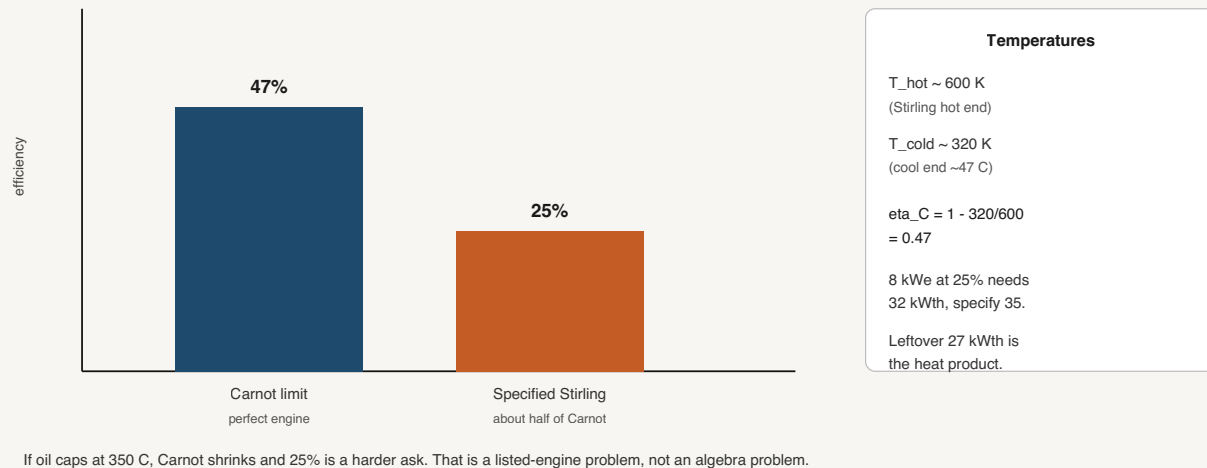
$$\begin{aligned} T_{\text{hot}} &\approx 600 \text{ K} && (\text{Stirling hot end, order of } 327^\circ\text{C}) \\ T_{\text{cold}} &\approx 320 \text{ K} && (\text{about } 47^\circ\text{C cool end}) \\ \eta_{\text{Carnot}} &= 1 - 320/600 = 0.47 && (47\%) \end{aligned}$$

A real Stirling at **25%** is about **half of that ceiling**. That is a normal real engine, not a calculation error. Heat and electricity are both products. The 27 kWth leftover is not a defect.

If the oil loop caps at 350°C , the hot-end Kelvin number is lower and Carnot shrinks. Then 25% is a harder ask, or the convertor runs a bit less efficiently and you size five 2 kWe units with margin (10 kWe installed for 8 kWe net).

Figure 25. Carnot ceiling versus a real Stirling

Perfect engine between 600 K hot and 320 K cold: 47%. Specified Stirling: 25%, about half that ceiling. Heat and electricity are both products.



Pass (bound): 25% is under 47% at this temperature pair. **Pass (hardware):** a listed convertor at the oil temperature the chase actually delivers, five units, 8 kWe net, 10 to 15 year swap in the utility room.

Status: Bound for Carnot. **Open** for the measured 25%.

From “hotter means less splitting” to the Doppler requirement

No coefficient is invented here. The chain is the requirement the lattice model must meet.

House sentence. If the core tries to run away, heat itself applies a brake in milliseconds, without a person or a motor. That brake lives in the uranium.

Step 1. A fission dumps most of its energy as fragments stopping inside the kernel. Kernel temperature rises first. Graphite and the heat pipes lag.

Step 2. U-238 (and U-235) have narrow energy traps, **resonances**, where a neutron is very likely to be captured instead of going on to split. Between the traps, a neutron can slip through.

Step 3. Heat is motion of the nuclei. To a neutron, a jiggling trap looks wider. That widening is **Doppler broadening**. More neutrons fall into U-238 capture. Fewer are left to split U-235.

Step 4. So as fuel temperature rises, k-effective falls. The fuel-temperature coefficient is negative. That is the millisecond brake.

Step 5. Graphite, getting hotter, can go slightly the other way in some layouts (spectrum shifts, less density, a few more neutrons in the wrong band). That is a **moderator-temperature** coefficient, and it can be slightly positive.

Step 6. The design rule is not “Doppler exists.” The design rule is: **from 20°C to 750°C, the net power coefficient is negative.** Hotter still means less splitting overall, after fuel and graphite have both moved. That is a hard SCALE/MCNP criterion on the real lattice (packing fraction, carbon-to-uranium, enrichment, poisons, drums).

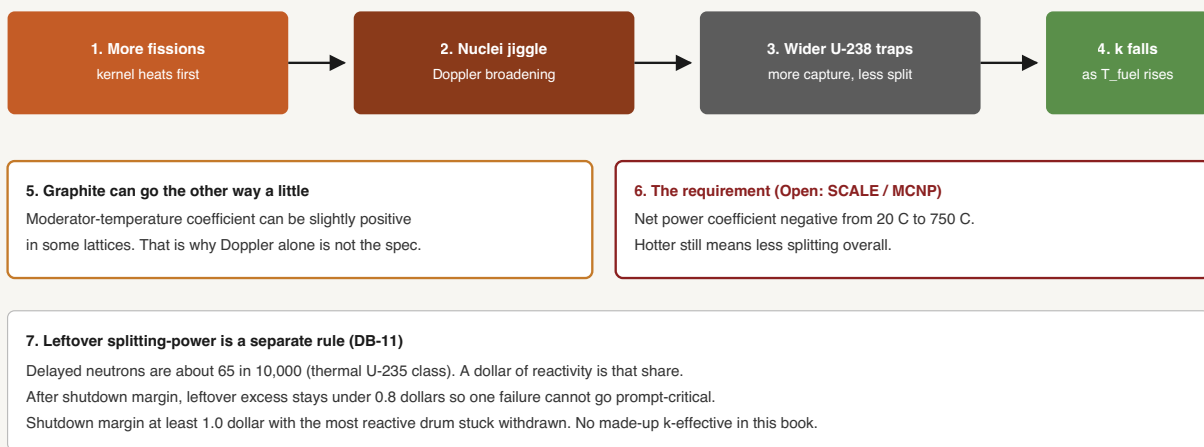
Step 7. Doppler does not get a free pass on leftover splitting-power. Delayed neutrons are about **65 in 10,000** of the new neutrons (thermal U-235 class). If you add more splitting-power than that tiny share all at once, the chain can speed up on the immediate neutrons alone. **DB-11:** after shutdown margin, leftover excess stays

under 0.8 dollars of reactivity so a single failure cannot make the core prompt-critical. Shutdown margin at least **1.0 dollar** with the most reactive drum stuck withdrawn.

A dollar of reactivity is that delayed-neutron share. This file does not print a made-up k-effective, a made-up pcm per degree, or a made-up drum worth.

Figure 26. From hotter fuel to the Doppler requirement

Logic is Closed. The coefficient is Open. Do not invent pcm per degree or a k-effective.



House sentence

If the core tries to run away, heat itself applies the brake in milliseconds, without a person or a motor. That brake lives in the uranium, not in an app.

What would count as shown

A lattice run with packing fraction, C/U, enrichment, poisons, and drums: negative net power coefficient, plus the Chapter 8 section 3.4 k-effective table.

This drawing is the chain. It is not the coefficient.

Pass: the lattice model shows a negative net power coefficient 20°C to 750°C, and the §3.4 k-effective table, including DB-11. **Not a pass:** this paragraph.

Status: Open (MCNP/SCALE). The chain is Closed as logic. The number is not.

Proof items

Status is **Closed**, **Bound**, or **Open**. Pass is what a reviewer should see.

Power, inventory, size

Claim	Status	Method	Pass	Home
8.0 kWe needs 35 kWth at 25% conversion with 10% margin	Closed	Algebra	$8/0.25 = 32$; $\times 1.10 \rightarrow$ specify 35; reject 27 kWth	DB-1, DB-2, §2.1
70,080 kWh/y nameplate; 63,072 kWh/y at 90% CF	Closed	Algebra	$8 \times 8,760$; $\times 0.90$	§2.2
1,278 MWd in 100 years at 35 kWth	Closed	Algebra	$0.035 \text{ MW} \times 365.25 \times 100$	§2.3
16 kg heavy metal consumed at 80 GWd/t	Closed	Algebra	$1,278.4 / 80,000 \text{ t}$	§2.3
30 kg at first start is enough leftover fissile at year 100	Open	SCALE/ORIGEN + lattice	$k\text{-effective(EOL)} \geq 1$ with drums; peak compact $\leq 80 \text{ GWd/t}$	§2.3, §3.4, §16
1.35 kg U-235-equivalent fissioned in 100 years	Closed	200 MeV/fission	$1.092 \times 10^{15} \text{ fissions/s} \rightarrow 13.45 \text{ g/y} \times 100$	§2.4
5.93 kg U-235 at first start (19.75% of 30 kg)	Closed	Algebra	0.1975×30	§2.4
Enrichment stays below 20% (HALEU, not HEU)	Closed as spec	Assay at fabrication	$19.75 \pm 0.20 \text{ wt\% U-235}$; no lot above 20%	§3.2
Decay heat is heater-class, not plant-class	Bound	Way fit; then ANS-5.1	$\sim 2.3 \text{ kW}$ at 1 s; $\sim 0.45 \text{ kW}$ at 1 h; $\sim 0.24 \text{ kW}$ at 1 d	§2.5, DB-10

Claim	Status	Method	Pass	Home
350 kJ in a 10 s jump at 35 kW	Closed	Algebra	35 kW × 10 s	§7.4
Core 62.8 L, 0.56 kW/L, ~200× less dense than a PWR	Closed	Algebra	$\pi \times 0.20^2 \times 0.50$; 35/62.8	§3.3
Fluence $\sim 3 \times 10^{20}$ to 7×10^{20} n/cm ² in 100 years	Bound	Fission density / assumed Σ_f	Order of research-reactor cladding, not a PWR barrel	§3.5
Graphite shrinkage and dpa are acceptable at 100 years	Open	Materials memo vs AGR/HTGR and graphite data	Allowables on dimensional change, strength, conductivity	§3.5, §16, §17

Neutronics (do not invent k-effective)

Claim	Status	Method	Pass	Home
Factory / truck: subcritical with drums in	Open	MCNP6 or Serpent + SCALE/KENO	$k\text{-effective} \leq 0.95$ including 3σ and tolerances	§3.4, DB-9
Hot, xenon equilibrium, beginning of life	Open	Same	$1.000 \leq k\text{-effective} \leq 1.005$, drums not on a stop	§3.4
End of life, poisons depleted	Open	Depletion + lattice	$k\text{-effective} \geq 1.000$ with drums at max remaining worth	§3.4
Most reactive drum stuck out, others in	Open	Lattice	$k\text{-effective} \leq 0.99$	§3.4
Shutdown margin ≥ 1.0 dollar, most reactive drum stuck withdrawn	Open	Lattice, worth in dollars	≥ 1.0 dollar	§3.4
Leftover excess after shutdown margin under 0.8 dollars	Open	Lattice	Under 0.8 dollars so one failure cannot go prompt	DB-11
All drums out, cold, clean (beyond design) still below prompt-critical	Open	Lattice	Below prompt-critical; DB-11	§3.4
Flooded inner vessel: k-effective decreases or stays subcritical	Open	Lattice, void and flooded	Analyze both ways; must not go prompt	§3.4, DB-8

Claim	Status	Method	Pass	Home
Net power coefficient negative, 20°C to 750°C	Open	Temperature-dependent lattice	$dk/dP < 0$ over that range (Doppler plus graphite)	§7.1
Six B ₄ C drums, mechanical locks for transport	Open as hardware	Factory test + I&C	Locks pin absorbing face in; owner key does not exist	§3.4, §10
Burnable poison holds the century (drums move a few degrees per decade)	Open	Depletion	Drum motion slow; no weekly chase	§3.2, §3.4
Exact heat-pipe and compact lattice	Open	MCNP/SCALE geometry	Meets the table above at 24 pipes, 80 to 120 compact channels	§3.3

Shielding and dose

Claim	Status	Method	Pass	Home
Unshielded neutron source $\sim 2.62 \times 10^{15}$ n/s	Closed	$2.4 \text{ n/fission} \times \text{fission rate}$	Order 10^{15} n/s	§6.1
Unshielded flux at 2 m $\sim 5 \times 10^9 \text{ n/cm}^2 \cdot \text{s}$	Closed	$1/r^2$ point source	Shield is mandatory	§6.1
Radial stack ~ 4 TVL neutrons, $\sim 4,000$ gamma before soil	Bound	Handbook TVL	Geometry for MCNP, not a lawn number	§6.4
Extra dose at grass $\leq 5 \text{ mrem/y}$, dry soil, running, 8,760 h	Open	MCNP6	$\leq 0.57 \text{ } \mu\text{rem/h}$; DB-5	DB-5, §6.2
Same 5 mrem/y with lid bare (scour)	Open	MCNP6, skyshine	DB-6	DB-6
Dry sand 5 wt% water still meets DB-5	Open	MCNP6	Dry and wet (15 wt%) cases	§6.4, §11
Playset 1 h/d $\sim 0.21 \text{ mrem/y}$	Closed	$5 \times 365/8,760$	Follows if DB-5 holds	§6.5
Chase after two bends $\leq 50 \text{ } \mu\text{rem/h}$ contact	Open	MCNP chase model	$\leq 50 \text{ } \mu\text{rem/h}$ house-end pipe	§9
Bare-shell 1-hour unplanned exposure $< 100 \text{ mrem}$	Open	MCNP tally on outer shell	Worker number, not a playset number	§11
Public dose also under 10 CFR 20.1301	Open	10 CFR 20 + MCNP	100 mrem/y TEDE is the federal cap; design is 5 mrem/y extra at the grass	DB-5; 10 CFR 20.1301

Heat transport

Claim	Status	Method	Pass	Home
$24 \times 1.6 \text{ kW} = 38.4 \text{ kW} > 35 \text{ kWth}$	Closed	Algebra	Count with two pipes out still ≥ 35	§4.1
Each pipe actually carries 1.6 kW at $\sim 600^\circ\text{C}$ vapor	Open	Transient heat-pipe model + test	1.6 kW rating; startup from freeze; one failed-pipe test	§4.1, §16
100-year sealed heat-pipe life	Open	Life test / qualified extrapolation	Design life 100 y; fail-in-place + N+2 is the operational net	§4.1, §17
Sodium stays in the tube (single envelope; inner vessel is second barrier)	Bound as design	Helium leak on vessel; pipe QA	He leak $\leq 1 \times 10^{-8}$ mbar·L/s on inner vessel; pipe fail freezes in place	§4.1
Oil 0.40 kg/s, 350°C , double-wall, leak-detected	Closed for flow; Open for listing	Algebra + NFPA / local fire	0.40 kg/s; annulus detected; trip on leak	§4.3
No vault pump required for core cooling or decay heat	Bound	Heat pipes + 42 t + earth $\Delta T \sim 28 \text{ K}$ at 0.5 kW	DB-10; do not dump 35 kW to soil	DB-10, §2.5
Oil pump off / Stirling off trips drums	Open	Analog two-channel test	Condenser T high $\rightarrow$ springs in, no software, no phone	§4.4, §7.2, §10.2
Startup from cold sodium (melts 98°C)	Open	Trace heat or NaK subset test	Licensed startup only; then lock out heater	§4.1, §10

Mechanical, civil, transport package

Claim	Status	Method	Pass	Home
Vault ~42 t, 2.46 m OD × 2.60 m H	Bound	Takeoff in §8.1	Shop drawing closes mass and geometry	§8.1
Bearing ~1.8 ksf on 4.75 m ²	Closed as arithmetic	42,100 kg × 9.81 / 4.75 m ²	Pad 200 mm stone or thin mat if geotech says no	§8.2
Will not float in full flood (12 t buoyant vs 42 t)	Closed	Displaced 12.3 m ³	No buoyancy straps	§8.2, DB-8
Inner vessel ≤0.5 MPa, SS316L, 12 mm	Open	ASME BPVC VIII or III (Div. 5 if high-T nuclear)	Stress < allowables; He leak test	§3.3, §16
Pad, lift lugs, seismic box	Open	ASME + ASCE 7 / ASCE 43, IBC site class D, 0.3 g	Stress < allowables; 4 trunnions, 2 × 25 t WLL redundant	§8.3, §11
Fueled vault is a Type B package	Open	10 CFR 71 (71.51, 71.73)	Drop, puncture, 800°C / 30 min fire, immersion; k-effective stays down	§8.3, §11, §16
Outer shell IP68 / submarine weld	Open	Shop hydro / helium	Indefinite immersion, inner boundary leak-tight	§11, DB-8
3.0 m (10 ft) soil above lid	Closed as spec	Civil	Cover is the yard; dose still lid-bare	§8.4, DB-6
Penetrations floor only, two 90° bends	Closed as spec; Open for streaming	Drawings + MCNP	No lid penetration; chase ≤50 µrem/h	DB-7, §9

Conversion, house, fire

Claim	Status	Method	Pass	Home
5 × 2.0 kWe Stirling, 4 of 5 for 8 kWe	Open	UL listing of convertor class; factory test	8 kWe net at DC bus; 10 to 15 y swap in utility room	\$5, DB-1
Inverter / gateway	Open	UL 1741, IEEE 1547	Listing; islanding needs the gateway	\$5, \$13
Battery 27 to 40 kWh, 10 to 15 kW peak	Open	Listed storage	Peaks are the battery	\$5, \$13
27 kW dry cooler	Open	HVAC coil rating	Summer reject; do not dump 35 kW to dirt	\$4.4, DB-4
Oil fire / leak in yard or utility room	Open	NFPA 30 (flammable liquids) / local fire; double-wall	Detected annulus; spill is hazmat, not a core event	\$4.3, \$16
Hydronic 80 to 120°F isolated from potable	Open	Plumbing code + listed plate HX	27 kWth house product	DB-4, \$13
Convertor is not in the vault	Closed as design	Architecture	100-year Stirling inside the vault is not claimed	\$5, \$17

Fuel, glass dump, cold critical

Claim	Status	Method	Pass	Home
TRISO-UCO 19.75% is a qualified fuel class for this burnup and temperature	Open	AGR/HTGR irradiation data + lot tests; NRC coated-particle fuel qualification	Failure fraction vs source term at 650 to 750°C compact, 80 GWd/t	§3.1, §3.2, §16
Fission gas stays in the TRISO buffer	Open	Fuel QAP + particle tests	No vented fuel; sealed vault	§3.2
Glass dump flows, sets, and kills reactivity	Open	Furnace mockup on a dummy lattice	Plug 750 ± 20°C; boron worth; no vessel cut	DB-12, §7.3
Glass dump is not credited until that test exists	Closed as policy	Spec	Not in the licensing basis until tested	DB-12, §7.3
Cold critical at the factory matches the model	Open	Licensed cell, zero-power physics test	Measured k-effective and drum worth vs model	§12, §16
Fresh core is cold on the truck	Closed as spec; Open as procedure	Locks + transport criticality (10 CFR 71)	FACTORY_SHUTDOWN / TRANSPORT, drums locked in	DB-9, §10

Safety functions (the stack)

Claim	Status	Method	Pass	Home
No water in the core to lose	Closed as design	Architecture	No LOCA path of that type	§3, §7
One or two dead heat pipes are ordinary faults	Closed for count; Open for test	Algebra + failed-pipe test	22 pipes still ≥ 35 kW	§4.1, §7.2
Analog drum trip, two channels, no phone	Open	I&C qualification (IEEE 323/344 class if claimed nuclear)	Condenser T, drum disagree, licensed key trip	§10.2
Owner cannot move drums	Closed as spec; Open as hardware	Locks, no owner key, app status only	App is not a control room	§10.2
Last-ditch glass is last	Bound as design	750°C plug vs 600°C vapor	Expected use: once in never	§7.3

What still needs outside testing, codes, or a stamp

This is the work Chapter 8 §16 listed as a checklist, written so a reader can see **what, with what, against which rule, and what pass looks like**. Until these rows are green, the book is a design specification, not a signed reactor.

Neutronics and depletion

What. k-effective in every §3.4 state; drum worth; flood; temperature coefficients; 100-year depletion.

Software. MCNP6 or Serpent for transport. SCALE/KENO as the independent check. ORIGEN (in SCALE) or Serpent depletion for burnup, poisons, and leftover fissile. A real lattice: compact channels, 24 heat pipes, 6 drums, reflector, inner vessel.

What you cannot use instead. This chapter. A handbook six-factor formula. A copied k-effective from Kilopower or eVinci (wrong size, wrong life, wrong packing).

Pass. The §3.4 table, DB-11 (under 0.8 dollars leftover), shutdown margin ≥ 1.0 dollar with the most reactive drum stuck withdrawn, net power coefficient negative 20°C to 750°C, k-effective(EOL) ≥ 1 with drums, peak compact ≤ 80 GWd/t.

Who. A nuclear-analysis shop with those codes, an independent peer check, then a cold-critical in a licensed cell.

Shielding and public dose

What. Dose at grass, lid bare, dry and wet soil, chase streaming, skyshine, bare-shell worker tally.

Software. MCNP6 or equivalent with the §6.3 stack, floor labyrinth, and chase. Running and shutdown sources from the depletion inventory, not from a single 2.4 n/fission number alone.

Regulation. Design target DB-5 / DB-6 (5 mrem/y live-on-grass). Federal public cap **10 CFR 20.1301** (100 mrem/y TEDE).

Occupational numbers for set and take-back under **10 CFR 20** subpart C. If someone cheats the BPE, dirt is not the shield: DB-6 is the enforcement.

Pass. $\leq 0.57 \mu\text{rem/h}$ at the grass in the running dry-sand lid-bare case; chase $\leq 50 \mu\text{rem/h}$; 1-hour bare-shell $< 100 \text{ mrem}$.

Who. Shielding analyst plus a survey plan for first units (you cannot skip the meter after the model).

Heat pipes

What. Startup from frozen sodium, steady 1.6 kW, one or two pipes failed in a block, freeze/thaw, long-life extrapolation.

Equipment. Heat-pipe vendor stand (ACT / Thermacore class): calorimetry, wick dryout, non-condensable gas. A graphite or electrically heated mock lattice for the failed-pipe test. Sodium handling in a qualified lab.

There is no 100-year certificate on the shelf. Tests of this class have run for years, not a century in a buried house appliance. Qualification is a test program plus conservative life models, plus fail-in-place and four plugged spares.

Pass. 35 kWth with two pipes out; startup procedure that does not over-heat the compact; documented life case for 100 years or an honest derate of the life claim.

Vessel, pad, lift, seismic

What. Inner vessel, outer shell, trunnions, pad, soil, earthquake.

Codes. **ASME Boiler and Pressure Vessel Code** Section VIII (shop vessel) or Section III (nuclear), including **Section III Division 5** if the high-temperature graphite and metallic core support is in the

nuclear class. Welds **ASME IX**. **ASCE 7** and **IBC** for the site; **ASCE 43** if the vault is treated as a nuclear structure. Soil: geotech report, 200 mm crushed stone or a thin mat.

Pass. Stresses under allowables at ≤ 0.5 MPa inner design pressure, lift, 0.3 g site class D, flood external pressure. Helium leak $\leq 1 \times 10^{-8}$ mbar·L/s on the inner vessel.

Type B transport of a fueled vault

What. The vault on the highway is a radioactive-material package, not a septic tank with a permit.

Regulation. 10 CFR 71. Type B tests under **71.73**: 9 m drop, puncture, 800°C for 30 minutes, immersion. Containment and shielding after the tests: **71.51**. Criticality safety of the package: **71.55 / 71.59**. International twin is IAEA SSR-6 if it ever leaves the US pattern.

Equipment. Drop pad, furnace or pool fire, immersion tank. A certified package design, not a paper claim.

Pass. After the hypothetical accident tests, containment holds, dose rates meet 71.51, and k-effective stays down (glass dump may fire; that is acceptable if the core stays subcritical).

This is a real cost and a real year-count. Do not skip it.

Fuel qualification

What. Particle failure fraction at this temperature, burnup, and fluence; source term if particles fail; lot tests at the fabricator.

Data and rules. Existing **AGR / HTGR** irradiation and safety-test data for TRISO-UCO. NRC path for coated-particle fuel (fuel qualification for advanced reactors). Fabrication **quality assurance** (NQA-1 / 10 CFR 50 Appendix B if the license is Part 50/52). Assay and enrichment control so no lot is HEU.

Equipment. Fuel line (BWXT / X-energy class), post-irradiation examination at a hot cell, compact and particle QA (radiography, leach, burn-leach).

Pass. Failure fraction low enough that the vault source term still meets DB-5 and 10 CFR 20 even in the bounding failed-particle case. Peak compact 80 GWd/t is inside the qualified envelope, or the envelope is extended with new irradiations.

Glass dump

What. 80 kg borosilicate frit + 15 kg B₄C, fusible plug 750 ± 20°C, gravity pour into channels, glass wets and sets, boron worth.

Equipment. Furnace mockup of a dummy lattice. Not a computer sketch.

Regulation. No existing “glass dump” rule. Treat it as a unique safety-related structure: test, then take credit. Until the furnace test exists, **DB-12 is not credited**.

Pass. Plug melts in the window, flow fills the intended channels, measured boron worth, inner vessel not cut, debris cannot jam drums in the out position.

Cold critical

What. Zero-power physics test of the real core cartridge: k-effective, drum worth, temperature coefficient at low power.

Where. A **licensed** facility (10 CFR 50 or 70), not the backyard, not a warehouse without a license.

Pass. Measurements match the model within the agreed band. Then drums lock in for shipping.

Stirling, inverter, oil loop

What. Electricity and fire on the house side.

Rules. Convertors: listing of the engine class (UL or equivalent for a hermetic 2 kWe free-piston unit). Inverter and interconnection: **UL 1741, IEEE 1547**. Gateway required to island. Oil: **NFPA 30** and the local fire marshal for a 350°C organic fluid in a buried double-wall chase and a utility room. Leak detection on the annulus. Hydronic isolation from potable water: plumbing code.

Pass. 8 kWe net, listed interconnection, detected double-wall, spill is a trench or room cleanup.

Material control, security, HALEU

What. 30 kg of 19.75% U is HALEU. Attractive. Not direct-use HEU. About 5.93 kg of U-235 contained.

Rules (high level, not a how-to). 10 CFR 70 (special nuclear material). **10 CFR 73** (physical protection). **10 CFR 74** (material control and accounting). Classification of the quantity and enrichment is an NRC (or Agreement State) determination. This section does not design a guard force.

Product facts that feed that determination: 42 t monolith, no owner hatch, IAEA-style seals, drums locked, GPS/tamper on transport, two-person drum key, licensed startup only.

Pass. A written MC&A and protection plan the regulator accepts for a distributed house fleet, not a one-off lab. That plan does not exist in this book.

License path for the object itself

What. Permission to manufacture, ship, set, start, and take back a furnace-scale sealed core as a **product**, many times, rather than a custom plant on every lot.

Rules that exist today. 10 CFR 50 and **52** are plant licenses and manufacturing licenses aimed at plants. **10 CFR 53** (final rule effective 29 April 2026) is an optional, risk-informed, technology-inclusive framework for a **commercial nuclear plant**, including microreactors. It is the right family of paper for a non-water core. It is still a plant license, not a house-appliance stamp. Transport is **10 CFR 71** on any path. Environmental review: **NEPA, 10 CFR 51**. Chapter 6 is the policy ask: size-based manufactured appliance, approved once, installed many times. **That stamp does not exist yet.**

Pass. A product certificate, general license, or exemption that matches Chapter 6: one certified design, licensed hands at fuel, transport, set, start, and take-back, owner is not the operator. Until then every early unit is a special case.

Who is the licensee (factory vs owner vs operating company) is a legal design choice. This book does not pick it.

Quality assurance

What. Traceability from a TRISO lot to a vault serial number to a lawn.

Rules. NQA-1. 10 CFR 50 Appendix B if the nuclear license is Part 50/52. Shop stamps (ASME N or equivalent) as the QA program dictates.

Pass. A QA program a regulator will audit. Not a markdown file.

What this book has shown, and what it has not

Shown (Closed). Energy balance 8 kWe / 35 kWth / 27 kWth. Annual kilowatt-hours. 1,278 MWd and 16 kg consumed at 80 GWd/t. Fission rate and 1.35 kg U-235-equivalent. 5.93 kg U-235 at first start. Pipe **count** versus 35 kW. Oil **flow**. Vault **will not float**. Playset **ratio**. Core volume and 0.56 kW/L. Unshielded source magnitude. 350 kJ in 10 s. Carnot is 47% at the example temperatures, so 25% is not a physics error. Doppler as a mechanism. Decay heat as a heater by the Way fit.

Bounded, not signed. Shield TVLs and $1/r^2$. Fluence order of magnitude. Earth ΔT at decay heat. Type-unit mass takeoff. Heat-pipe architecture (single envelope, vessel is the second wall).

Not shown. k-effective in any state. Lawn dose as a result. 100-year depletion. 100-year heat pipes, graphite, and TRISO at this flux. Type B tests. ASME stresses. Glass-dump furnace. Cold critical. Listed Stirling at 25% for this oil temperature. HALEU contract. A house-appliance nuclear stamp. Who the licensee is.

That split is the honest version of a Chapter 10. A later revision can turn Open rows green when the runs and tests exist. It should not turn them green with invented output.

What this chapter covered

This is the proof plan, not a proof. Each design claim is a proof item: Closed, Bound, or Open, with a method and a pass number. The Closed arithmetic is on the page: power, inventory, fission rate, decay-heat fit, pipe count, oil flow, float, playset ratio, fluence order of magnitude. Handbook TVLs, inverse square, and Carnot size the shield and the engine; they do not replace MCNP or a listed Stirling. Hotter means less splitting is traced to a negative net power coefficient and to DB-11, without a fake coefficient.

What still needs specialized software or equipment is named with the spec or test: MCNP/SCALE/ORIGEN, heat-pipe stands, ASME and ASCE, 10 CFR 71 Type B, AGR/HTGR fuel data, a glass-dump furnace, a licensed cold-critical cell, UL/IEEE/NFPA on the house side, 10 CFR 20/70/73/74, and a license path that does not yet exist for a house appliance. Chapter 8 is the design basis. Chapter 9 is the fuel. Chapter 6 is the stamp. Until the Open rows are green, this is a specification you can start a sign-off package from, not a signed reactor.